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Delivery of a gene-editing system with a single retroviral particle and methods of generation and use

Abstract

The invention provides a recombinant RNA molecule comprising (i) a sequence of a gene-editing molecule mRNA, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of a coding or non-coding enrichment RNA, or a sequence of a functional fragment or derivative thereof, wherein the enrichment RNA, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing molecule mRNA, or functional fragment or derivative thereof, into a retroviral particle. The invention provides a method of producing the retroviral particles of the invention, the method comprising culturing a packaging cell in conditions sufficient for the production of a plurality of retroviral particles.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage application of International Application No. PCT/US2018/049547, filed Sep. 5, 2018, which claims priority to U.S. Provisional Patent Application No. 62/554,500 filed Sep. 5, 2017, all of which is hereby incorporated by reference in their entirety.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Nov. 2, 2018, is named 250298_000061_SL.txt and is 337,425 bytes in size.

TECHNICAL FIELD

(2) Described herein are retroviral particles for delivery of gene-editing molecules to a target cell.

BACKGROUND

(3) The Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR)/CRISPR-associated (Cas) system is a powerful genome editing technique allowing for a permanent disruption, deletion, repair, mutation, addition, alteration, or modification of a gene sequence at a target locus in a gene after a single administration. Current methods rely on two essential elements: a component for targeted DNA recognition (e.g., guide RNA) and a component for nuclease activity to induce a single or double strand break at the specific site in the genomic DNA (e.g., Cas9 nucleases). See Mout et al., (2017) *Bioconjugate Chem* 25:880-884. In instances where there is a repair or genetic alteration to be made, a repair template with the desired repair/mutation may be present. Id. One way to deliver the CRISPR/Cas system to the cell of interest is by using a viral packaging system. Virol genomes, however, are compact and there is typically no room for a Cas protein coding sequence together with the guide RNA and repair template in the same viral particle genome. For example, the size limit of a cargo gene in adeno-associated viral particles is ˜4.7 kbp, while the size of SpCas9 by itself is ˜4.3 kbp. Id. For lentiviral particles, the packaging limit is ˜8.5 kbp—however, inserts larger than ˜3 kbp are packaged less efficiently. See e.g., Komor et al., (2017) *Cell* 168:20-36. Thus, the existing delivery systems suffer from inefficiencies because the different CRISPR/Cas components are split into different delivery units (e.g., via two viral particles or via a viral particle and a lipid nanoparticle) or smaller Cas proteins would need to be used. Moreover, many of the current systems require the CRISPR genes to remain in the target cells once delivered, resulting in unwanted gene editing and potential immunogenic reactions due to the constitutive expression. See e.g., Mout et al., (2017) *ACS Nano* 11:2452-2458.

SUMMARY

(4) As specified in the Background section, above, there is a need in the art for a single recombinant viral particle system and methods for generating such particles that allow for the efficient delivery of all components of a gene-editing system (e.g., the CRISPR/Cas system) required for gene modification at a target locus in a genome. The present invention addresses this and other needs by providing recombinant retroviral particles for delivery of gene-editing molecules to target cells. The retroviral particles described herein are capable of delivering together all components of a gene-editing system (e.g., the CRISPR/Cas system) required for gene modification in a target cell and for the treatment of various diseases.

(5) In one aspect, the invention provides a recombinant RNA molecule comprising (i) a sequence of a gene-editing molecule mRNA, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of a coding or non-coding enrichment RNA, or a sequence of a functional fragment or derivative thereof, wherein the enrichment RNA, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing molecule mRNA, or functional fragment or derivative thereof, into a retroviral particle. In some embodiments, the gene-editing molecule mRNA, or a fragment or derivative thereof, is codon optimized for expression in a eukaryotic cell (e.g., a non-human mammalian cell, a human cell, a rodent cell, a pluripotent cell, a one-cell stage embryo, a differentiated cell, or a combination thereof). In some embodiments, the

sequence of the gene-editing molecule mRNA, or a sequence of a fragment or derivative thereof, and the sequence of the small RNA, or a sequence of a fragment or derivative thereof are separated by a linker sequence. In some embodiments, the gene-editing molecule is a Cas protein (e.g., Cpf1, CasX, CasY, C2C2, Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas5e (CasD), Cash, Cas6e, Cas6f, Cas7, Cas8a1, Cas8a2, Cas8b, Cas8c, Cas9 (Csn1 or Csx12), Cas10, Cas10d, CasF, CasG, CasH, Csy1, Csy2, Csy3, Cse1 (CasA), Cse2 (CasB), Cse3 (CasE), Cse4 (CasC), Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4, or Cu1966). In some embodiments, the Cas protein is a Cas9 protein (e.g., wild type Cas9, a Cas9 nickase, a dead Cas9 (dCas9), or a split Cas9). In some embodiments, the Cas9 protein is a *Streptococcus pyogenes* Cas9 protein or *Staphylococcus aureus* Cas9 protein. In some embodiments, the enrichment RNA is a small non-coding or coding RNA. In some embodiments, the small non-coding RNA is 7SL RNA, tRNA (including primer tRNAs), 5S rRNA, U1 snRNA, U2 snRNA, U6 snRNA, Y1 RNA, Y3 RNA, B1 RNA, VL30 RNA, 7SK RNA, Alu RNA, miRNA, snoRNA, or cytoplasmic vault ncRNA. In some embodiments, the small non-coding RNA is 7SL RNA. In some embodiments, the gene-editing molecule mRNA is Cas9 mRNA and the non-coding RNA is 7SL RNA. In some embodiments, the retroviral particle is a lentiviral particle.
(6) In a related aspect, the invention provides nucleic acid molecules encoding the recombinant RNA molecules of the invention as well as vectors comprising the nucleic acid molecules, wherein the nucleic acid molecule encoding a recombinant RNA molecule is operably linked to a promoter.
(7) In a further aspect, the invention provides host cells comprising the RNA molecules of the invention or the nucleic acid molecules encoding such RNA molecules or the vectors comprising the nucleic acid molecules.
(8) In another aspect, the invention provides recombinant retroviral particles comprising the RNA molecules of the invention. In some embodiments, the retroviral particles are lentiviral particles. In some embodiments, the retroviral particles further comprise a nucleic acid molecule encoding one or more guide RNAs (gRNA) and/or a nucleic acid molecule comprising one or more sequences corresponding to one or more repair templates (RT). In some embodiments, the nucleic acid sequence(s) encoding the gRNA and the nucleic acid sequence(s) corresponding to the RT are located within the same nucleic acid molecule. In some embodiments, in addition to the sequence(s) encoding the gRNA and/or the sequence(s) corresponding to the RT, the nucleic acid molecule comprises one or more retroviral elements. In some embodiments, the retroviral particles are replication deficient.
(9) In some embodiments, the retroviral particles further comprise a gene-editing molecule fusion protein. In some embodiments, the gene-editing molecule fusion protein comprises (i) a sequence of a gene-editing protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of an enrichment protein, or a sequence of a functional fragment or derivative thereof, wherein the enrichment protein, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing protein, or functional fragment or derivative thereof, into the retroviral particle. In some embodiments, (i) the sequence of the gene-editing molecule, or a sequence of a fragment or derivative thereof, and (ii) the at least one sequence of the enrichment protein, or a sequence of a fragment or derivative thereof, are separated by a linker sequence (e.g., (G.sub.4S).sub.3 (SEQ ID NO: 89)). In some embodiments, the gene-editing molecule is a Cas protein (e.g., Cpf1, CasX, CasY, C2C2, Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas5e (CasD), Cash, Cas6e, Cas6f, Cas7, Cas8a1, Cas8a2, Cas8b, Cas8c, Cas9 (Csn1 or Csx12), Cas10, Cas10d, CasF, CasG, CasH, Csy1, Csy2, Csy3, Cse1 (CasA), Cse2 (CasB), Cse3 (CasE), Cse4 (CasC), Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4, Cu1966, or homologs or modified versions thereof). In some embodiments the Cas protein is a Cas9 protein (e.g., a wild type Cas9, a Cas9 nickase, a dead Cas9 (dCas9), or a split Cas9). In

some embodiments, the Cas9 protein is a *Streptococcus pyogenes* Cas9 protein or *Staphylococcus aureus* Cas9 protein. In some embodiments, the enrichment protein is cyclophilin A (CypA) protein and/or a viral protein R (Vpr).

(10) In another aspect, the invention provides a method of producing the retroviral particles of the invention, the method comprising culturing a packaging cell in conditions sufficient for the production of a plurality of retroviral particles, wherein the packaging cell comprises one or more plasmids comprising (i) one or more retroviral elements involved in the assembly of the retroviral particle, and (ii) at least one nucleic acid sequence encoding the RNA molecule of the invention. In some embodiments, the packaging cell further comprises a plasmid encoding one or more guide RNAs (gRNA) and/or comprising one or more sequences corresponding to one or more repair templates (RT). In some embodiments, the packaging cell comprises (a) GAG, (b) POL, and (c) TAT and/or REV retroviral (e.g., lentiviral) elements. In some embodiments, the method of producing the retroviral particles of the invention further comprises collecting the retroviral particles. In some embodiments, the method of producing the retroviral particles of the invention comprises one or more of the following steps: (a) clearing cell debris, (b) treating a supernatant containing the retroviral particles with DNase I and MgCl.sub.2, (c) concentrating the retroviral particles, and (d) purifying the retroviral particles.

(11) In a related embodiment, the invention provides the retroviral particle made by any of the above methods.

(12) In another aspect, the invention provides pharmaceutical compositions comprising any of the retroviral particles of the invention and a pharmaceutically acceptable carrier or excipient.

(13) In a further aspect, the invention provides pharmaceutical dosage forms comprising any of the retroviral particles of the invention.

(14) In yet another aspect, the invention provides a method for modifying a genome of a target cell comprising introducing into the cell any of the retroviral particles of the invention. In some embodiments, the genome modification is a disruption, deletion, repair, mutation, addition, alteration, or modification of a gene.

(15) In yet another aspect, the invention provides a method for modulating an activity of a gene in a target cell comprising introducing into the cell any of the retroviral particles of the invention. In some embodiments, the method inhibits, suppresses, down regulates, knocks down, knocks out, or silences the expression of a gene product. In some embodiments, the gene product is a protein or RNA.

(16) In some embodiments of any of the above methods, the target cell is in a subject and the retroviral particle is administered to the subject.

(17) In some embodiments of any of the above methods, the method further comprises harvesting the target cell from a subject prior to introducing the retroviral particle into the target cell, introducing the retroviral particle into the target cell ex vivo and returning the target cell to the subject.

(18) In some embodiments of any of the above methods, the method further comprises introducing into the target cell one or more gRNA molecules (e.g., as a complex with proteins). In some embodiments, such separate introduction of gRNA molecules is conducted when retroviral particles comprise no gRNA. In some embodiments, retroviral particles comprise one or more gRNA and separately introduced gRNA provide additional gRNA molecules.

(19) In a further aspect, the invention provides a method for treating a disease in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of any of the retroviral particles of the invention or a pharmaceutical composition comprising such retroviral particles, wherein the retroviral particles target a cell in the subject.

(20) In some embodiments of any of the above methods involving administration to the subject, the retroviral particle or the pharmaceutical composition is administered intravenously, subcutaneously, intramuscularly, transdermally, intranasally, orally, or mucosally.

(21) In a further aspect, the invention provides a method for treating a disease in a subject in need thereof, the method comprising:
(22) a. harvesting a target cell from the subject;
(23) b. introducing into the target cell from step a) ex vivo a therapeutically effective amount of any of the retroviral particles of the invention or a pharmaceutical composition comprising such retroviral particles; and
(24) c. returning the target cell from step b) to the subject.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A-C provide a schematic representation of concepts and four exemplary Cas9 constructs of the invention. FIG. **1**A shows recruiting the Cas9 fusion proteins or Cas9 mRNAs into lentiviral particles for transduction of functional Cas9 proteins into infected cells. The gRNA is contained in the retroviral genome. FIG. **1**B shows 7SL RNA constructs, wherein 7SL RNA is fused before the start codon or after stop codon of Cas9 mRNA. FIG. **1**C shows Cas9 protein constructs: cyclophilin A (CypA) fused to the N-terminus of Cas9 protein or viral protein R (Vpr) fused to the C-terminus of the Cas9 protein via a linker (e.g., (G.sub.4S).sub.3 (SEQ ID NO: 89)).
(2) FIG. **2** A-G show fluorescence-activated cell sorting (FACS) analysis evaluating expression upon infection of 293T pLVX EF1a eGFP IRES puro c12 cells with: (FIG. **2**A) mock infection; (FIG. **2**B) Cas9/non-gRNA: a lentiviral particle carrying randomly packaged Cas9 mRNA or proteins, but no gRNA on lentiviral genome; (FIG. **2**C) Cas9/GFP gRNA: a lentiviral particle carrying randomly packaged Cas9 proteins and mRNAs and gRNA; (FIG. **2**D) CypA-Cas9/GFP gRNA: a lentiviral particle carrying CypA-Cas9 fusion protein and gRNA; (FIG. **2**E) Cas9-Vpr/GFP gRNA: a lentiviral particle carrying Cas9-Vpr fusion protein and gRNA; (FIG. **2**F) a lentiviral particle carrying 7SL RNA-Cas9 mRNA fusion RNA and gRNA; and (FIG. **2**G) a lentiviral particle carrying Cas9 mRNA-7SL RNA fusion RNA and gRNA. All gRNAs are on the lentiviral genome driven by hU6 promoter. Lentiviral particles are produced by transfecting corresponding plasmids in Tables 2 and 3 into 293T cells and harvested by ultracentrifugation. 1E9 vg of viruses were added to 50,000 of 293T pLVX EF1a eGFP IRES puro c12 cells. FACS was performed 7 days after infection.
(3) FIG. **3** shows Sybr green quantitative PCR (qPCR) data (Thermofisher scientific Cat #11762100) demonstrating the enrichment of Cas9 mRNA in the lentiviral particles upon fusion to 7SL RNA. Depicted are the qPCR results obtained using RNA extracted from: (i) a standard lentiviral RNA from the Lenti-X qPCR titration kit (Clontech #631235), lentiviral particles randomly packaged; (ii) Cas9 mRNA; and/or (iii) 7SL RNA-Cas9 mRNA fusion RNA molecule. Two primers detecting lentiviral genome and Cas9 mRNA were used (sequences: Cas9F GACAGGCACAGCATCAAGAA (SEQ ID NO: 69) Cas9R TTCTGGCGGTTCTCTTCAGT (SEQ ID NO: 70)). The Cas9 mRNA level was normalized to lentiviral genome level.
(4) FIG. **4** is a schematic representation of a nucleic acid molecule comprising the wild type (WT) LacZ and Zombie LacZ (Q585E mutation), which has abolished enzymatic activity. A functional Cas9/gRNA system can convert the Zombie LacZ to WT LacZ. FIG. **4** discloses SEQ ID NOS 146-149, respectively, in order of appearance.
(5) FIG. **5**A is a schematic representation of LV gRNA/RT/Cas9 7SL, a lentivirus carrying Zombie LacZ gRNA and repair template on its genome and Cas9-7SL fusion RNA. FIG. **5**B is a graph showing the results of luminescence assay analysis evaluating β-galactosidase expression (i.e., the ability to convert the Zombie LacZ to WT LacZ) upon infection of pShuttle CMV Zombie lacZ transfected 293T cells with (i) a mock lentiviral vector system; (ii) gRNA/RT: a lentiviral particle carrying Zombie lacZ gRNA and repair template (RT); and (iii) gRNA/RT/Cas9 7SL: a lentiviral

particle carrying gRNA and RT in its genome and Cas9 mRNA-7SL RNA fusion RNA. Lentiviral particles are produced by transfecting corresponding plasmids in Table 4 into 293T cells and harvested by ultracentrifugation.
(6) FIG. **6**A shows the secondary structure of an exemplar 7SL RNA, including its Alu and S domains. FIG. **6**A discloses SEQ ID NO: 81. FIG. **6**B shows an example of the Alu domain of a 7SL RNA. FIG. **6**B discloses SEQ ID NO: 150. FIGS. **6**C and **6**D show non-limiting examples of fragments of the Alu domain of a 7SL RNA. FIG. **6**C discloses SEQ ID NO: 151. FIG. **6**D discloses SEQ ID NO: 152. FIG. **6**E shows the S domain of a 7SL RNA. FIG. **6**E discloses SEQ ID NO: 153. FIG. **6**F shows a non-limiting example of a fragment of the S domain of a 7SL RNA. FIG. **6**F discloses SEQ ID NO: 154. FIG. **6**G shows a non-limiting example of a derivative of the S domain of a 7SL RNA. FIG. **6**G discloses SEQ ID NO: 155. FIGS. **6**H and **6**I show non-limiting examples of the 5c domain of a 7SL RNA. FIG. **6**H discloses SEQ ID NO: 156. FIG. **6**I discloses SEQ ID NO: 90. The numbers in FIGS. **6**A-**6**G represent established nomenclature for 7SL RNA helices and loops. See also Keene and Telesnitsky *J Virol.* 2012 August; 86(15): 7934-7942, incorporated by reference herein in its entirety for all purposes.

DETAILED DESCRIPTION

(7) The present invention provides recombinant retroviral particles for the delivery of a gene-editing fusion molecule, and optionally a guide RNA (gRNA) and/or a repair template, in one retroviral particle. The retroviral particles described herein are capable of delivery of all components required for gene modification and/or modulating an activity of a gene in a target cell and are therefore useful for treatment of various diseases treatable by modification and/or modulating an activity of a gene.

I. Definitions

(8) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs.
(9) The terms “protein,” “polypeptide,” and “peptide,” used interchangeably herein, include polymeric forms of amino acids of any length, including coded and non-coded amino acids and chemically or biochemically modified or derivatized amino acids. The terms also include polymers that have been modified, such as polypeptides having modified peptide backbones.
(10) Proteins are said to have an “N-terminus” and a “C-terminus.” The term “N-terminus” relates to the start of a protein or polypeptide, terminated by an amino acid with a free amine group (—NH2). The term “C-terminus” relates to the end of an amino acid chain (protein or polypeptide), terminated by a free carboxyl group (—COOH).
(11) The terms “nucleic acid” and “polynucleotide,” used interchangeably herein, include polymeric forms of nucleotides of any length, including ribonucleotides, deoxyribonucleotides, or analogs or modified versions thereof. They include single-, double-, and multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, and polymers comprising purine bases, pyrimidine bases, or other natural, chemically modified, biochemically modified, non-natural, or derivatized nucleotide bases.
(12) Nucleic acids are said to have “5′ ends” and “3′ ends” because mononucleotides are reacted to make oligonucleotides in a manner such that the 5′ phosphate of one mononucleotide pentose ring is attached to the 3′ oxygen of its neighbor in one direction via a phosphodiester linkage. An end of an oligonucleotide is referred to as the “5′ end” if its 5′ phosphate is not linked to the 3 oxygen of a mononucleotide pentose ring. An end of an oligonucleotide is referred to as the “3′ end” if its 3′ oxygen is not linked to a 5′ phosphate of another mononucleotide pentose ring. A nucleic acid sequence, even if internal to a larger oligonucleotide, also may be said to have 5′ and 3′ ends. In either a linear or circular DNA molecule, discrete elements are referred to as being “upstream” or 5′ of the “downstream” or 3′ elements.
(13) The term “wild type” includes entities having a structure and/or activity as found in a normal (as contrasted with mutant, diseased, altered, or so forth) state or context. Wild type genes and

polypeptides often exist in multiple different forms (e.g., alleles).
(14) “Exogenous” molecules or sequences include molecules or sequences that are not normally present in a cell in that form. Normal presence includes presence with respect to the particular developmental stage and environmental conditions of the cell. An exogenous molecule or sequence, for example, can include a mutated version of a corresponding endogenous sequence within the cell, such as a humanized version of the endogenous sequence, or can include a sequence corresponding to an endogenous sequence within the cell but in a different form (i.e., not within a chromosome). In contrast, endogenous molecules or sequences include molecules or sequences that are normally present in that form in a particular cell at a particular developmental stage under particular environmental conditions.
(15) “Codon optimization” takes advantage of the degeneracy of codons, as exhibited by the multiplicity of three-base pair codon combinations that specify an amino acid, and generally includes a process of modifying a nucleic acid sequence for enhanced expression in particular host cells (e.g., packaging cells) and/or target cells by replacing at least one codon of the native sequence with a codon that is more frequently or most frequently used in the genes of the host cells and/or target cells while maintaining the native amino acid sequence. For example, a nucleic acid encoding a Cas9 protein can be modified to substitute codons having a higher frequency of usage in a given prokaryotic or eukaryotic cell, including a bacterial cell, a yeast cell, a human cell, a non-human cell, a mammalian cell, a rodent cell, a mouse cell, a rat cell, a hamster cell, or any other host and/or target cell, as compared to the naturally occurring nucleic acid sequence. Codon usage tables are readily available, for example, at the “Codon Usage Database.” These tables can be adapted in a number of ways. See Nakamura et al. (2000) *Nucleic Acids Research* 28:292, herein incorporated by reference in its entirety for all purposes. Computer algorithms for codon optimization of a particular sequence for expression in a particular host and/or target are also available (see, e.g., Gene Forge).
(16) A “promoter” is a regulatory region of DNA usually comprising a TATA box capable of directing RNA polymerase II to initiate RNA synthesis at the appropriate transcription initiation site for a particular polynucleotide sequence. A promoter may additionally comprise other regions which influence the transcription initiation rate. As used herein, the term “promoter” encompasses enhancers. The promoter sequences disclosed herein modulate transcription of an operably linked polynucleotide. A promoter can be active in one or more of the cell types disclosed herein (e.g., a eukaryotic cell, a non-human mammalian cell, a human cell, a rodent cell, a pluripotent cell, a one-cell stage embryo, a differentiated cell, or a combination thereof). A promoter can be, for example, a constitutively active promoter, a conditional promoter, an inducible promoter, a temporally restricted promoter (e.g., a developmentally regulated promoter), or a spatially restricted promoter (e.g., a cell-specific or tissue-specific promoter). RNA Pol III promoters are frequently used to express small RNAs, such as small interfering RNA (siRNA)/short hairpin RNA (shRNA) and guide RNA sequences used in CRISPR-Cas9 systems. Examples of RNA Pol III promoters that can be used in the invention include, but are not limited to, the human U6 promoter, a rat U6 polymerase III promoter, or a mouse U6 polymerase III promoter, and the HI promoter, which are described in, for example Goomer and Kunkel, Nucl. Acids Res., 20 (18): 4903-4912 (1992), and Myslinski et al., Nucleic Acids Res., 29(12): 2502-9 (2001). Examples of promoters can be found, for example, in WO 2013/176772, herein incorporated by reference in its entirety for all purposes.
(17) Examples of inducible promoters include, for example, chemically regulated promoters and physically-regulated promoters. Chemically regulated promoters include, for example, alcohol-regulated promoters (e.g., an alcohol dehydrogenase (alcA) gene promoter), tetracycline-regulated promoters (e.g., a tetracycline-responsive promoter, a tetracycline operator sequence (tetO), a tet-On promoter, or a tet-Off promoter), steroid regulated promoters (e.g., a rat glucocorticoid receptor, a promoter of an estrogen receptor, or a promoter of an ecdysone receptor), or metal-regulated promoters (e.g., a metalloprotein promoter). Physically regulated promoters include, for example

temperature-regulated promoters (e.g., a heat shock promoter) and light-regulated promoters (e.g., a light-inducible promoter or a light-repressible promoter).
(18) Tissue-specific promoters can be, for example, neuron-specific promoters, glia-specific promoters, muscle cell-specific promoters, heart cell-specific promoters, kidney cell-specific promoters, bone cell-specific promoters, endothelial cell-specific promoters, or immune cell-specific promoters (e.g., a B cell promoter or a T cell promoter).
(19) Developmentally regulated promoters include, for example, promoters active only during an embryonic stage of development, or only in an adult cell.
(20) “Operable linkage” or being “operably linked” includes juxtaposition of two or more components (e.g., a promoter and another sequence element) such that both components function normally and allow the possibility that at least one of the components can mediate a function that is exerted upon at least one of the other components. For example, a promoter can be operably linked to a coding sequence if the promoter controls the level of transcription of the coding sequence in response to the presence or absence of one or more transcriptional regulatory factors. Operable linkage can include such sequences being contiguous with each other or acting in trans (e.g., a regulatory sequence can act at a distance to control transcription of the coding sequence).
(21) “Complementarity” or “complementary” of nucleic acids means that a nucleotide sequence in one strand of nucleic acid, due to orientation of its nucleobase groups, forms hydrogen bonds with another sequence on an opposing nucleic acid strand. The complementary bases in DNA are typically A with T and C with G. In RNA, they are typically C with G and U with A. Complementarity can be perfect or substantial/sufficient. Perfect complementarity between two nucleic acids means that the two nucleic acids can form a duplex in which every base in the duplex is bonded to a complementary base by Watson-Crick pairing. “Substantial” or “sufficient” complementary means that a sequence in one strand is not completely and/or perfectly complementary to a sequence in an opposing strand, but that sufficient bonding occurs between bases on the two strands to form a stable hybrid complex in set of hybridization conditions (e.g., salt concentration and temperature). Such conditions can be predicted by using the sequences and standard mathematical calculations to predict the Tm (melting temperature) of hybridized strands, or by empirical determination of Tm by using routine methods. Tm includes the temperature at which a population of hybridization complexes formed between two nucleic acid strands are 50% denatured (i.e., a population of double-stranded nucleic acid molecules becomes half dissociated into single strands). At a temperature below the Tm, formation of a hybridization complex is favored, whereas at a temperature above the Tm, melting or separation of the strands in the hybridization complex is favored. Tm may be estimated for a nucleic acid having a known G+C content in an aqueous 1 M NaCl solution by using, e.g., Tm=81.5+0.41(% G+C), although other known Tm computations take into account nucleic acid structural characteristics.
(22) “Hybridization condition” includes the cumulative environment in which one nucleic acid strand bonds to a second nucleic acid strand by complementary strand interactions and hydrogen bonding to produce a hybridization complex. Such conditions include the chemical components and their concentrations (e.g., salts, chelating agents, formamide) of an aqueous or organic solution containing the nucleic acids, and the temperature of the mixture. Other factors, such as the length of incubation time or reaction chamber dimensions may contribute to the environment. See, e.g., Sambrook et al., Molecular Cloning, A Laboratory Manual, 2.sup.nd ed., pp. 1.90-1.91, 9.47-9.51, 11.47-11.57 (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1989), herein incorporated by reference in its entirety for all purposes.
(23) Hybridization requires that the two nucleic acids contain complementary sequences, although mismatches between bases are possible. The conditions appropriate for hybridization between two nucleic acids depend on the length of the nucleic acids and the degree of complementation, variables which are well known. The greater the degree of complementation between two nucleotide sequences, the greater the value of the melting temperature (Tm) for hybrids of nucleic

acids having those sequences. For hybridizations between nucleic acids with short stretches of complementarity (e.g. complementarity over 35 or fewer, 30 or fewer, 25 or fewer, 22 or fewer, 20 or fewer, or 18 or fewer nucleotides) the position of mismatches becomes important (see Sambrook et al., supra, 11.7-11.8). Typically, the length for a hybridizable nucleic acid is at least about 10 nucleotides. Illustrative minimum lengths for a hybridizable nucleic acid include at least about 15 nucleotides, at least about 20 nucleotides, at least about 22 nucleotides, at least about 25 nucleotides, and at least about 30 nucleotides. Furthermore, the temperature and wash solution salt concentration may be adjusted as necessary according to factors such as length of the region of complementation and the degree of complementation.

(24) The sequence of polynucleotide need not be 100% complementary to that of its target nucleic acid/target locus to be specifically hybridizable. Moreover, a polynucleotide may hybridize over one or more segments such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure or hairpin structure). A polynucleotide (e.g., gRNA) can comprise at least 70%, at least 80%, at least 90%, at least 95%, at least 99%, or 100% sequence complementarity to a target region within the target nucleic acid/target locus sequence to which they are targeted. For example, a gRNA in which 18 of 20 nucleotides are complementary to a target region, and would therefore specifically hybridize, would represent 90% complementarity. In this example, the remaining noncomplementary nucleotides may be clustered or interspersed with complementary nucleotides and need not be contiguous to each other or to complementary nucleotides.

(25) Percent complementarity between particular stretches of nucleic acid sequences within nucleic acids can be determined routinely using BLAST programs (basic local alignment search tools) and PowerBLAST programs (Altschul et al. (1990) *J Mol. Biol.* 215:403-410; Zhang and Madden (1997) *Genome Res.* 7:649-656) or by using the Gap program (Wisconsin Sequence Analysis Package, Version 8 for Unix, Genetics Computer Group, University Research Park, Madison Wis.), using default settings, which uses the algorithm of Smith and Waterman (Adv. Appl. Math., 1981, 2, 482-489).

(26) The methods and compositions provided herein employ a variety of different components. Some components throughout the description can have active variants and fragments. Such components include, for example, Cas proteins, CRISPR RNAs, and guide RNAs. Biological activity for each of these components is described elsewhere herein. The term “functional” refers to the innate ability of a protein or nucleic acid (or a fragment or derivative thereof) to exhibit a biological activity or function. Such biological activities or functions can include, for example, the ability of a Cas protein, or functional fragment or derivative thereof, to bind to a guide RNA (gRNA), bind to a target DNA sequence, exhibit endonuclease activity, or to retain its ability to be incorporated into the retroviral particle. The biological functions of functional fragments or derivatives may be the same or may in fact be changed (e.g., with respect to their specificity or selectivity or efficacy) in comparison to the full-length wild type molecule or original counterpart, but with retention of the basic biological function of gene-editing.

(27) The term “fragment” when referring to a protein means a protein that is shorter or has fewer amino acids than the full-length protein. A fragment can be, for example, an N-terminal fragment (i.e., removal of a portion of the C-terminal end of the protein), a C-terminal fragment (i.e., removal of a portion of the N-terminal end of the protein), or an internal fragment. The term “fragment” when referring to a nucleic acid means a nucleic acid that is shorter or has fewer nucleotides than the full-length nucleic acid. A fragment can be, for example, a 5′ fragment (i.e., removal of a portion of the 3′ end of the nucleic acid), a 3′ fragment (i.e., removal of a portion of the 5′ end of the protein), or an internal fragment.

(28) The term “derivative” as used herein refers to a nucleic acid, peptide, or protein or a variant or analog thereof comprising one or more mutations and/or chemical modifications as compared to a corresponding full-length wild type nucleic acid, peptide or protein. Non-limiting examples of

chemical modifications involving nucleic acids include, for example, modifications to the base moiety, sugar moiety, phosphate moiety, phosphate-sugar backbone, or a combination thereof.

(29) “Sequence identity” or “identity” in the context of two polynucleotides or polypeptide sequences makes reference to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window. When percentage of sequence identity is used in reference to proteins, residue positions which are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have “sequence similarity” or “similarity.” Means for making this adjustment are well known. Typically, this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, California).

(30) “Percentage of sequence identity” includes the value determined by comparing two optimally aligned sequences (greatest number of perfectly matched residues) over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity. Unless otherwise specified (e.g., the shorter sequence includes a linked heterologous sequence), the comparison window is the full length of the shorter of the two sequences being compared.

(31) Unless otherwise stated, sequence identity/similarity values include the value obtained using GAP Version 10 using the following parameters: % identity and % similarity for a nucleotide sequence using GAP Weight of 50 and Length Weight of 3, and the nwsgapdna.cmp scoring matrix; % identity and % similarity for an amino acid sequence using GAP Weight of 8 and Length Weight of 2, and the BLOSUM62 scoring matrix; or any equivalent program thereof “Equivalent program” includes any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by GAP Version 10.

(32) The term “conservative amino acid substitution” refers to the substitution of an amino acid that is normally present in the sequence with a different amino acid of similar size, charge, or polarity. Examples of conservative substitutions include the substitution of a non-polar (hydrophobic) residue such as isoleucine, valine, or leucine for another non-polar residue. Likewise, examples of conservative substitutions include the substitution of one polar (hydrophilic) residue for another such as between arginine and lysine, between glutamine and asparagine, or between glycine and serine. Additionally, the substitution of a basic residue such as lysine, arginine, or histidine for another, or the substitution of one acidic residue such as aspartic acid or glutamic acid for another acidic residue are additional examples of conservative substitutions. Examples of non-conservative substitutions include the substitution of a non-polar (hydrophobic) amino acid residue such as isoleucine, valine, leucine, alanine, or methionine for a polar (hydrophilic) residue such as cysteine, glutamine, glutamic acid or lysine and/or a polar residue for a non-polar residue. Typical amino acid categorizations are summarized below.

(33) TABLE-US-00001 Alanine Ala A Nonpolar Neutral 1.8 Arginine Arg R Polar Positive −4.5 Asparagine Asn N Polar Neutral −3.5 Aspartic acid Asp D Polar Negative −3.5 Cysteine Cys C Nonpolar Neutral 2.5 Glutamic acid Glu E Polar Negative −3.5 Glutamine Gln Q Polar Neutral −3.5 Glycine Gly G Nonpolar Neutral −0.4 Histidine His H Polar Positive −3.2 Isoleucine Ile I Nonpolar Neutral 4.5 Leucine Leu L Nonpolar Neutral 3.8 Lysine Lys K Polar Positive −3.9 Methionine Met M Nonpolar Neutral 1.9 Phenylalanine Phe F Nonpolar Neutral 2.8 Proline Pro P Nonpolar Neutral −1.6 Serine Ser S Polar Neutral −0.8 Threonine Thr T Polar Neutral −0.7 Tryptophan Trp W Nonpolar Neutral −0.9 Tyrosine Tyr Y Polar Neutral −1.3 Valine Val V Nonpolar Neutral 4.2

(34) The term "enriched" as used herein in relation to fusion gene-editing molecule RNA or protein of the invention indicates that the retroviral particle population comprises a higher number or higher percentage of gene-editing molecules than is found when the gene-editing molecule is not fused to an "enrichment" molecule (e.g., Vpr or CypA protein or 7SL RNA) or a fragment or derivative thereof. The "enrichment molecule" or fragment or derivative thereof is capable of effectively incorporating (i.e., enhancing inclusion of) the gene-editing molecule into a retroviral particle. In certain embodiments, the number of gene-editing fusion molecules incorporated into the retroviral particle is at least 1%, 2%, 3%, 4%, 5%, 7%, 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90% or 95% higher than when compared to the same non-fused gene-editing molecule into the same type of retroviral particle.

(35) The terms "retroviral element" or "retroviral component" are used herein to refer to retroviral genes (e.g., genes encoding polymerase or structural proteins) or other elements of the retroviral genome (e.g., packaging signals, regulatory elements, LTRs, etc.).

(36) By "decreased" is intended any decrease in the level or activity of the gene/protein (e.g., encoded at the locus of interest). For example, a decrease in activity can comprise either (1) a statistically significant decrease in the overall level or activity of a given protein including, for example, a decreased level or activity of 0.5%, 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 120% or greater when compared to an appropriate control.

(37) By "increased" is intended any increase in the level or activity of the gene/protein (e.g., encoded at the locus of interest). For example, an increase in activity can comprise either (1) a statistically significant increase in the overall level or activity of a given protein including, for example, an increased level or activity of 0.5%, 1%, 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 120% or greater when compared to an appropriate control.

(38) The term "in vitro" includes artificial environments and to processes or reactions that occur within an artificial environment (e.g., a test tube). The term "in vivo" includes natural environments (e.g., a cell or organism or body) and to processes or reactions that occur within a natural environment. The term "ex vivo" includes cells that have been removed from the body of an individual and to processes or reactions that occur within such cells.

(39) Compositions or methods "comprising" or "including" one or more recited elements may include other elements not specifically recited. For example, a composition that "comprises" or "includes" a protein may contain the protein alone or in combination with other ingredients. The transitional phrase "consisting essentially of" means that the scope of a claim is to be interpreted to encompass the specified elements recited in the claim and those that do not materially affect the basic and novel characteristic(s) of the claimed invention. Thus, the term "consisting essentially of" when used in a claim of this invention is not intended to be interpreted to be equivalent to "comprising."

(40) An "individual" or "subject" or "animal" refers to humans, veterinary animals (e.g., cats, dogs, cows, horses, sheep, pigs, etc.) and experimental animal models of diseases (e.g., mice, rats). In a preferred embodiment, the subject is a human.

(41) The terms "treat" or "treatment" of a state, disorder or condition include: (1) preventing, delaying, or reducing the incidence and/or likelihood of the appearance of at least one clinical or

sub-clinical symptom of the state, disorder or condition developing in a subject that may be afflicted with or predisposed to the state, disorder or condition, but does not yet experience or display clinical or subclinical symptoms of the state, disorder or condition; or (2) inhibiting the state, disorder or condition, i.e., arresting, reducing or delaying the development of the disease or a relapse thereof or at least one clinical or sub-clinical symptom thereof; or (3) relieving the disease, i.e., causing regression of the state, disorder or condition or at least one of its clinical or sub-clinical symptoms. The benefit to a subject to be treated is either statistically significant or at least perceptible to the patient or to the physician.

(42) The term "effective" applied to dose or amount refers to that quantity of a compound or pharmaceutical composition that is sufficient to result in a desired activity upon administration to a subject in need thereof. Note that when a combination of active ingredients is administered, the effective amount of the combination may or may not include amounts of each ingredient that would have been effective if administered individually. The exact amount required will vary from subject to subject, depending on the species, age, and general condition of the subject, the severity of the condition being treated, the particular drug or drugs employed, the mode of administration, and the like.

(43) The phrase "pharmaceutically acceptable", as used in connection with compositions described herein, refers to molecular entities and other ingredients of such compositions that are physiologically tolerable and do not typically produce untoward reactions when administered to a mammal (e.g., a human). Preferably, the term "pharmaceutically acceptable" means approved by a regulatory agency of the Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in mammals, and more particularly in humans.

(44) In accordance with the disclosure herein, there may be employed conventional molecular biology, microbiology, and recombinant DNA techniques within the skill of the art. Such techniques are explained fully in the literature. See, e.g., Sambrook, Fritsch & Maniatis, Molecular Cloning: A Laboratory Manual, Second Edition. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press, 1989 (herein "Sambrook et al., 1989"); DNA Cloning: A Practical Approach, Volumes I and II (D. N. Glover ed. 1985); Oligonucleotide Synthesis (M. J. Gait ed. 1984); Nucleic Acid Hybridization [B. D. Hames & S. J. Higgins eds. (1985)]; Transcription And Translation [B. D. Hames & S. J. Higgins, eds. (1984)]; Animal Cell Culture [R. I. Freshney, ed. (1986)]; Immobilized Cells And Enzymes [IRL Press, (1986)]; B. Perbal, A Practical Guide To Molecular Cloning (1984); Ausubel, F. M. et al. (eds.). Current Protocols in Molecular Biology. John Wiley & Sons, Inc., 1994. These techniques include site directed mutagenesis as described in Kunkel, Proc. Natl. Acad. Sci. USA 82: 488-492 (1985), U.S. Pat. No. 5,071,743, Fukuoka et al., Biochem. Biophys. Res. Commun. 263: 357-360 (1999); Kim and Maas, BioTech. 28: 196-198 (2000); Parikh and Guengerich, BioTech. 24: 428-431 (1998); Ray and Nickoloff, BioTech. 13: 342-346 (1992); Wang et al., BioTech. 19: 556-559 (1995); Wang and Malcolm, BioTech. 26: 680-682 (1999); Xu and Gong, BioTech. 26: 639-641 (1999), U.S. Pat. Nos. 5,789,166 and 5,932,419, Hogrefe, Strategies 14. 3: 74-75 (2001), U.S. Pat. Nos. 5,702,931, 5,780,270, and 6,242,222, Angag and Schutz, Biotech. 30: 486-488 (2001), Wang and Wilkinson, Biotech. 29: 976-978 (2000), Kang et al., Biotech. 20: 44-46 (1996), Ogel and McPherson, Protein Engineer. 5: 467-468 (1992), Kirsch and Joly, Nucl. Acids. Res. 26: 1848-1850 (1998), Rhem and Hancock, J. Bacteriol. 178: 3346-3349 (1996), Boles and Miogsa, Curr. Genet. 28: 197-198 (1995), Barrenttino et al., Nuc. Acids. Res. 22: 541-542 (1993), Tessier and Thomas, Meths. Molec. Biol. 57: 229-237, and Pons et al., Meth. Molec. Biol. 67: 209-218.

(45) "Optional" or "optionally" means that the subsequently described event or circumstance may or may not occur and that the description includes instances in which the event or circumstance occurs and instances in which it does not.

(46) Designation of a range of values includes all integers within or defining the range, and all subranges defined by integers within the range.

(47) The term "about" or "approximately" includes being within a statistically meaningful range of a value. Such a range can be within an order of magnitude, preferably within 50%, more preferably within 20%, still more preferably within 10%, and even more preferably within 5% of a given value or range. The allowable variation encompassed by the term "about" or "approximately" depends on the particular system under study, and can be readily appreciated by one of ordinary skill in the art.
(48) The term "and/or" refers to and encompasses any and all possible combinations of one or more of the associated listed items, as well as the lack of combinations when interpreted in the alternative ("or").
(49) The term "or" refers to any one member of a particular list and also includes any combination of members of that list.
(50) Singular forms "a", "an", and "the" include plural references unless the context clearly dictates otherwise. Thus, for example, a reference to "a method" includes one or more methods, and/or steps of the type described herein, and/or which will become apparent to those persons skilled in the art upon reading this disclosure.
II. Recombinant Retroviral Particles
(51) In one aspect, described herein is a recombinant retroviral particle that is capable of introducing a gene-editing system (e.g., the CRISPR/Cas system) into a target cell, the recombinant retroviral particle comprising (i) a gene-editing fusion molecule and optionally (ii) a guide RNA (gRNA) and/or a repair template. In certain embodiments, the gene-editing molecule is a functional fragment or derivative thereof. In certain embodiments, the gene-editing fusion molecule is a recombinant RNA molecule. In certain embodiments, the recombinant RNA molecule is a fusion RNA molecule which comprises (i) a sequence of a gene-editing molecule mRNA, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of an enrichment coding or non-coding RNA (e.g., a small coding or non-coding RNA), or a sequence of a functional fragment or derivative thereof, wherein the enrichment RNA, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing molecule mRNA, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the recombinant RNA molecule comprises one or more sequences of an enrichment coding or non-coding RNA, or functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of enrichment coding or non-coding RNA, or functional fragment or derivative thereof, each enrichment coding or non-coding RNA, or functional fragment or derivative thereof, is the same or different. In certain embodiments, the gene-editing fusion molecule is a recombinant protein molecule. In certain embodiments, the recombinant protein molecule is a fusion protein molecule which comprises (i) a sequence of a gene-editing protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one enrichment protein sequence, or a sequence of functional fragment or derivative thereof, wherein the enrichment protein, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the fusion protein molecule comprises one or more sequences of an enrichment protein, or functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of enrichment protein, or functional fragment or derivative thereof, each enrichment protein, or functional fragment or derivative thereof, is the same or different. In certain embodiments, the retroviral particle comprises both the fusion RNA molecule and the fusion protein molecule.
(52) In certain embodiments, the recombinant retroviral particle comprises a sequence encoding gRNA. In certain embodiments, the recombinant retroviral particle comprises a sequence corresponding to a repair template (RT). In certain embodiments, the recombinant retroviral particle comprises both a sequence encoding gRNA and a sequence corresponding to a repair template (RT). In certain embodiments, upon introduction into a target cell of the recombinant retroviral particle, the sequence encoding the gRNA and/or the sequence corresponding to the

repair template (RT) integrate into the cell genome. In certain embodiments, upon introduction into a target cell of the recombinant retroviral particle, the sequence encoding the gRNA and/or the sequence corresponding to the repair template (RT) do not integrate into the cell genome. In certain embodiments, the gRNA can comprise a single RNA molecule (single RNA polynucleotide). In certain embodiments, the gRNA can comprise two separate RNA molecules: an activator-RNA (e.g., tracrRNA) and a targeter-RNA (e.g., CRISPR RNA or crRNA).
(53) In certain embodiments, the target cell for the retroviral particle is a eukaryotic cell, e.g., a mammalian cell (including, e.g., human, veterinary animal or rodent cell) or amphibian cell, avian cell, insect cell, and yeast.
(54) The term “retroviral particle” as used herein refers to a recombinant retroviral particle generated in a packaging cell that is able to deliver a gene-editing fusion molecule, and optionally a guide RNA (gRNA) and/or a repair template into a suitable target cell. In one embodiment, the retroviral particles of the invention are replication deficient. In certain embodiments, the retroviral particle is a retroviral particle (e.g., a lentiviral particle). In some embodiments, the retroviral particle is lacking one or more of the wild type envelope proteins. In some embodiments, the retroviral particle is a pseudotyped particle. The term “pseudotyped” in connection with enveloped retroviral particles described herein refers to retroviral particles comprising in their lipid envelope molecules, e.g., proteins, glycoproteins, etc., which are mutated and/or heterologous compared to molecules typically found on the surface of the retrovirus from which the particles are derived, and which may affect, contribute to, direct, redirect and/or completely change the tropism of the retroviral particle in comparison to a reference wild type retrovirus from which the retroviral particle is derived. In some embodiments, a retroviral particle is pseudotyped such that it recognizes, binds and/or infects a target (ligand or cell) that is different to that of a reference wild type retrovirus from which the retroviral particle is derived. In some embodiments, a retroviral particle is pseudotyped such that it does not recognize, bind, and/or infect a target (ligand or cell) of the reference wild type retrovirus from which the retroviral particle is derived.
(55) A “gene-editing molecule” is a molecule (e.g., a protein or mRNA encoding such protein) used for modifying a genomic locus of interest (i.e., target) in a cell (e.g., eukaryotic, mammalian, human, or non-human cell). Such modifications include, but are not limited to a disruption, deletion, repair, mutation, addition, alteration, or modification of a gene sequence at a target locus in a gene. Examples of gene-editing molecules include, but are not limited to, endonucleases. Endonucleases are enzymes that cleave the phosphodiester bond within a polynucleotide chain, but they only break internal phosphodiester bonds. Examples of gene-editing endonucleases useful in the compositions and methods of the present invention include, but are not limited to, zinc finger nucleases (ZFns), transcription activator-like effector nucleases (TALENs), meganucleases, restriction endonucleases, recombinases, and Clustered Regularly Interspersed Short Palindromic Repeats, (CRISPR)/CRISPR-associated (Cas) proteins.
(56) A. Cas Fusion Molecule
(57) The methods and compositions disclosed herein can utilize the Clustered Regularly Interspersed Short Palindromic Repeats (CRISPR)/CRISPR-associated (Cas) systems or components of such systems to modify a genome within a cell. CRISPR/Cas systems include transcripts and other elements involved in the expression of, or directing the activity of, Cas genes. A CRISPR/Cas system can be, for example, a type I, a type II, or a type III system. Alternatively, a CRISPR/Cas system can be a type V system (e.g., subtype V-A or subtype V-B). The methods and compositions disclosed herein can employ CRISPR/Cas systems by utilizing CRISPR complexes (comprising a guide RNA (gRNA) complexed with a Cas protein) for site-directed cleavage of nucleic acids.
(58) CRISPR/Cas systems used in the compositions and methods disclosed herein can be non-naturally occurring. A “non-naturally occurring” system includes anything indicating the involvement of the hand of man, such as one or more components of the system being altered or

mutated from their naturally occurring state, being at least substantially free from at least one other component with which they are naturally associated in nature, or being associated with at least one other component with which they are not naturally associated. For example, some CRISPR/Cas systems employ non-naturally occurring CRISPR complexes comprising a gRNA and a Cas protein that do not naturally occur together, employ a Cas protein that does not occur naturally, or employ a gRNA that does not occur naturally.
(59) In one aspect, described herein are Cas fusion molecules to be delivered to a target cell. In certain aspects, the Cas portion of the Cas fusion molecule is a Cas molecule, or a functional fragment or derivative thereof.
(60) In certain embodiments, the Cas fusion molecule is a recombinant RNA molecule. In certain embodiments, the recombinant RNA molecule is a Cas fusion RNA which comprises (i) a sequence of a Cas mRNA, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of a coding or non-coding RNA, or a sequence of a functional fragment or derivative thereof, wherein the coding or non-coding RNA, or functional fragment or derivative thereof is capable of enhancing inclusion of the Cas mRNA, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas fusion RNA molecule comprises one or more sequences of an enrichment coding or non-coding RNA, or functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of enrichment coding or non-coding RNA, or functional fragment or derivative thereof, each enrichment coding or non-coding RNA, or functional fragment or derivative thereof, is the same or different. In certain embodiments, the coding or non-coding RNA is a small coding or non-coding RNA. The term “small RNA” as used herein refers to RNA molecules that are <350 nucleotides that can be coding or non-coding RNA molecules. The term “coding RNA” refers to a functional RNA molecule that can translate into a protein. The terms “non-coding RNA (ncRNA)”, “non-protein-coding RNA (npcRNA)”, “non-messenger RNA (nmRNA)”, and “functional RNA (fRNA)” refer to a functional RNA molecule that is not translated into a protein. The nucleic acid sequence from which a non-coding RNA is transcribed is often called an RNA gene. Non-limiting examples of non-coding RNAs include, e.g., 7SL RNA, tRNAs (including primer tRNAs), 5S rRNA, U1 snRNA, U2 snRNA, U6 snRNA, Y1 RNA, Y3 RNA, B1 RNA, VL30 RNA, 7SK RNA, Alu RNA, miRNA, snoRNA, and cytoplasmic vault ncRNA. Small coding or non-coding RNA are discussed in greater detail below.
(61) In certain embodiments, the Cas fusion molecule is a recombinant protein molecule. In certain embodiments, the recombinant protein molecule is a Cas fusion protein which comprises (i) a sequence of a Cas protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one enrichment protein sequence, or a sequence of a functional fragment or derivative thereof, wherein the enrichment protein, or functional fragment or derivative thereof, is capable of enhancing inclusion of the gene-editing protein, or functional fragment or derivative thereof, into the retroviral particle. In certain embodiments, the Cas fusion protein comprises one or more sequences of an enrichment protein, or functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of and enrichment protein, or functional fragment or derivative thereof, each enrichment protein, or functional fragment or derivative thereof, is the same or different. In one embodiment, the enrichment protein is cyclophilin A (CypA) protein or a viral protein R (Vpr). Additional enrichment proteins are discussed in greater detail below.
(62) In certain embodiments, the retroviral particle comprises at least one Cas fusion RNA. In certain embodiments, the retroviral particle comprises at least one Cas fusion protein. In certain embodiments, the retroviral particle comprises at least one Cas fusion RNA and at least one Cas fusion protein.
(63) The Cas fusion molecule, or functional fragment or derivative thereof, can be associated with a retroviral particle. The Cas fusion molecule, or functional fragment or derivative thereof, is

"associated" with the retroviral particle if it is incorporated in, physically or chemically linked or bound to the retroviral particle, such that a complex between the Cas fusion molecule and retroviral particle is formed. The Cas fusion molecule can be associated with the retroviral particle using any suitable method for nucleic acid-nucleic acid, nucleic acid-protein, protein-protein linking, nucleic acid-virus, or protein-virus linking known in the art. In certain embodiments, the Cas fusion molecule is associated with a capsid protein of the retroviral particle. In certain embodiments, the Cas fusion molecule is associated with a nucleocapside domain of the retroviral particle. In other embodiments, the Cas fusion molecule can be packaged into the retroviral particle.

(64) (i) Cas Molecule

(65) "Cas molecules", "Cas proteins" or "Cas nucleases" useful in the compositions and methods of the invention generally comprise at least one RNA recognition or binding domain that can interact with guide RNAs (gRNAs, described in more detail below). Cas proteins can also comprise nuclease domains (e.g., DNase or RNase domains), DNA binding domains, helicase domains, protein-protein interaction domains, dimerization domains, and other domains. A nuclease domain possesses catalytic activity for nucleic acid cleavage, which includes the breakage of the covalent bonds of a nucleic acid molecule. Cleavage can produce blunt ends or staggered ends, and it can be single-stranded or double-stranded. For example, a wild type Cas9 protein will typically create a blunt cleavage product. Alternatively, a wild type Cpf1 protein (e.g., FnCpf1) can result in a cleavage product with a 5-nucleotide 5′ overhang, with the cleavage occurring after the 18th base pair from the PAM sequence on the non-targeted strand and after the 23rd base on the targeted strand. A Cas protein can have full cleavage activity to create a double-strand break at a target genomic locus (e.g., a double-strand break with blunt ends), or it can be a nickase that creates a single-strand break at a target genomic locus.

(66) Examples of Cas proteins useful in the compositions and methods of the invention include Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas5e (CasD), Cash, Cas6e, Cas6f, Cas7, Cas8a1, Cas8a2, Cas8b, Cas8c, Cas9 (Csn1 or Csx12), Cas10, Cas10d, CasF, CasG, CasH, Csy1, Csy2, Csy3, Cse1 (CasA), Cse2 (CasB), Cse3 (CasE), Cse4 (CasC), Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15, Csf1, Csf2, Csf3, Csf4, and Cu1966, and homologs or modified versions thereof.

(67) An exemplary Cas protein is a Cas9 protein or a protein derived from Cas9 from a type II CRISPR/Cas system. Cas9 proteins are from a type II CRISPR/Cas system and typically share four key motifs with a conserved architecture. Motifs 1, 2, and 4 are RuvC-like motifs, and motif 3 is an HNH motif. Exemplary Cas9 proteins are from *Streptococcus pyogenes, Streptococcus thermophilus, Streptococcus* sp., *Staphylococcus aureus, Nocardiopsis dassonvillei, Streptomyces pristinaespiralis, Streptomyces viridochromogenes, Streptomyces viridochromogenes, Streptosporangium roseum, Streptosporangium roseum, Alicyclobacillus acidocaldarius, Bacillus pseudomycoides, Bacillus selenitireducens, Exiguobacterium sibiricum, Lactobacillus delbrueckii, Lactobacillus salivarius, Microscilla marina, Burkholderiales bacterium, Polaromonas naphthalenivorans, Polaromonas* sp., *Crocosphaera watsonii, Cyanothece* sp., *Microcystis aeruginosa, Synechococcus* sp., *Acetohalobium arabaticum, Ammonifex degensii, Caldicelulosiruptor becscii, Candidatus Desulforudis, Clostridium botulinum, Clostridium difficile, Finegoldia magna, Natranaerobius thermophilus, Pelotomaculum thermopropionicum, Acidithiobacillus caldus, Acidithiobacillus ferrooxidans, Allochromatium vinosum, Marinobacter* sp *Nitrosococcus halophilus, Nitrosococcus watsoni, Pseudoalteromonas haloplanktis, Ktedonobacter racemifer, Methanohalobium evestigatum, Anabaena variabilis, Nodularia spumigena, Nostoc* sp., *Arthrospira maxima, Arthrospira platensis, Arthrospira* sp., *Lyngbya* sp., *Microcoleus chthonoplastes, Oscillatoria* sp., *Petrotoga mobilis, Thermosipho africanus, Acaryochloris marina, Neisseria meningitidis,* or *Campylobacter jejuni*. Additional examples of the Cas9 family members are described in WO 2014/131833, herein incorporated by reference in its

entirety for all purposes. Cas9 from *S. pyogenes* (SpCas9) (assigned SwissProt accession number Q99ZW2) is an exemplary Cas9 protein. Cas9 from *S. aureus* (SaCas9) (assigned UniProt accession number J7RUA5) is another exemplary Cas9 protein. Cas9 from *Campylobacter jejuni* (CjCas9) (assigned UniProt accession number Q0P897) is another exemplary Cas9 protein. See, e.g., Kim et al. (2017) *Nat. Comm.* 8:14500, herein incorporated by reference in its entirety for all purposes. SaCas9 is smaller than SpCas9, and CjCas9 is smaller than both SaCas9 and SpCas9.
(68) Another example of a Cas protein is a Cpf1 (CRISPR from *Prevotella* and *Francisella* 1) protein. Cpf1 is a large protein (about 1300 amino acids) that contains a RuvC-like nuclease domain homologous to the corresponding domain of Cas9 along with a counterpart to the characteristic arginine-rich cluster of Cas9. However, Cpf1 lacks the HNH nuclease domain that is present in Cas9 proteins, and the RuvC-like domain is contiguous in the Cpf1 sequence, in contrast to Cas9 where it contains long inserts including the HNH domain. See, e.g., Zetsche et al. (2015) *Cell* 163(3):759-771, herein incorporated by reference in its entirety for all purposes. Exemplary Cpf1 proteins are from *Francisella tularensis* 1, *Francisella tularensis* subsp. *novicida, Prevotella albensis*, Lachnospiraceae bacterium MC2017 1, *Butyrivibrio proteoclasticus*, Peregrinibacteria bacterium GW2011_GWA2_33_10, Parcubacteria bacterium GW2011_GWC2_44_17, *Smithella* sp. *SCADC, Acidaminococcus* sp. BV3L6, Lachnospiraceae bacterium MA2020, *Candidatus Methanoplasma termitum, Eubacterium eligens, Moraxella bovoculi* 237, *Leptospira inadai*, Lachnospiraceae bacterium ND2006, *Porphyromonas crevioricanis* 3, *Prevotella disiens*, and *Porphyromonas macacae*. Cpf1 from *Francisella novicida* U112 (FnCpf1; assigned UniProt accession number A0Q7Q2) is an exemplary Cpf1 protein.
(69) Cas proteins can be wild type proteins (i.e., those that occur in nature), modified Cas proteins (i.e., Cas protein variants), or fragments of wild type or modified Cas proteins. Cas proteins can also be active variants or fragments with respect to catalytic activity of wild type or modified Cas proteins. Active variants or fragments with respect to catalytic activity can comprise at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more sequence identity to the wild type or modified Cas protein or a portion thereof, wherein the active variants retain the ability to cut at a desired cleavage site and hence retain nick-inducing or double-strand-break-inducing activity. Assays for nick-inducing or double-strand-break-inducing activity are known and generally measure the overall activity and specificity of the Cas protein on DNA substrates containing the cleavage site.
(70) One example of a modified Cas protein is the modified SpCas9-HF1 protein, which is a high-fidelity variant of *Streptococcus pyogenes* Cas9 harboring alterations (N497A/R661A/Q695A/Q926A) designed to reduce non-specific DNA contacts. See, e.g., Kleinstiver et al. (2016) *Nature* 529(7587):490-495, herein incorporated by reference in its entirety for all purposes. Another example of a modified Cas protein is the modified eSpCas9 variant (K848A/K1003A/R1060A) designed to reduce off-target effects. See, e.g., Slaymaker et al. (2016) *Science* 351(6268):84-88, herein incorporated by reference in its entirety for all purposes. Other SpCas9 variants include K855A and K810A/K1003A/R1060A.
(71) Cas proteins can be modified to increase or decrease one or more of nucleic acid binding affinity, nucleic acid binding specificity, and enzymatic activity. Cas proteins can also be modified to change any other activity or property of the protein, such as stability. For example, one or more nuclease domains of the Cas protein can be modified, deleted, or inactivated, or a Cas protein can be truncated to remove domains that are not essential for the function of the protein or to optimize (e.g., enhance or reduce) the activity of the Cas protein.
(72) Cas proteins can comprise at least one nuclease domain, such as a DNase domain. For example, a wild type Cpf1 protein generally comprises a RuvC-like domain that cleaves both strands of target DNA, perhaps in a dimeric configuration. Cas proteins can also comprise at least two nuclease domains, such as DNase domains. For example, a wild type Cas9 protein generally comprises a RuvC-like nuclease domain and an HNH-like nuclease domain. The RuvC and HNH

domains can each cut a different strand of double-stranded DNA to make a double-stranded break in the DNA. See, e.g., Jinek et al. (2012) *Science* 337:816-821, herein incorporated by reference in its entirety for all purposes.
(73) One or more of the nuclease domains can be deleted or mutated so that they are no longer functional or have reduced nuclease activity. For example, if one of the nuclease domains is deleted or mutated in a Cas9 protein, the resulting Cas9 protein can be referred to as a nickase and can generate a single-strand break at a guide RNA recognition sequence within a double-stranded DNA but not a double-strand break (i.e., it can cleave the complementary strand or the non-complementary strand, but not both). If both of the nuclease domains are deleted or mutated, the resulting Cas protein (e.g., Cas9) will have a reduced ability to cleave both strands of a double-stranded DNA (e.g., a nuclease-null or nuclease-inactive Cas protein, or a catalytically dead Cas protein (dCas)). An example of a mutation that converts Cas9 into a nickase is a D10A (aspartate to alanine at position 10 of Cas9) mutation in the RuvC domain of Cas9 from *S. pyogenes*. Likewise, H939A (histidine to alanine at amino acid position 839), H840A (histidine to alanine at amino acid position 840), or N863A (asparagine to alanine at amino acid position N863) in the HNH domain of Cas9 from *S. pyogenes* can convert the Cas9 into a nickase. Other examples of mutations that convert Cas9 into a nickase include the corresponding mutations to Cas9 from *S. thermophilus*. See, e.g., Sapranauskas et al. (2011) *Nucleic Acids Research* 39:9275-9282 and WO 2013/141680, each of which is herein incorporated by reference in its entirety for all purposes. Such mutations can be generated using methods such as site-directed mutagenesis, PCR-mediated mutagenesis, or total gene synthesis. Examples of other mutations creating nickases can be found, for example, in WO 2013/176772 and WO 2013/142578, each of which is herein incorporated by reference in its entirety for all purposes. If all of the nuclease domains are deleted or mutated in a Cas protein (e.g., both of the nuclease domains are deleted or mutated in a Cas9 protein), the resulting Cas protein (e.g., Cas9) will have a reduced ability to cleave both strands of a double-stranded DNA (e.g., a nuclease-null or nuclease-inactive Cas protein). One specific example is a D10A/H840A *S. pyogenes* Cas9 double mutant or a corresponding double mutant in a Cas9 from another species when optimally aligned with *S. pyogenes* Cas9. Another specific example is a D10A/N863A *S. pyogenes* Cas9 double mutant or a corresponding double mutant in a Cas9 from another species when optimally aligned with *S. pyogenes* Cas9.
(74) Examples of inactivating mutations in the catalytic domains of *Staphylococcus aureus* Cas9 proteins are also known. For example, the *Staphyloccocus aureus* Cas9 enzyme (SaCas9) may comprise a substitution at position N580 (e.g., N580A substitution) and a substitution at position D10 (e.g., D10A substitution) to generate a nuclease-inactive Cas protein. See, e.g., WO 2016/106236, herein incorporated by reference in its entirety for all purposes.
(75) Examples of inactivating mutations in the catalytic domains of Cpf1 proteins are also known. With reference to Cpf1 proteins from *Francisella novicida* U112 (FnCpf1), *Acidaminococcus* sp. BV3L6 (AsCpf1), Lachnospiraceae bacterium ND2006 (LbCpf1), and *Moraxella bovoculi* 237 (MbCpf1 Cpf1), such mutations can include mutations at positions 908, 993, or 1263 of AsCpf1 or corresponding positions in Cpf1 orthologs, or positions 832, 925, 947, or 1180 of LbCpf1 or corresponding positions in Cpf1 orthologs. Such mutations can include, for example one or more of mutations D908A, E993A, and D1263A of AsCpf1 or corresponding mutations in Cpf1 orthologs, or D832A, E925A, D947A, and D1180A of LbCpf1 or corresponding mutations in Cpf1 orthologs. See, e.g., US 2016/0208243, herein incorporated by reference in its entirety for all purposes.
(76) Cas fusion proteins can also be tethered to labeled nucleic acids. Such tethering (i.e., physical linking) can be achieved through covalent interactions or noncovalent interactions, and the tethering can be direct (e.g., through direct fusion or chemical conjugation, which can be achieved by modification of cysteine or lysine residues on the protein or intein modification), or can be achieved through one or more intervening linkers or adapter molecules such as streptavidin or aptamers. See, e.g., Pierce et al. (2005) *Mini Rev. Med. Chem.* 5(1):41-55; Duckworth et al. (2007)

Angew. Chem. Int. Ed. Engl. 46(46):8819-8822; Schaeffer and Dixon (2009) *Australian J. Chem.* 62(10): 1328-1332; Goodman et al. (2009) *Chembiochem.* 10(9): 1551-1557; and Khatwani et al. (2012) *Bioorg. Med. Chem.* 20(14):4532-4539, each of which is herein incorporated by reference in its entirety for all purposes. Noncovalent strategies for synthesizing protein-nucleic acid conjugates include biotin-streptavidin and nickel-histidine methods. Covalent protein-nucleic acid conjugates can be synthesized by connecting appropriately functionalized nucleic acids and proteins using a wide variety of chemistries. Some of these chemistries involve direct attachment of the oligonucleotide to an amino acid residue on the protein surface (e.g., a lysine amine or a cysteine thiol), while other more complex schemes require post-translational modification of the protein or the involvement of a catalytic or reactive protein domain. Methods for covalent attachment of proteins to nucleic acids can include, for example, chemical cross-linking of oligonucleotides to protein lysine or cysteine residues, expressed protein-ligation, chemoenzymatic methods, and the use of photoaptamers. The labeled nucleic acid can be tethered to the C-terminus, the N-terminus, or to an internal region within the Cas protein. Preferably, the labeled nucleic acid is tethered to the C-terminus or the N-terminus of the Cas protein. Likewise, the Cas protein can be tethered to the 5′ end, the 3′ end, or to an internal region within the labeled nucleic acid. That is, the labeled nucleic acid can be tethered in any orientation and polarity. Preferably, the Cas protein is tethered to the 5′ end or the 3′ end of the labeled nucleic acid.

(77) In some embodiments, the nucleic acids encoding the Cas proteins of the invention, or functional fragments or derivatives thereof, can be codon optimized for efficient translation into protein in a particular cell or organism. For example, the nucleic acid encoding a Cas protein, or functional fragment or derivative thereof, can be modified to substitute codons having a higher frequency of usage in a bacterial cell, a yeast cell, a human cell, a non-human cell, a mammalian cell, a rodent cell, a mouse cell, a rat cell, or any other host (e.g. packaging) and/or target cell of interest. When a fusion RNA encoding a Cas protein, or a functional fragment or derivative thereof, is introduced into the cell, the Cas protein, or functional fragment or derivative thereof, can be transiently or conditionally expressed in the cell.

(78) (ii) Cas Fusion RNA

(79) In one aspect, described herein is a recombinant RNA molecule that is capable of being translated into a gene-editing protein, or functional fragment or derivative thereof, in a target cell. In certain embodiments, the recombinant RNA molecule is a Cas fusion RNA that is capable of being translated into a Cas protein, or functional fragment or derivative thereof, in a target cell, the Cas fusion RNA, which comprises (i) a sequence of a Cas mRNA, or a sequence of a functional fragment or derivative thereof, and (ii) a sequence of at least one small coding or non-coding RNA, or a sequence of a functional fragment or derivative thereof, wherein the small coding or non-coding RNA, or functional fragment or derivative thereof, is capable of enhancing inclusion of the Cas mRNA, or functional fragment or derivative thereof, into a retroviral particle.

(80) In certain embodiments, the Cas fusion RNA molecule comprises one or more sequences of an enrichment coding or non-coding RNA, or functional fragment or derivative thereof. In certain embodiments, the Cas fusion RNA comprises at least about 1, at least about 2, at least about 3, at least about 4, at least about 5, at least about 6, at least about 7, at least about 8, at least about 9, or at least about 10 sequences of a small coding or non-coding RNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas fusion RNA comprises multiple copies of the same sequence of the small coding or non-coding RNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas fusion RNA comprises different sequences of small coding or non-coding RNA, or sequences of functional fragments or derivatives thereof.

(81) In certain embodiments, the Cas mRNA, or functional fragment or derivative thereof, is operably linked to at least one small coding or non-coding RNA molecule, or fragment or derivative thereof. In certain embodiments, the at least one sequence of small coding or non-coding

RNA, or a sequence of a fragment or derivative thereof, is operably linked to the 5′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of small coding or non-coding RNA, or a sequence of a fragment or derivative thereof, is fused before the start codon of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of small coding or non-coding RNA, or a sequence of a fragment or derivative thereof, is operably linked to the 3′ end of the sequence of Cas mRNA, or a sequence of a functional fragment of derivative thereof. In certain embodiments, the at least one sequence of small coding or non-coding RNA, or a sequence of a fragment or derivative thereof, is fused after the stop codon of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of small coding or non-coding RNA, or a fragment or derivative thereof, at least one sequence of small coding or non-coding RNA, or a fragment or derivative thereof, is operably linked to the 5′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof and at least one sequence of small coding or non-coding RNA, or a fragment or derivative thereof, is operably linked to the 3′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof.

(82) The Cas mRNA and small coding or non-coding RNA are linked such that there is no interruption of the initiation, elongation, and/or termination of the translation of the Cas mRNA in the target cell. The Cas mRNA, or functional fragment or derivative thereof, and small coding or non-coding RNA, or functional fragment or derivative thereof, are linked such that there is no interruption of the packaging of the small coding or non-coding RNA, or functional fragment or derivative thereof, into the retroviral particle. In certain embodiments, the enhancement of inclusion in to the retroviral particle can occur by interaction or binding of the small coding or non-coding RNA, or functional fragment or derivative thereof, to a component of the retroviral particle. In certain embodiments, the enhancement or inclusion into the retroviral particle can occur by interaction or binding of the small coding or non-coding RNA, or functional fragment or derivative thereof, to a nucleocapsid (NC) domain of the retroviral particle. In certain embodiments, the NC domain is the NC domain of a Gag molecule.

(83) In certain embodiments, the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof, and sequence of non-coding RNA, or a sequence of a functional fragment or derivative thereof, are separated by a linker. In certain embodiments, when there is more than one enrichment coding or non-coding RNA, or functional fragment or derivative thereof, each enrichment coding or non-coding RNA, or functional fragment or derivative thereof, can be separated by a linker. Suitable linkers used in the Cas fusion RNA can be of any of a number of suitable lengths, such as, e.g., from 1 to 200 nucleotides or even longer. In certain embodiments, the linker is not cleavable (e.g., by a restriction enzyme). In certain embodiments, the linker is cleavable. In certain embodiments, the linker is a selectively cleavable linker. A selectively cleavable linker is a linker that is cleaved under selected conditions, such as a photocleavable linker, a chemically cleavable linker and an enzymatically cleavable linker (i.e., a restriction endonuclease site or a ribonucleotide/RNase digestion). The stability of a nucleic acid-based biodegradable linker molecule can be modulated by using various chemistries, for example combinations of ribonucleotides, deoxyribonucleotides, and chemically-modified nucleotides, such as 2′-O-methyl, 2′-fluoro, 2′-amino, 2′-O-amino, 2′-C-allyl, 2′-O-allyl, and other 2′-modified or base modified nucleotides. The linker molecule can also comprise nucleic acid backbone, nucleic acid sugar, or nucleic acid base modifications.

(84) The compositions and methods take advantage of the fact that small coding or non-coding RNA can be effectively incorporated ("enriched") from the packaging cell into the retroviral particle. As an example, retroviruses are ribonucleoprotein complexes produced by cells as orchestrated by genetic information contained in virion RNA. While the most abundant nucleic acid in a retroviral particle is its genomic RNA (>50% of the RNA mass), the remainder of the

RNA in the retroviral particle is host (e.g., packaging cell) encoded. See Onafuwa-Nuga et al., (2006) RNA 12:542-546. One of the first host encoded RNA packaged into a retrovirus was a molecule with 7S sedimentation properties, identified as 7SL RNA. Id. 7SL is the non-coding RNA component of signal recognition particle (SRP), which is the ribonucleoprotein complex promoting co-translational protein transport into the endoplasmic reticulum. See Water and Blobel (1982) Nature 299:691-698 and Doudna and Batey (2004) Annu. Rev. Biochem. 74:3046-3057, both herein incorporated by reference in their entirety for all purposes. In fact, 7SL RNA is the most abundant non-viral derived RNA found in lentiviruses (Eckwahl et al., (2016) mBio 7:e02025-15) and it is 7SL RNA is found in higher copy number in HIV viral particles than its own viral genomic RNA (Onafuwa-Nuga et al., (2006) *RNA* 12:542-546). Such data indicates that 7SL RNA could be selectively enriched into retroviral particles, and it is possible that the 7SL RNAs are not passively included into budding particles, but are actively packaged via interactions with Gag molecules. See Tian et al., (2007) *Nucleic Acids Res.* 35:7288-7302, which is incorporated by reference in its entirety for all purposes. In addition to 7SL RNA, several other small coding or non-coding RNA molecules are known to be enriched into retroviral particles. Examples of such small coding or non-coding RNA useful in the compositions and methods of the present invention include, but are not limited to, tRNAs (including primer tRNAs, SEQ ID NO: GCCCGGATAGCTCAGTCGGTAGAGCATCA GACTTTTAATCTGAGGGTCCAGGGTTCAAGTCCCTGTTCGGGCG), 5S rRNA, U1 snRNA, U2 snRNA, U6 snRNA, Y1 RNA, Y3 RNA, B1 RNA, VL30 RNA, 7SK RNA, Alu RNA, miRNA, snoRNA, and cytoplasmic vault ncRNA. See Linial and Miller (1990) *Curr. Top. Microbiol. Immunol* 157:125-152; Berkowitz, et al., (1996) *Curr. Top. Microbiol. Immunol* 214:177-218; Giles et al., (2004) *RNA* 10:299-307; Onafuwa-Nuga et al., (2005) *J. Virol.* 79:13528-13537; Tian et al., (2007) *Nucleic Acids Res.* 35:7288-7302, each of which herein incorporated by reference in their entirety for all purposes. Enrichment into the retroviral particles can occur by any means. For example, Primer tRNAs are selectively enriched via an interaction with retroviral reverse transcriptase. See Levin and Seidman (1979) *J. Virol.* 29:328-335; Kleiman (2002) *IUBMB Life* 53:107-114, each of which are incorporated in their entirety for all purposes. tRNA(Lys.sub.3) is also selected by forming a complex with the capsid domain of the Gag molecule. See Cen et al., (2002) *J. Virol.* 76:13111-13115, incorporated in its entirety for all purposes. At least some nascent RNAs (e.g., pre-tRNAs) are packaged as well. See Eckwahl et al., (2016) *mBio* 7:e02025-15, incorporated by reference in its entirety for all purposes.

(85) In certain embodiments, the non-coding RNA can be, but is not limited to, 7SL RNA, tRNAs (including primer tRNAs), 5S rRNA, U1 snRNA, U2 snRNA, U6 snRNA, Y1 RNA, Y3 RNA, B1 RNA, VL30 RNA, 7SK RNA, Alu RNA, miRNA, snoRNA, cytoplasmic vault ncRNA, or a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas mRNA into a retroviral particle. In certain embodiments, primer tRNAs include, but are not limited to, tRNA(Trp), tRNA(Pro), tRNA(Lys.sub.1,2), tRNA(Lys.sub.3), tRNA(iMet), tRNA(Gln), tRNA(Leu), tRNA(Ser), tRNA(Asn), tRNA(Ile), and tRNA(Arg). In certain embodiments, the primer tRNA is tRNA(Lys.sub.3). In certain embodiments, the non-coding RNA is 7SL RNA.

(86) In certain embodiments, the Cas fusion RNA molecule comprises (i) a sequence of Cas mRNA encoding a Cas protein, or functional fragment or derivative thereof, and (ii) at least one sequence of 7SL RNA, or a sequence of a functional fragment or derivative thereof, that is capable of enhancing inclusion of the Cas mRNA, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the 7SL RNA fragment or derivative comprises the 7SLrem domain (e.g., see below; SEQ ID NO: 85); the Alu domain (e.g., see below; SEQ ID NOs: 83&87 and 86&88; and FIG. **6**B), or fragment or derivative thereof (e.g., FIGS. **6**C and **6**D); the S domain (e.g., see below; SEQ ID NO: 84; and FIG. **6**E), or fragment or derivative thereof (e.g., FIGS. **6**F and **6**G); or a fragment or derivative of 7SL RNA or Alu domain comprising the 5c helix (See FIGS. **6**A, **6**B, **6**H, and **6**I). See e.g., Keene et al., (2010) *J. of Virology* 84:9070-9077 and

Keene and Telesnitsky *J Virol.* 2012 August; 86(15): 7934-7942, both incorporated by reference in their entirety for all purposes. In certain embodiments, fragments or derivatives can encompass a 7SL RNA sequence, 7SLrem domain, Alu domain, S domain, or 5c helix with the same or similar secondary structure. In certain embodiments, exemplar fragments or derivatives of 7SL RNA sequence, 7SLrem domain, Alu domain, S domain, or 5c helix are those disclosed above and/or below or similar sequences with additional or fewer nucleotides. In certain embodiments, the additional or fewer nucleotides are those present in the 7SL RNA sequence. In certain embodiments, the additional nucleotides are not those that normally appear next in the 7SL RNA sequence. Both the Alu domain and the S domain are sufficient on their own to mediate packaging when expressed as separate truncations of 7SL. Inclusion of the 5c helix aids in packaging efficiency. See Keene (2012). In certain embodiments, the 7SL RNA, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the 7SL RNA, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(87) In certain embodiments, the Cas mRNA, or functional fragment or derivative thereof, is operably linked to at least one 7SL RNA, or fragment or derivative thereof. In certain embodiments, the at least one sequence of 7SL RNA, or the sequence of a functional fragment or derivative thereof, is operably linked to the 5′ of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the 7SL RNA, or a sequence of fragment of derivative thereof, is fused before the start codon of the sequence encoding the Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of 7SL RNA, or a sequence of functional fragment or derivative thereof, is operably linked to the 3′ of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the 7SL RNA, or a sequence of fragment of derivative thereof, is fused after the stop codon of the sequence encoding the Cas mRNA or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of 7SL RNA, or a fragment or derivative thereof, at least one sequence of 7SL RNA, or a fragment or derivative thereof, is operably linked to the 5′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof and at least one sequence of 7SL RNA, or a fragment or derivative thereof, is operably linked to the 3′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof.
(88) In certain embodiments, the Cas fusion RNA comprises a linker between the Cas mRNA, or functional fragment or derivative thereof, and the 7SL RNA, or functional fragment or derivative thereof. The Cas mRNA, or functional fragment or derivative thereof, and 7SL RNA, or functional fragment or derivative thereof, are linked such that there is no interruption of the translation of the Cas mRNA into Cas protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, when there is more than one 7SL RNA, or functional fragment or derivative thereof, each 7SL RNA, or functional fragment or derivative thereof, can be separated by a linker. Suitable linkers are disclosed above.
(89) In certain embodiments, the 7SL RNA is encoded by the nucleic acid sequence of SEQ ID NO: 1, SEQ ID NO: 81, or SEQ ID NO:82. In certain embodiments, the 7SL RNA fragment comprises the 7SLrem domain; or the Alu domain, or functional fragment or derivative thereof; the S domain, or functional fragment or derivative thereof or a fragment or derivative of 7SL RNA or Alu domain comprising the 5c helix. See e.g., FIGS. **6**A-**6**I; SEQ ID NOs: 83&87 and 84-85; and Keene et al., (2010) 1 of Virology 84:9070-9077 and Keene and Telesnitsky *J Virol.* 2012 August; 86(15): 7934-7942, both incorporated by reference in their entirety for all purposes. In certain embodiments, the 7SL RNA fragment is encoded by one of the following nucleic acid sequences:
(90) TABLE-US-00002 (SEQ ID NO: 1)
GCCGGGCGCGGTGGCGCGTGCCTGTAGTCCCAGCTACTCGGGAGGCTGAGG
CTGGAGGATCGCTTGAGTCCAGGAGTTCTGGGCTGTAGTGCGCTATGC*CGA*

TCGGGTGTCCGCACTAAGTTCGGCATCAATATGGTGACCTCCCGGGAGCGG
GGGACCACCAGGTTGCCTAAGGAGGGGTGAACCGGCCCAGGTCGGAAACGG
*AGCAGGTCAAAACTCCCGTGCTGATCAGTAGTGGGATCGCGCCTGT*GAATA
GCCACTGCACTCCAGCCTGGGC**AACATAGCGAGACCCCGTCTCT**
(underlined is the sequence of the 7SLrem domain the sequences corresponding to the Alu domain are shown in bold; the sequence corresponding to the S domain is italicized).

(91) In certain embodiments, the Cas mRNA, or functional fragment or derivative thereof, is operably linked to at least one tRNA(Lys.sub.3), or fragment or derivative thereof. In certain embodiments, the at least one sequence of tRNA(Lys.sub.3), or the sequence of a functional fragment or derivative thereof, is operably linked to the 5′ of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the tRNA(Lys.sub.3), or a sequence of fragment of derivative thereof, is fused before the start codon of the sequence encoding the Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of tRNA(Lys.sub.3), or a sequence of functional fragment or derivative thereof, is operably linked to the 3′ of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the tRNA(Lys.sub.3), or a sequence of fragment of derivative thereof, is fused after the stop codon of the sequence encoding the Cas mRNA or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, at least one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, is operably linked to the 5′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof and at least one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, is operably linked to the 3′ end of the sequence of Cas mRNA, or a sequence of a functional fragment or derivative thereof.

(92) In certain embodiments, the Cas fusion RNA comprises a linker between the Cas mRNA, or functional fragment or derivative thereof, and the tRNA(Lys.sub.3), or functional fragment or derivative thereof. The Cas mRNA, or functional fragment or derivative thereof, and tRNA(Lys.sub.3), or functional fragment or derivative thereof, are linked such that there is no interruption of the translation of the Cas mRNA into Cas protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, when there is more than one tRNA(Lys.sub.3), or functional fragment or derivative thereof, each tRNA(Lys.sub.3), or functional fragment or derivative thereof, can be separated by a linker. Suitable linkers are disclosed above.

(93) In certain embodiments, the tRNA(Lys.sub.3) is encoded by the nucleic acid sequence of SEQ ID NO: 80.

(94) In certain embodiments, the Cas molecule is a Cas9 molecule, or a functional fragment or derivative thereof. In certain embodiments, the Cas9 can be wild type Cas9, a Cas9 nickase, a dead Cas9 (dCas9) a split Cas9, and a Cas9 fusion protein. In certain embodiments, the Cas9 is a *Streptococcus pyogenes* or *Staphylococcus aureus* Cas9. In certain embodiments, the sequence of the Cas9 mRNA is codon optimized for expression in a eukaryotic cell. In certain embodiments, the Cas9 mRNA molecule is encoded by the nucleic acid sequence of SEQ ID NO: 2 (mouse codon optimized).

(95) In certain embodiments, the Cas fusion RNA molecule comprises (i) a sequence of Cas9 mRNA encoding a Cas9 protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of 7SL RNA, or a sequence of a functional fragment or derivative thereof, that is capable of enhancing inclusion of the Cas9 mRNA, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas9 mRNA, or functional fragment or derivative thereof, is operably linked to the at least one 7SL RNA molecule, or fragment or derivative thereof. In certain embodiments, the at least one sequence of 7SL RNA, or sequence of a functional fragment or derivative thereof, is operably linked to the 5′ of the sequence of Cas9

mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the 7SL RNA, or a sequence of fragment of derivative thereof, is fused before the start codon of the sequence encoding the Cas9 mRNA or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of 7SL RNA, or a sequence of a functional fragment or derivative thereof, is operably linked to the 3′ of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the 7SL RNA, or a sequence of a fragment of derivative thereof, is fused after the stop codon of the sequence encoding the Cas9 mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of 7SL RNA, or a fragment or derivative thereof, at least one sequence of 7SL RNA, or a fragment or derivative thereof, is operably linked to the 5′ end and at least one sequence 7SL RNA, or a fragment or derivative thereof, 3′ end of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof

(96) In certain embodiments, the Cas9 fusion RNA comprises a linker between the Cas9 mRNA, or functional fragment or derivative thereof, and the 7SL RNA, or functional fragment or derivative thereof. The Cas9 mRNA and 7SL RNA are linked such that there is no interruption of the initiation, elongation, and/or termination of the translation of the Cas9 mRNA in the target cell. In certain embodiments, when there is more than one 7SL RNA, or functional fragment or derivative thereof, each 7SL RNA, or functional fragment or derivative thereof, can be separated by a linker.

(97) In certain embodiments, the Cas9 mRNA, or functional fragment or derivative thereof, is operably linked to at least one tRNA(Lys.sub.3), or fragment or derivative thereof. In certain embodiments, the at least one sequence of tRNA(Lys.sub.3), or the sequence of a functional fragment or derivative thereof, is operably linked to the 5′ of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the tRNA(Lys.sub.3), or a sequence of fragment of derivative thereof, is fused before the start codon of the sequence encoding the Cas9 mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of tRNA(Lys.sub.3), or a sequence of functional fragment or derivative thereof, is operably linked to the 3′ of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence encoding the tRNA(Lys.sub.3), or a sequence of fragment of derivative thereof, is fused after the stop codon of the sequence encoding the Cas9 mRNA or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, at least one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, is operably linked to the 5′ end of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof and at least one sequence of tRNA(Lys.sub.3), or a fragment or derivative thereof, is operably linked to the 3′ end of the sequence of Cas9 mRNA, or a sequence of a functional fragment or derivative thereof.

(98) In certain embodiments, the Cas9 fusion RNA comprises a linker between the Cas9 mRNA, or functional fragment or derivative thereof, and the tRNA(Lys.sub.3), or functional fragment or derivative thereof. The Cas9 mRNA, or functional fragment or derivative thereof, and tRNA(Lys.sub.3), or functional fragment or derivative thereof, are linked such that there is no interruption of the translation of the Cas9 mRNA into Cas9 protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, when there is more than one tRNA(Lys.sub.3), or functional fragment or derivative thereof, each tRNA(Lys.sub.3), or functional fragment or derivative thereof, can be separated by a linker. Suitable linkers are disclosed above.

(99) In certain embodiments, the nucleic acid sequence encoding Cas9 fusion RNA is SEQ ID NO: 3 (i.e., the nucleic acid sequence encoding 7SL attached to the 5′ end of the Cas9 mRNA before the start codon). In certain embodiments, the nucleic acid sequence encoding Cas9 fusion RNA is SEQ ID NO: 4 (i.e., the nucleic acid sequence encoding 7SL attached to the 3′ end of the Cas9 mRNA

after the stop codon).
(100) The Cas fusion RNA can be produced using routine molecular biology techniques, such as those described in He et al., (2014) *Gene Therapy,* 21:759-766. Optionally, the Cas mRNA can be codon optimized for efficient translation into the Cas protein, or functional fragment or derivative thereof, in a particular cell or organism. For example, the nucleic acid sequence encoding the Cas protein, or a functional fragment or derivative thereof, can be modified to substitute codons having a higher frequency of usage in a bacterial cell, a yeast cell, a human cell, a non-human cell, a mammalian cell, a rodent cell, a mouse cell, a rat cell, or any other host (e.g., packaging) and/or target cell of interest, as compared to the naturally occurring polynucleotide sequence.
(101) In certain embodiments, the nucleic acid molecule encoding the Cas fusion RNA further comprises a regulatory element, including for example, a promoter (e.g., including an enhancer), or a transcriptional repressor-binding element. Exemplary expression control sequences are known in the art and described in, for example, Goeddel, (1990) Gene Expression Technology: Methods in Enzymology, Vol. 185, Academic Press, San Diego, Calif., incorporated by reference in its entirety for all purposes.
(102) (iii) Cas Fusion Protein
(103) In one aspect, described herein is a recombinant Cas fusion protein that is capable of being delivered directly to a target cell via the recombinant retroviral particle, the Cas fusion protein comprising (i) a sequence of a Cas protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of an enrichment protein, or a sequence of a functional fragment or derivative thereof, that is capable of effectively incorporating (i.e., enhancing inclusion of) the Cas protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas protein, or functional fragment or derivative thereof, is operably linked to the enrichment protein, or fragment or derivative thereof. In certain embodiments, the Cas fusion protein comprises one or more sequences of an enrichment protein, or functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of enrichment protein, or functional fragment or derivative thereof, each enrichment protein, or functional fragment or derivative thereof, is the same or different.
(104) In certain embodiments, the at least one sequence of the enrichment protein, or a sequence of a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of the enrichment protein, or a sequence of a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of enrichment protein, or a fragment or derivative thereof, at least one sequence of enrichment protein, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of enrichment protein, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof.
(105) In certain embodiments, the Cas protein, or functional fragment or derivative thereof, and the at least one enrichment protein are separated by a linker. In certain embodiments, when there is more than one enrichment protein, or functional fragment or derivative thereof, each enrichment protein, or functional fragment or derivative thereof, can be separated by a linker. In certain embodiments, the Cas fusion protein is associated with a capsid protein of the retroviral particle. In certain embodiments, the Cas fusion protein is associated with a nucleocapside domain of the retroviral particle. In other embodiments, the Cas fusion protein can be packaged into the retroviral particle.
(106) The Cas fusion protein can be provided as a fusion protein comprising a Cas protein, or functional fragment or derivative thereof, and a Virol Protein R (Vpr), or a functional fragment or derivative thereof, which is capable of enhancing inclusion of the Cas protein, or functional

fragment or derivative thereof, into a retroviral particle. Vpr is a 96 amino acid 14-kDa protein within the HIV genome that plays a significant role in regulating nuclear import of the HIV-1 pre-integration complex. See Bukrinsky and Adzhubei (1999) *Rev. Med. Virol.* 9:39-49, incorporated by reference in its entirety for all purposes. Vpr is also required for retrovirus replication in non-dividing cells such as macrophages.
(107) The Cas fusion protein can be provided as a fusion protein comprising a Cas protein, or functional fragment or derivative thereof, and a Virol Protein X (Vpx), or a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas protein, or functional fragment or derivative thereof, into a retroviral particle. Vpx shares a close sequence similarity with Vpr. See Wu et al., (1994) *J. Virol.* 68:6161-6169, incorporated by reference in its entirety for all purposes.
(108) Vpr and Vpx are packaged into retroviral particles in quantities comparable with retroviral Gag proteins. See Bukrinsky and Adzhubei, supra. The packaging is likely to occur via interaction with p6.sup.Gag region of the Pr55.sup.Gag precursor protein. See Lu et al., (1993) *J. Virol.* 67:6542-6550; Paxton et al., (1993) *J. Virol.* 67:7229-7237; Wu et al., (1994) *J. Virol.* 68:6161-6169, each of which incorporated by reference in their entirety for all purposes.
(109) In certain embodiments, the Cas fusion protein molecule comprises (i) a sequence of Cas protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of Vpr, or a sequence of a functional fragment or derivative thereof, that is capable of enhancing inclusion of the Cas fusion protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas protein, or functional fragment or derivative thereof, is operably linked to at least one Vpr, or fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpr, or a sequence of a functional fragment or derivative thereof, is operably linked to the N-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpr, or a sequence of functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of Vpr, or a fragment or derivative thereof, at least one sequence of Vpr, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of Vpr, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas fusion protein comprises a linker between the Cas protein, or functional fragment or derivative thereof, and the Vpr, or functional fragment or derivative thereof. In certain embodiments, when there is more than one Vpr, or functional fragment or derivative thereof, each Vpr, or functional fragment or derivative thereof, can be separated by a linker. The Cas protein, or functional fragment or derivative thereof, and Vpr, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, the Vpr, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the Vpr, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(110) In certain embodiments, the Vpr is encoded by the nucleic acid sequence of SEQ ID NO: 5. In certain embodiment, the amino acid sequence of Vpr is SEQ ID NO: 6.
(111) In certain embodiments, the Cas fusion protein molecule comprises (i) a sequence of Cas protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas fusion protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas protein, or functional fragment or derivative thereof, is operably linked to at least one Vpx, or fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof, is

operably linked to the N-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of Vpx, or a fragment or derivative thereof, at least one sequence of Vpx, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of Vpx, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas fusion protein comprises a linker between the Cas protein, or functional fragment or derivative thereof, and the Vpx, or functional fragment or derivative thereof. In certain embodiments, when there is more than one Vpx, or functional fragment or derivative thereof, each Vpx, or functional fragment or derivative thereof, can be separated by a linker. The Cas protein, or functional fragment or derivative thereof, and Vpx, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, the Vpx, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the Vpx, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.

(112) In certain embodiments, the Vpx is encoded by the nucleic acid sequence of SEQ ID NO: 7. In certain embodiment, the amino acid sequence of Vpx is SEQ ID NO: 8.

(113) The Cas fusion protein can be provided as a fusion protein comprising a Cas protein, or functional fragment or derivative thereof, and a Cyclophilin A (CypA), or a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas protein, or functional fragment or derivative thereof, into a retroviral particle. CypA is an 18 kDa, 165-amino acid long cytosolic protein that regulates many biological processes, including intracellular signaling, transcription, inflammation, and apoptosis. CypA can also interact with several HIV proteins, and has been shown to interact with lentiviral Gag proteins and become highly enriched in lentiviral particles. See e.g., Luban et al., (1993) *Cell* 73:1067-78, incorporated by reference in its entirety for all purposes.

(114) In certain embodiments, the Cas fusion protein molecule comprises (i) a sequence of Cas protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of CypA or a sequence of a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas fusion protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas protein, or functional fragment or derivative thereof, is operably linked to at least one CypA, or fragment or derivative thereof. In certain embodiments, the at least one sequence of CypA, or a sequence of a functional fragment or derivative thereof, is operably linked to the N-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of CypA, or a sequence of a functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of CypA, or a fragment or derivative thereof, at least one sequence of CypA, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of CypA, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas fusion protein comprises a linker between the Cas protein, or functional fragment or derivative thereof, and the CypA, or functional fragment or derivative thereof. In certain embodiments, when there is more than one CypA, or functional fragment or derivative thereof, each CypA, or functional fragment or derivative thereof, can be separated by a

linker. The Cas protein, or functional fragment or derivative thereof, and CypA, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, the CypA, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the CypA, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(115) In certain embodiments, the CypA is encoded by the nucleic acid sequence of SEQ ID NO: 9. In certain embodiment, the amino acid sequence of CypA is SEQ ID NO: 10.
(116) In certain embodiments, the Cas protein is Cas9, or functional fragment or derivative thereof. In certain embodiments, the Cas9 is selected from the group consisting of wild type Cas9, a Cas9 nickase, a dead Cas9 (dCas9), a split Cas9, an inducible Cas9, and a Cas9 fusion protein. In certain embodiments, the Cas9 is a *Streptococcus pyogenes* or *Staphylococcus aureus* Cas9. In certain embodiments, the sequence of the Cas9 mRNA is codon optimized for expression in a eukaryotic cell. In certain embodiments, the Cas9 protein is encoded by the nucleic acid sequence of SEQ ID NO: 2. In certain embodiments, the Cas9 protein is encoded by the nucleic acid sequence of SEQ ID NO: 11.
(117) In certain embodiments, the Cas fusion protein molecule comprises (i) a sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of Vpr, or a sequence of a functional fragment or derivative thereof, that is capable of enhancing inclusion of the Cas9 fusion protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas9 protein, or functional fragment or derivative thereof, is operably linked to at least one Vpr, or fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpr, or a sequence of a functional fragment or derivative thereof, is operably linked to the N-terminus of the sequence of the Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpr, or a sequence of a functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of Vpr, or a fragment or derivative thereof, at least one sequence of Vpr, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of Vpr, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas9 fusion protein comprises a linker between the Cas9 protein, or functional fragment or derivative thereof, and the Vpr, or functional fragment or derivative thereof. In certain embodiments, when there is more than one Vpr, or functional fragment or derivative thereof, each Vpr, or functional fragment or derivative thereof, can be separated by a linker. The Cas9 protein, or functional fragment or derivative thereof, and Vpr, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas9 protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, the Vpr, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the Vpr, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(118) In certain embodiments, the Cas fusion protein comprises (i) a sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof that is capable of enhancing inclusion of the Cas9 fusion protein, or functional fragment or derivative thereof, into a retroviral particle. In certain embodiments, the Cas9 protein, or functional fragment or derivative thereof, is operably linked to at least one Vpx, or fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof, is operably linked to the N-terminus of the sequence of the Cas9 protein, or a sequence of a functional

fragment or derivative thereof. In certain embodiments, the at least one sequence of Vpx, or a sequence of a functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of Vpx, or a fragment or derivative thereof, at least one sequence of Vpx, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of Vpx, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas9 fusion protein comprises a linker between the Cas9 protein, or functional fragment or derivative thereof, and the Vpx, or functional fragment or derivative thereof. In certain embodiments, when there is more than one Vpx, or functional fragment or derivative thereof, each Vpx, or functional fragment or derivative thereof, can be separated by a linker. The Cas9 protein, or functional fragment or derivative thereof, and Vpx, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas9 protein in the target cell. In certain embodiments, the Vpx, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the Vpx, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(119) In certain embodiments, the Cas fusion protein molecule comprises (i) a sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof, and (ii) at least one sequence of CypA, or a sequence of a functional fragment or derivative thereof, that is capable of enhancing inclusion of the Cas9 fusion protein into a retroviral particle. In certain embodiments, the Cas9 protein, or functional fragment or derivative thereof, is operably linked to at least one CypA, or fragment or derivative thereof. In certain embodiments, the at least one sequence of CypA, or a sequence of a functional fragment or derivative thereof, is operably linked to the N-terminus of the sequence of the Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the at least one sequence of CypA, or a sequence of a functional fragment or derivative thereof, is operably linked to the C-terminus of the sequence of the Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, when there is more than one sequence of CypA, or a fragment or derivative thereof, at least one sequence of CypA, or a fragment or derivative thereof, is operably linked to the N-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof and at least one sequence of CypA, or a fragment or derivative thereof, is operably linked to the C-terminus of the sequence of Cas9 protein, or a sequence of a functional fragment or derivative thereof. In certain embodiments, the Cas9 fusion protein comprises a linker between the Cas9 protein, or functional fragment or derivative thereof, and the CypA, or functional fragment or derivative thereof. In certain embodiments, when there is more than one CypA, or functional fragment or derivative thereof, each CypA, or functional fragment or derivative thereof, can be separated by a linker. The Cas9 protein, or functional fragment or derivative thereof, and CypA, or functional fragment or derivative thereof, are linked such that there is no decrease in activity of the Cas9 protein, or functional fragment or derivative thereof, in the target cell. In certain embodiments, the CypA, or fragment or derivative thereof, is able to interact or bind with an NC domain. In certain embodiments, the CypA, or fragment or derivative thereof, is able to interact or bind with a Gag molecule.
(120) In certain embodiments, the nucleic acid sequence encoding Cas9 fusion protein can be SEQ ID NOs: 12, 14, 16, 18, 20, or 22. In certain embodiments, the amino acid sequence encoding Cas9 fusion protein can be SEQ ID NOs: 13, 15, 17, 19, 21, or 23.
(121) In certain embodiments, the sequence of Cas protein, or a sequence of a functional fragment or derivative thereof, and the sequence of the enrichment protein (e.g., CypA, Vpr and/or Vpx), or a sequence of a fragment or derivative thereof, are separated by a linker.
(122) Suitable peptide linkers used in the Cas fusion proteins can be of any of a number of suitable

lengths, such as, for example, from 1 amino acid (e.g., Gly) to 20 amino acids, from 2 amino acids to 15 amino acids, from 3 amino acids to 12 amino acids, including 4 amino acids to 10 amino acids, 5 amino acids to 9 amino acids, 6 amino acids to 8 amino acids, or 7 amino acids to 8 amino acids, and can be 1, 2, 3, 4, 5, 6, or 7 amino acids. Linkers can be cleavable or non-cleavable. Non-limiting examples of linkers include, e.g., glycine polymers (G)n, glycine-serine polymers (including, for example, (GS)n, (GSGGS) (SEQ ID NO: 91) and (GGGS) (SEQ ID NO: 92) (TABLE 1), where n is an integer of at least one), glycine-alanine polymers, alanine-serine polymers, and other flexible linkers known in the art. See, e.g., Chichili et al, (2013) *Protein Science*, 22:153-167. Glycine and glycine-serine polymers can be used; both Gly and Ser are relatively unstructured, and therefore can serve as a neutral tether between components. Glycine polymers can be used; glycine accesses significantly more phi-psi space than even alanine, and is much less restricted than residues with longer side chains. See Scheraga, (1992) *Rev. Computational Chem.* 1:1173-142. Exemplary linkers can comprise amino acid sequences including, but not limited to, GGSG (SEQ ID NO: 93), GGSGG (SEQ ID NO: 94), GSGSG (SEQ ID NO: 95), GSGGG (SEQ ID NO: 96), GGGSG (SEQ ID NO: 97), GSSSG (SEQ ID NO: 98), GCGASGGGGSGGGGS (SEQ ID NO: 99), GCGASGGGGSGGGGS (SEQ ID NO: 99), GGGASGGGGSGGGGS (SEQ ID NO: 100), GGGASGGGGS (SEQ ID NO: 101), G.sub.3S (SEQ ID NO: 92), (G.sub.3S).sub.2 (SEQ ID NO: 102), (G.sub.3S).sub.3 (SEQ ID NO: 103), G.sub.4S (SEQ ID NO: 104), (G.sub.4S).sub.2 (SEQ ID NO: 105), (G.sub.4S).sub.3 (SEQ ID NO: 89), (G.sub.4S).sub.4 (SEQ ID NO: 106), (TABLE 1), SGGSGGS (SEQ ID NO: 107); EFGNM (SEQ ID NO: 108); EFGGNM (SEQ ID NO: 109); EFGGNGGNM (SEQ ID NO: 110); or GGSNMAG (SEQ ID NO: 111) and the like. In certain embodiments, the linker sequence is (G.sub.4S).sub.3 (SEQ ID NO: 89).

(123) TABLE-US-00003 TABLE 1 Linker codon- Linker codon- Linker optimized optimized amino acids nucleotide nucleotide sequence sequence (human) sequence (mouse) GGGS (SEQ GGGGGTGGTTCC GGTGGCGGTAGT ID NO: 92) (SEQ ID NO: 112) (SEQ ID NO: 128) GGSG (SEQ GGTGGGTCTGGG GGGGGATCTGGT ID NO: 93) (SEQ ID NO: 113) (SEQ ID NO: 129) GSGGS (SEQ GGGTCCGGGGGCTCC GGCAGTGGCGGTAGC ID NO: 91) (SEQ ID NO: 114) (SEQ ID NO: 130) GGSGG GGTGGGAGCGGTGGT GGAGGGAGTGGAGGG (SEQ ID NO: (SEQ ID NO: 115) (SEQ ID NO: 131) 94) GSGSG (SEQ GGCAGCGGAAGCGGA GGGTCTGGCTCAGGC ID NO: 95) (SEQ ID NO: 116) (SEQ ID NO: 132) GSGGG GGGAGTGGGGGAGGT GGTTCTGGCGGAGGT (SEQ ID NO: (SEQ ID NO: 117) (SEQ ID NO: 133) 96) GGGSG GGTGGGGGAAGTGGA GGTGGTGGGAGTGGA (SEQ ID NO: (SEQ ID NO: 118) (SEQ ID NO: 134) 97) GSSSG (SEQ GGCAGCTCATCTGGT GGCTCAAGCAGTGGA ID NO: 98) (SEQ ID NO: 119) (SEQ ID NO: 135) GCGASGGG GGATGTGGTGCATCTG GGCTGTGGGGCTAGTG GSGGGGS GAGGGGGAGGCTCTGG GGGGAGGTGGTAGTGG (SEQ ID NO: GGGGGGTGGATCT TGGTGGCGGTTCC 99) (SEQ ID NO: 120) (SEQ ID NO: 136) GCGASGGG GGGTGTGGTGCTAGTG GGATGTGGGGCCTCAG GSGGGGS GGGGTGGCGGATCAGG GTGGGGGTGGCAGCGG (SEQ ID NO: TGGAGGCGGGAGC TGGTGGAGGGTCA 99) (SEQ ID NO: 121) (SEQ ID NO: 137) GGGGSGGG GGGGGCGGAGGATCTGGG GGTGGCGGGGGCTCTG GS (SEQ ID GGAGGGGGATCA GTGGAGGAGGATCT NO: 105) (SEQ ID NO: 122) (SEQ ID NO: 138) GGGASGGG GGGGGGGGCGCTTCAGGCG GGAGGCGGCGCTTCTG GSGGGGS GAGGTGGAAGTGGTGGAG GGGGCGGGGGTAGTGG (SEQ ID NO: GAGGT GGGTGGAGGT 100) (SEQ ID NO: 123) (SEQ ID NO: 139) GGGGSGGG GGAGGGGGAGGTTCTGGCG GGTGGAGGTGGAAGTG GSGGGGS GCGGGGGATCAGGAGGCG GAGGAGGGGGATCAGG (SEQ ID NO: GTGGGAGC CGGAGGCGGGAGC 89) (SEQ ID NO: 124) (SEQ ID NO: 140) GGGASGGG

GGTGGGGGGGCGTCAGGTG GGAGGGGGAGCCTCTG GS (SEQ ID GAGGCGGAAGT GCGGTGGAGGATCA NO: 101) (SEQ ID NO: 125) (SEQ ID NO: 141) GGGGSGGG GGCGGCGGAGGTTCTGGTG GGGGGAGGAGGCAGTG GSGGGGS GGGGTGGCAGTGGAGGAG GAGGTGGGGGAAGTGG (SEQ ID NO: GAGGCAGC TGGAGGGGGGTCT 89) (SEQ ID NO: 126) (SEQ ID NO: 142) GGGGSGGG GGAGGTGGAGGTAGTGGC GGGGGGTGGAGGATCA GSGGGGSG GTGGTGGGTCAGGGGGAGG GGAGGCGGTGGTTCTG GGGS (SEQ CGGGTCCGGTGGCGGTGGG GGGGAGGTGGATCCGG ID NO: 106) AGT (SEQ ID NO: 127) CGGGGGTGGTAGT (SEQ ID NO: 143)

(124) In certain embodiments, the linker is a cleavable linker. Preferably, such cleavable linker is susceptible to cleavage under intracellular conditions in the target cell but not in the packaging cell. Suitable cleavable linkers include, for example, a peptide linker cleavable by an intracellular protease, such as lysosomal protease or an endosomal protease. In exemplar embodiments, the linker can be a dipeptide linker, such as a valine-citrulline (val-cit) or a phenylalanine-lysine (phe-lys) linker. Other suitable linkers include linkers hydrolyzable at a pH of less than 5.5, such as a hydrazone linker. Additional suitable cleavable linkers include disulfide linkers.

(125) Cas proteins, or functional fragment or derivative thereof, also be operably linked to other heterologous polypeptides as fusion proteins. For example, a Cas protein, or functional fragment or derivative thereof, can be fused to a cleavage domain, an epigenetic modification domain, a transcriptional activation domain, or a transcriptional repressor domain. See WO 2014/089290, herein incorporated by reference in its entirety for all purposes. Examples of transcriptional activation domains include a herpes simplex virus VP16 activation domain, VP64 (which is a tetrameric derivative of VP16), a NFκB p65 activation domain, p53 activation domains 1 and 2, a CREB (cAMP response element binding protein) activation domain, an E2A activation domain, and an NFAT (nuclear factor of activated T-cells) activation domain. Other examples include activation domains from Oct1, Oct-2A, SP1, AP-2, CTF1, P300, CBP, PCAF, SRC1, PvALF, ERF-2, OsGAI, HALF-1, C1, AP1, ARF-5, ARF-6, ARF-7, ARF-8, CPRF1, CPRF4, MYC-RP/GP, TRAB1PC4, and HSF1. See, e.g., US 2016/0237456, EP3045537, and WO 2011/145121, each of which is incorporated by reference in its entirety for all purposes. In some cases, a transcriptional activation system can be used comprising a dCas9-VP64 fusion protein paired with MS2-p65-HSF1. Guide RNAs in such systems can be designed with aptamer sequences appended to sgRNA tetraloop and stem-loop 2 designed to bind dimerized MS2 bacteriophage coat proteins. See, e.g., Konermann et al. (2015) *Nature* 517(7536):583-588, herein incorporated by reference in its entirety for all purposes. Examples of transcriptional repressor domains include inducible cAMP early repressor (ICER) domains, Kruppel-associated box A (KRAB-A) repressor domains, YY1 glycine rich repressor domains, Sp1-like repressors, E(spl) repressors, IκB repressor, and MeCP2. Other examples include transcriptional repressor domains from A/B, KOX, TGF-beta-inducible early gene (TIEG), v-erbA, SID, SID4X, MBD2, MBD3, DNMT1, DNMG3A, DNMT3B, Rb, ROM2, See, e.g., EP3045537 and WO 2011/145121, each of which is incorporated by reference in its entirety for all purposes. Cas proteins can also be fused to a heterologous polypeptide providing increased or decreased stability. The fused domain or heterologous polypeptide can be located at the N-terminus, the C-terminus, or internally within the Cas protein, or functional fragment or derivative thereof.

(126) As one example, a Cas protein, or functional fragment or derivative thereof, can be fused to one or more heterologous polypeptides that provide for subcellular localization. Such heterologous polypeptides can include, for example, one or more nuclear localization signals (NLS) such as the SV40 NLS and/or an alpha-importin NLS for targeting to the nucleus, a mitochondrial localization signal for targeting to the mitochondria, an ER retention signal, and the like. See, e.g., Lange et al. (2007) *J. Biol. Chem.* 282:5101-5105, herein incorporated by reference in its entirety for all purposes. Such subcellular localization signals can be located at the N-terminus, the C-terminus, or

anywhere within the Cas protein, or functional fragment or derivative thereof. An NLS can comprise a stretch of basic amino acids, and can be a monopartite sequence or a bipartite sequence. Optionally, the Cas protein, or functional fragment or derivative thereof, comprises two or more NLSs, including an NLS (e.g., an alpha-importin NLS) at the N-terminus and/or an NLS (e.g., an SV40 NLS) at the C-terminus.

(127) Cas proteins, or functional fragment or derivative thereof, can also be operably linked to a heterologous polypeptide for ease of tracking or purification, such as a fluorescent protein, a purification tag, or an epitope tag. Examples of fluorescent proteins include green fluorescent proteins (e.g., GFP, GFP-2, tagGFP, turboGFP, eGFP, Emerald, Azami Green, Monomeric Azami Green, CopGFP, AceGFP, ZsGreenl), yellow fluorescent proteins (e.g., YFP, eYFP, Citrine, Venus, YPet, PhiYFP, ZsYellow1), blue fluorescent proteins (e.g. eBFP, eBFP2, Azurite, mKalama1, GFPuv, Sapphire, T-sapphire), cyan fluorescent proteins (e.g. eCFP, Cerulean, CyPet, AmCyanl, Midoriishi-Cyan), red fluorescent proteins (mKate, mKate2, mPlum, DsRed monomer, mCherry, mRFP1, DsRed-Express, DsRed2, DsRed-Monomer, HcRed-Tandem, HcRed1, AsRed2, eqFP611, mRaspberry, mStrawberry, Jred), orange fluorescent proteins (mOrange, mKO, Kusabira-Orange, Monomeric Kusabira-Orange, mTangerine, tdTomato), and any other suitable fluorescent protein. Examples of tags include glutathione-S-transferase (GST), chitin binding protein (CBP), maltose binding protein, thioredoxin (TRX), poly(NANP), tandem affinity purification (TAP) tag, myc, AcV5, AU1, AUS, E, ECS, E2, FLAG, hemagglutinin (HA), nus, Softag 1, Softag 3, Strep, SBP, Glu-Glu, HSV, KT3, S, 51, T7, V5, VSV-G, histidine (His), biotin carboxyl carrier protein (BCCP), and calmodulin.

(128) The Cas fusion protein can be produced using routine molecular biology techniques, such as those described above or in He et al., supra. Alternatively, Cas fusion protein can be prepared by various other methods.

(129) In certain embodiments, the nucleic acid encoding the Cas fusion protein comprises a regulatory element, including for example, a promoter, an enhancer, or a transcriptional repressor-binding element. Exemplary expression control sequences are known in the art and described in, for example, Goeddel, (1990) Gene Expression Technology: Methods in Enzymology, Vol. 185, Academic Press, San Diego, Calif., incorporated by reference in its entirety for all purposes.

(130) B. Transcription Activator-Like Effector Nucleases, Zinc Figure Nucleases, Meganucleases, and Restriction Endonucleases

(131) In certain embodiments, the gene-editing molecule can be zinc finger nucleases (ZFns), transcription activator-like effector nucleases (TALENs), meganucleases, and/or restriction endonucleases. Fusion RNA and fusion protein molecules using these gene-editing molecules, or functional fragment or derivative thereof, for use in the compositions and methods of the invention can be made in the same fashion and structure as that disclosed above for Cas molecules, or functional fragment or derivative thereof

(132) Transcription activator-like effector nucleases (TALEN) are restriction enzymes that can be engineered to cut target sequences of DNA. They are made by fusing a TAL effector DNA-binding domain to a DNA cleavage domain (a nuclease which cuts DNA strands). TAL effector nucleases are a class of sequence-specific nucleases that can be used to make double-strand breaks at specific target sequences in the genome of a prokaryotic or eukaryotic organism. TAL effector nucleases are created by fusing a native or engineered transcription activator-like (TAL) effector, or functional part thereof, to the catalytic domain of an endonuclease, such as, for example, FokI. The unique, modular TAL effector DNA binding domain allows for the design of proteins with potentially any given DNA recognition specificity. Thus, the DNA binding domains of the TAL effector nucleases can be engineered to recognize specific DNA target sites and thus, used to make double-strand breaks at desired target sequences. See, WO 2010/079430; Morbitzer et al. (2010) *PNAS* 10.1073/pnas.1013133107; Scholze & Boch (2010) *Virulence* 1:428-432; Christian et al. *Genetics* (2010) 186:757-761; Li et al. (2010) *Nuc. Acids Res.* doi: 10.1093/nar/gkq704; and Miller et al.

(2011) *Nature Biotechnology* 29:143-148; all of which are herein incorporated by reference in their entirety and for all purposes.

(133) Examples of suitable TAL nucleases, and methods for preparing suitable TAL nucleases, are disclosed, e.g., in US Patent Application No. 2011/0239315 A1, 2011/0269234 A1, 2011/0145940 A1, 2003/0232410 A1, 2005/0208489 A1, 2005/0026157 A1, 2005/0064474 A1, 2006/0188987 A1, and 2006/0063231 A1 (each hereby incorporated by reference in their entirety and for all purposes). In various embodiments, TAL effector nucleases are engineered that cut in or near a target nucleic acid sequence in, e.g., a genomic locus of interest, wherein the target nucleic acid sequence is at or near a sequence to be modified by a targeting vector. The TAL nucleases suitable for use with the various methods and compositions provided herein include those that are specifically designed to bind at or near target nucleic acid sequences to be modified by targeting vectors.

(134) In one embodiment, each monomer of the TALEN comprises 12-25 TAL repeats, wherein each TAL repeat binds a 1 bp subsite. In certain embodiments, the gene-editing molecule is a chimeric protein comprising a TAL repeat-based DNA binding domain operably linked to an independent nuclease. In certain embodiments, the independent nuclease is a FokI endonuclease. In one embodiment, the gene-editing molecule comprises a first TAL-repeat-based DNA binding domain and a second TAL-repeat-based DNA binding domain, wherein each of the first and the second TAL-repeat-based DNA binding domain is operably linked to a FokI nuclease, wherein the first and the second TAL-repeat-based DNA binding domain recognize two contiguous target DNA sequences in each strand of the target DNA sequence separated by about 6 bp to about 40 bp cleavage site, and wherein the FokI nucleases dimerize and make a double strand break at a target sequence.

(135) In certain embodiments, the gene-editing molecule comprises a first TAL-repeat-based DNA binding domain and a second TAL-repeat-based DNA binding domain, wherein each of the first and the second TAL-repeat-based DNA binding domain is operably linked to a FokI nuclease, wherein the first and the second TAL-repeat-based DNA binding domain recognize two contiguous target DNA sequences in each strand of the target DNA sequence separated by a 5 bp or 6 bp cleavage site, and wherein the FokI nucleases dimerize and make a double strand break.

(136) The gene-editing molecule employed in the various methods and compositions disclosed herein can further comprise a zinc-finger nuclease (ZFN). Zinc finger nucleases (ZFNs) are a class of engineered DNA-binding proteins that assist targeted editing of the genome by creating double strand breaks (DSBs) in DNA at targeted locations. ZFNs comprise two functional domains: i) a DNA-binding domain comprising a chain of two-finger modules (each recognizing a unique hexamer (6 bp) sequence of DNA—two-finger modules are stitched together to form a Zinc Finger Protein, each with specificity of >24 bp) and ii) a DNA-cleaving domain comprising a nuclease domain of Fok I. When the DNA-binding and -cleaving domains are fused together, a highly-specific pair of “genomic scissors” are created.

(137) In certain embodiments, each monomer of the ZFN comprises 3 or more zinc finger-based DNA binding domains, wherein each zinc finger-based DNA binding domain binds to a 3 bp subsite. In other embodiments, the ZFN is a chimeric protein comprising a zinc finger-based DNA binding domain operably linked to an independent nuclease. In certain embodiments, the independent endonuclease is a FokI endonuclease. In certain embodiments, the gene-editing molecule comprises a first ZFN and a second ZFN, wherein each of the first ZFN and the second ZFN is operably linked to a FokI nuclease, wherein the first and the second ZFN recognize two contiguous target DNA sequences in each strand of the target DNA sequence separated by about 6 bp to about 40 bp cleavage site or about a 5 bp to about 6 bp cleavage site, and wherein the FokI nucleases dimerize and make a double strand break. See, e.g., US20060246567; US20080182332; US20020081614; US20030021776; WO/2002/057308A2; US20130123484; 0520100291048; and, WO/2011/017293A2, each of which is herein incorporated by reference in their entirety for all

purposes.
(138) In certain embodiments of the compositions and methods provided herein, the gene-editing molecule comprises (a) a chimeric protein comprising a zinc finger-based DNA binding domain fused to a FokI endonuclease; or (b) a chimeric protein comprising a Transcription Activator-Like Effector Nuclease (TALEN) fused to a FokI endonuclease.
(139) In still another embodiment, the gene-editing molecule is a meganuclease. Meganucleases have been classified into four families based on conserved sequence motifs, the families are the LAGLIDADG (SEQ ID NO: 24), GIY-YIG, H—N—H, and His-Cys box families. These motifs participate in the coordination of metal ions and hydrolysis of phosphodiester bonds. HEases are notable for their long recognition sites, and for tolerating some sequence polymorphisms in their DNA substrates. Meganuclease domains, structure and function are known, see e.g., Guhan and Muniyappa (2003) Crit Rev Biochem Mol Biol 38:199-248; Lucas et al., (2001) Nucleic Acids Res 29:960-9; Jurica and Stoddard, (1999) Cell Mol Life Sci 55:1304-26; Stoddard, (2006) Q Rev Biophys 38:49-95; and Moure et al., (2002) Nat Struct Biol 9:764. In some examples a naturally occurring variant, and/or engineered derivative meganuclease is used. Methods for modifying the kinetics, cofactor interactions, expression, optimal conditions, and/or recognition site specificity, and screening for activity are known, see e.g., Epinat et al., (2003) Nucleic Acids Res 31:2952-62; Chevalier et al., (2002) Mol Cell 10:895-905; Gimble et al., (2003) Mol Biol 334:993-1008; Seligman et al., (2002) Nucleic Acids Res 30:3870-9; Sussman et al., (2004) J Mol Biol 342:31-41; Rosen et al., (2006) Nucleic Acids Res 34:4791-800; Chames et al., (2005) Nucleic Acids Res 33:e178; Smith et al., (2006) Nucleic Acids Res 34:e149; Gruen et al., (2002) Nucleic Acids Res 30:e29; Chen and Zhao, (2005) Nucleic Acids Res 33:e154; WO2005105989; WO2003078619; WO2006097854; WO2006097853; WO2006097784; and WO2004031346.
(140) Any meganuclease can be used herein, including, but not limited to, I-SceI, I-SceII, I-SceIII, I-SceIV, I-SceV, I-SceVI, I-SceVII, I-CeuI, I-CeuAIIP, I-CreI, I-CrepsblP, I-CrepsbllP, I-CrepsbIIIP, I-CrepsblVP, I-TliI, I-PpoI, PI-PspI, F-SceI, F-SceII, F-SuvI, F-TeeI, F-TevII, I-Aural, I-AniI, I-ChuI, I-CmoeI, I-CpaI, I-CpaII, I-CsmI, I-CvuI, I-CvuAIP, I-DdiI, I-DdiII, I-DirI, I-DmoI, I-HmuI, I-HmuII, I-HsNIP, I-LlaI, I-MsoI, I-NaaI, I-NanI, I-NcIIP, I-NgrIP, I-NitI, I-NjaI, I-Nsp236IP, I-PakI, I-PboIP, I-PcuIP, I-PcuAI, I-PcuVI, I-PgrIP, I-PobIP, I-PorI, I-PorIIP, I-PbpIP, I-SpBetaIP, I-ScaI, I-SexIP, I-SneIP, I-SpomI, I-SpomCP, I-SpomIP, I-SpomIIP, I-SquIP, I-Ssp6803I, I-SthPhiJP, I-SthPhiST3P, I-SthPhiSTe3bP, I-TdeIP, I-TeeI, I-TevII, I-TevIII, I-UarAP, I-UarHGPAIP, I-UarHGPA13P, I-VinIP, I-ZbiIP, PI-MtuI, PI-MtuHIP PI-MtuHIIP, PI-PfuI, PI-PfuII, PI-PkoI, PI-PkoII, PI-Rma43812IP, PI-SpBetaIP, PI-SceI, PI-TfuI, PI-TfuII, PI-ThyI, PI-TliI, PI-TliII, or any active variants or fragments thereof.
(141) In one embodiment, the meganuclease recognizes double-stranded DNA sequences of 12 to 40 base pairs. In one embodiment, the meganuclease recognizes one perfectly matched target sequence in the genome. In one embodiment, the meganuclease is a homing nuclease. In one embodiment, the homing nuclease is a LAGLIDADG (SEQ ID NO: 24) family of homing nuclease. In one embodiment, the LAGLIDADG (SEQ ID NO: 24) family of homing nuclease is selected from I-SceI, I-CreI, and I-DmoI.
(142) Gene-editing molecules can further comprise restriction endonucleases, which include Type I, Type II, Type III, and Type IV endonucleases. Type I and Type III restriction endonucleases recognize specific recognition sites, but typically cleave at a variable position from the nuclease binding site, which can be hundreds of base pairs away from the cleavage site (recognition site). In Type II systems the restriction activity is independent of any methylase activity, and cleavage typically occurs at specific sites within or near to the binding site. Most Type II enzymes cut palindromic sequences, however Type IIa enzymes recognize non-palindromic recognition sites and cleave outside of the recognition site, Type IIb enzymes cut sequences twice with both sites outside of the recognition site, and Type IIs enzymes recognize an asymmetric recognition site and cleave on one side and at a defined distance of about 1-20 nucleotides from the recognition site.

Type IV restriction enzymes target methylated DNA. Restriction enzymes are further described and classified, for example in the REBASE database (webpage at rebase.neb.com; Roberts et al., (2003) Nucleic Acids Res 31:418-20), Roberts et al., (2003) Nucleic Acids Res 31:1805-12, and Belfort et al., (2002) in Mobile DNA II, pp. 761-783, Eds. Craigie et al., (ASM Press, Washington, D.C.).
(143) ZFNs and TALENs introduce DSBs in a target genomic sequence and activate non-homologous end-joining (NHEJ)-mediated DNA repair, which generates a mutant allele comprising an insertion or a deletion of a nucleic acid sequence at the genomic locus of interest and thereby causes disruption of the genomic locus of interest in a cell. DSBs also stimulate homology-directed repair (HDR) by homologous recombination if a repair template is provided. HDR can result in a perfect repair that restores the original sequence at the broken site, or it can be used to direct a designed modification, such as a deletion, insertion, or replacement of the sequence at the site of the double strand break.
(144) C. Guide RNAs
(145) In one aspect, the retroviral particle comprises a guide RNA (gRNA). In certain embodiments, the nucleic acid sequence of gRNA is incorporated into the genomic plasmid of the retroviral particle.
(146) A "guide RNA" or "gRNA" is an RNA molecule that binds to a Cas protein (e.g., Cas9 protein), or functional fragment or derivative thereof, and targets the Cas protein to a specific location within a target DNA. Guide RNAs can comprise two segments: a "DNA-targeting segment" and a "protein-binding segment". "Segment" includes a section or region of a molecule, such as a contiguous stretch of nucleotides in an RNA. Some gRNAs, such as those for Cas9, can comprise two separate RNA molecules: an "activator-RNA" (e.g., tracrRNA) and a "targeter-RNA" (e.g., CRISPR RNA or crRNA). Other gRNAs are a single RNA molecule (single RNA polynucleotide), which can also be called a "single-molecule gRNA," a "single-guide RNA," or an "sgRNA". See, e.g., WO 2013/176772, WO 2014/065596, WO 2014/089290, WO 2014/093622, WO 2014/099750, WO 2013/142578, and WO 2014/131833, each of which is herein incorporated by reference in its entirety for all purposes. For Cas9, for example, a single-guide RNA can comprise a crRNA fused to a tracrRNA (e.g., via a linker). For Cpf1, for example, only a crRNA is needed to achieve binding to a target sequence. The terms "guide RNA" and "gRNA" include both double-molecule (i.e., modular) gRNAs and single-molecule gRNAs.
(147) An exemplary two-molecule gRNA comprises a crRNA-like ("CRISPR RNA" or "targeter-RNA" or "crRNA" or "crRNA repeat") molecule and a corresponding tracrRNA-like ("trans-acting CRISPR RNA" or "activator-RNA" or "tracrRNA") molecule. A crRNA comprises both the DNA-targeting segment (single-stranded) of the gRNA and a stretch of nucleotides that forms one half of the dsRNA duplex of the protein-binding segment of the gRNA.
(148) A corresponding tracrRNA (activator-RNA) comprises a stretch of nucleotides that forms the other half of the dsRNA duplex of the protein-binding segment of the gRNA. A stretch of nucleotides of a crRNA are complementary to and hybridize with a stretch of nucleotides of a tracrRNA to form the dsRNA duplex of the protein-binding domain of the gRNA. As such, each crRNA can be said to have a corresponding tracrRNA.
(149) In systems in which both a crRNA and a tracrRNA are needed, the crRNA and the corresponding tracrRNA hybridize to form a gRNA. In systems in which only a crRNA is needed, the crRNA can be the gRNA. The crRNA additionally provides the single-stranded DNA-targeting segment that hybridizes to a guide RNA recognition sequence. If used for modification within a cell, the exact sequence of a given crRNA or tracrRNA molecule can be designed to be specific to the species in which the RNA molecules will be used. See, e.g., Mali et al. (2013) *Science* 339:823-826; Jinek et al. (2012) *Science* 337:816-821; Hwang et al. (2013) *Nat. Biotechnol.* 31:227-229; Jiang et al. (2013) *Nat. Biotechnol.* 31:233-239; and Cong et al. (2013) *Science* 339:819-823, each of which is herein incorporated by reference in its entirety for all purposes.
(150) The DNA-targeting segment (crRNA) of a given gRNA comprises a nucleotide sequence that

is complementary to a sequence (i.e., the guide RNA recognition sequence) in a target DNA. The DNA-targeting segment of a gRNA interacts with a target DNA in a sequence-specific manner via hybridization (i.e., base pairing). As such, the nucleotide sequence of the DNA-targeting segment may vary and determines the location within the target DNA with which the gRNA and the target DNA will interact. The DNA-targeting segment of a subject gRNA can be modified to hybridize to any desired sequence within a target DNA. Naturally occurring crRNAs differ depending on the CRISPR/Cas system and organism but often contain a targeting segment of between 21 to 72 nucleotides length, flanked by two direct repeats (DR) of a length of between 21 to 46 nucleotides (see, e.g., WO 2014/131833, herein incorporated by reference in its entirety for all purposes). In the case of *S. pyogenes*, the DRs are 36 nucleotides long and the targeting segment is 30 nucleotides long. The 3′ located DR is complementary to and hybridizes with the corresponding tracrRNA, which in turn binds to the Cas protein.

(151) The DNA-targeting segment can have a length of at least about 12 nucleotides, at least about 15 nucleotides, at least about 17 nucleotides, at least about 18 nucleotides, at least about 19 nucleotides, at least about 20 nucleotides, at least about 25 nucleotides, at least about 30 nucleotides, at least about 35 nucleotides, or at least about 40 nucleotides. Such DNA-targeting segments can have a length from about 12 nucleotides to about 100 nucleotides, from about 12 nucleotides to about 80 nucleotides, from about 12 nucleotides to about 50 nucleotides, from about 12 nucleotides to about 40 nucleotides, from about 12 nucleotides to about 30 nucleotides, from about 12 nucleotides to about 25 nucleotides, or from about 12 nucleotides to about 20 nucleotides. For example, the DNA targeting segment can be from about 15 nucleotides to about 25 nucleotides (e.g., from about 17 nucleotides to about 20 nucleotides, or about 17 nucleotides, about 18 nucleotides, about 19 nucleotides, or about 20 nucleotides). See, e.g., US 2016/0024523, herein incorporated by reference in its entirety for all purposes. For Cas9 from *S. pyogenes*, a typical DNA-targeting segment is between 16 and 20 nucleotides in length or between 17 and 20 nucleotides in length. For Cas9 from *S. aureus*, a typical DNA-targeting segment is between 21 and 23 nucleotides in length. For Cpf1, a typical DNA-targeting segment is at least 16 nucleotides in length or at least 18 nucleotides in length.

(152) TracrRNAs can be in any form (e.g., full-length tracrRNAs or active partial tracrRNAs) and of varying lengths. They can include primary transcripts or processed forms. For example, tracrRNAs (as part of a single-guide RNA or as a separate molecule as part of a two-molecule gRNA) may comprise or consist of all or a portion of a wild type tracrRNA sequence (e.g., about or more than about 20, 26, 32, 45, 48, 54, 63, 67, 85, or more nucleotides of a wild type tracrRNA sequence). Examples of wild type tracrRNA sequences from *S. pyogenes* include 171-nucleotide, 89-nucleotide, 75-nucleotide, and 65-nucleotide versions. See, e.g., Deltcheva et al. (2011) *Nature* 471:602-607; WO 2014/093661, each of which is herein incorporated by reference in its entirety for all purposes. Examples of tracrRNAs within single-guide RNAs (sgRNAs) include the tracrRNA segments found within +48, +54, +67, and +85 versions of sgRNAs, where “+n” indicates that up to the +n nucleotide of wild type tracrRNA is included in the sgRNA. See U.S. Pat. No. 8,697,359, herein incorporated by reference in its entirety for all purposes.

(153) The percent complementarity between the DNA-targeting sequence and the guide RNA recognition sequence within the target DNA can be at least 60% (e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 98%, at least 99%, or 100%). The percent complementarity between the DNA-targeting sequence and the guide RNA recognition sequence within the target DNA can be at least 60% over about 20 contiguous nucleotides. As an example, the percent complementarity between the DNA-targeting sequence and the guide RNA recognition sequence within the target DNA is 100% over the 14 contiguous nucleotides at the 5′ end of the guide RNA recognition sequence within the complementary strand of the target DNA and as low as 0% over the remainder. In such a case, the DNA-targeting sequence can be considered to be 14 nucleotides in length. As another example, the percent

complementarity between the DNA-targeting sequence and the guide RNA recognition sequence within the target DNA is 100% over the seven contiguous nucleotides at the 5′ end of the guide RNA recognition sequence within the complementary strand of the target DNA and as low as 0% over the remainder. In such a case, the DNA-targeting sequence can be considered to be 7 nucleotides in length. In some guide RNAs, at least 17 nucleotides within the DNA-target sequence are complementary to the target DNA. For example, the DNA-targeting sequence can be 20 nucleotides in length and can comprise 1, 2, or 3 mismatches with the target DNA (the guide RNA recognition sequence). Preferably, the mismatches are not adjacent to a protospacer adjacent motif (PAM) sequence (e.g., the mismatches are in the 5′ end of the DNA-targeting sequence, or the mismatches are at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, or 19 base pairs away from the PAM sequence).

(154) The protein-binding segment of a gRNA can comprise two stretches of nucleotides that are complementary to one another. The complementary nucleotides of the protein-binding segment hybridize to form a double-stranded RNA duplex (dsRNA). The protein-binding segment of a subject gRNA interacts with a Cas protein, or functional fragment or derivative thereof, and the gRNA directs the bound Cas protein, or functional fragment or derivative thereof, to a specific nucleic acid sequence within target DNA via the DNA-targeting segment.

(155) Guide RNAs can include modifications or sequences that provide for additional desirable features (e.g., modified or regulated stability; subcellular targeting; tracking with a fluorescent label; a binding site for a protein or protein complex; and the like). Examples of such modifications include, for example, a 5′ cap (e.g., a 7-methylguanylate cap (m7G)); a 3′ polyadenylated tail (i.e., a 3′ poly(A) tail); a riboswitch sequence (e.g., to allow for regulated stability and/or regulated accessibility by proteins and/or protein complexes); a stability control sequence; a sequence that forms a dsRNA duplex (i.e., a hairpin); a modification or sequence that targets the RNA to a subcellular location (e.g., nucleus, mitochondria, chloroplasts, and the like); a modification or sequence that provides for tracking (e.g., direct conjugation to a fluorescent molecule, conjugation to a moiety that facilitates fluorescent detection, a sequence that allows for fluorescent detection, and so forth); a modification or sequence that provides a binding site for proteins (e.g., proteins that act on DNA, including transcriptional activators, transcriptional repressors, DNA methyltransferases, DNA demethylases, histone acetyltransferases, histone deacetylases, and the like); and combinations thereof. Other examples of modifications include engineered stem loop duplex structures, engineered bulge regions, engineered hairpins 3′ of the stem loop duplex structure, or any combination thereof. See, e.g., US 2015/0376586, herein incorporated by reference in its entirety for all purposes. A bulge can be an unpaired region of nucleic acids within the duplex made up of the crRNA-like region and the minimum tracrRNA-like region. A bulge can comprise, on one side of the duplex, an unpaired 5′-XXXY-3′ where X is any purine and Y can be a nucleotide that can form a wobble pair with a nucleotide on the opposite strand, and an unpaired nucleotide region on the other side of the duplex.

(156) In some cases, a transcriptional activation system can be used comprising a dCas9-VP64 fusion protein paired with MS2-p65-HSF1. Guide RNAs in such systems can be designed with aptamer sequences appended to sgRNA tetraloop and stem-loop 2 designed to bind dimerized MS2 bacteriophage coat proteins. See, e.g., Konermann et al. (2015) *Nature* 517(7536):583-588, herein incorporated by reference in its entirety for all purposes.

(157) Guide RNAs can be provided in any form. For example, the gRNA can be provided in the form of RNA, either as two molecules (separate crRNA and tracrRNA) or as one molecule (sgRNA). The gRNA can also be provided in the form of DNA encoding the gRNA. The DNA encoding the gRNA can encode a single RNA molecule (sgRNA) or separate RNA molecules (e.g., separate crRNA and tracrRNA). In the latter case, the DNA encoding the gRNA can be provided as one DNA molecule or as separate DNA molecules encoding the crRNA and tracrRNA, respectively.

(158) When a gRNA is provided in the form of DNA, the gRNA can be transiently, conditionally, or constitutively expressed in the cell. DNAs encoding gRNAs can be stably integrated into the genome of the cell and operably linked to a promoter active in the cell. Alternatively, DNAs encoding gRNAs can be operably linked to a promoter in an expression construct. For example, the DNA encoding the gRNA can be in a vector comprising a heterologous nucleic acid. Promoters that can be used in such expression constructs include promoters active, for example, in one or more of a eukaryotic cell, a human cell, a non-human cell, a mammalian cell, a non-human mammalian cell, a rodent cell, a mouse cell, a rat cell, a hamster cell, a rabbit cell, a pluripotent cell, an embryonic stem (ES) cell, an adult stem cell, a developmentally restricted progenitor cell, an induced pluripotent stem (iPS) cell, or a one-cell stage embryo. Such promoters can be, for example, conditional promoters, inducible promoters, constitutive promoters, or tissue-specific promoters. Such promoters can also be, for example, bidirectional promoters. In certain embodiments, an RNA Pol III promoter can be operatively linked to a gRNA sequence (if included in the lentivirus vector) to control expression of such sequence. RNA Pol III promoters are frequently used to express small RNAs, such as small interfering RNA (siRNA)/short hairpin RNA (shRNA) and guide RNA sequences used in CRISPR-Cas9 systems. Examples of RNA Pol III promoters that can be used in the invention include, but are not limited to, the human U6 promoter, a rat U6 polymerase III promoter, or a mouse U6 polymerase III promoter, and the H1 promoter, which are described in, for example Goomer and Kunkel, Nucl. Acids Res., 20 (18): 4903-4912 (1992), and Myslinski et al., Nucleic Acids Res., 29(12): 2502-9 (2001).
(159) D. Guide RNA Recognition Sequences
(160) The term “guide RNA recognition sequence” includes nucleic acid sequences present in a target DNA to which a DNA-targeting segment of a gRNA will bind, provided sufficient conditions for binding exist. For example, gRNA recognition sequences include sequences to which a gRNA is designed to have complementarity, where hybridization between a guide RNA recognition sequence and a DNA targeting sequence promotes the formation of a CRISPR complex. Full complementarity is not necessarily required, provided that there is sufficient complementarity to cause hybridization and promote formation of a CRISPR complex. Guide RNA recognition sequences also include cleavage sites for Cas proteins, described in more detail below. A gRNA recognition sequence can comprise any polynucleotide, which can be located, for example, in the nucleus or cytoplasm of a cell or within an organelle of a cell, such as a mitochondrion or chloroplast.
(161) The gRNA recognition sequence within a target DNA can be targeted by (i.e., be bound by, or hybridize with, or be complementary to) a Cas protein or a gRNA. Suitable DNA/RNA binding conditions include physiological conditions normally present in a cell. Other suitable DNA/RNA binding conditions (e.g., conditions in a cell-free system) are known in the art (see, e.g., Molecular Cloning: A Laboratory Manual, 3rd Ed. (Sambrook et al., Harbor Laboratory Press 2001), herein incorporated by reference in its entirety for all purposes). The strand of the target DNA that is complementary to and hybridizes with the Cas protein or gRNA can be called the “complementary strand,” and the strand of the target DNA that is complementary to the “complementary strand” (and is therefore not complementary to the Cas protein or gRNA) can be called “noncomplementary strand” or “template strand.”
(162) The Cas protein can cleave the nucleic acid at a site within or outside of the nucleic acid sequence present in the target DNA to which the DNA-targeting segment of a gRNA will bind. The “cleavage site” includes the position of a nucleic acid at which a Cas protein produces a single-strand break or a double-strand break. For example, formation of a CRISPR complex (comprising a gRNA hybridized to a guide RNA recognition sequence and complexed with a Cas protein) can result in cleavage of one or both strands in or near (e.g., within 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 50, or more base pairs from) the nucleic acid sequence present in a target DNA to which a DNA-targeting segment of a gRNA will bind. If the cleavage site is outside of the nucleic acid sequence to which

the DNA-targeting segment of the gRNA will bind, the cleavage site is still considered to be within the "guide RNA recognition sequence." The cleavage site can be on only one strand or on both strands of a nucleic acid. Cleavage sites can be at the same position on both strands of the nucleic acid (producing blunt ends) or can be at different sites on each strand (producing staggered ends (i.e., overhangs)). Staggered ends can be produced, for example, by using two Cas proteins, each of which produces a single-strand break at a different cleavage site on a different strand, thereby producing a double-strand break. For example, a first nickase can create a single-strand break on the first strand of double-stranded DNA (dsDNA), and a second nickase can create a single-strand break on the second strand of dsDNA such that overhanging sequences are created. In some cases, the guide RNA recognition sequence of the nickase on the first strand is separated from the guide RNA recognition sequence of the nickase on the second strand by at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 40, 50, 75, 100, 250, 500, or 1,000 base pairs.
(163) Site-specific binding and cleavage of target DNA by Cas proteins can occur at locations determined by both (i) base-pairing complementarity between the gRNA and the target DNA and (ii) a short motif, called the protospacer adjacent motif (PAM), in the target DNA. The PAM can flank the guide RNA recognition sequence. Optionally, the guide RNA recognition sequence can be flanked on the 3′ end by the PAM. Alternatively, the guide RNA recognition sequence can be flanked on the 5′ end by the PAM. For example, the cleavage site of Cas proteins can be about 1 to about 10 or about 2 to about 5 base pairs (e.g., 3 base pairs) upstream or downstream of the PAM sequence. In some cases (e.g., when Cas9 from *S. pyogenes* or a closely related Cas9 is used), the PAM sequence of the non-complementary strand can be 5′-N.sub.1GG-3′, where N.sub.1 is any DNA nucleotide and is immediately 3′ of the guide RNA recognition sequence of the non-complementary strand of the target DNA. As such, the PAM sequence of the complementary strand would be 5′-CCN.sub.2-3′, where N2 is any DNA nucleotide and is immediately 5′ of the guide RNA recognition sequence of the complementary strand of the target DNA. In some such cases, N.sub.1 and N.sub.2 can be complementary and the N.sub.1-N.sub.2 base pair can be any base pair (e.g., N.sub.1=C and N.sub.2=G; N.sub.1=G and N.sub.2=C; N.sub.1=A and N.sub.2=T; or N.sub.1=T, and N.sub.2=A). In the case of Cas9 from *S. aureus*, the PAM can be NNGRRT or NNGRR, where N can A, G, C, or T, and R can be G or A. In the case of Cas9 from *C. jejuni*, the PAM can be, for example, NNNNACAC or NNNNRYAC, where N can be A, G, C, or T, and R can be G or A. In some cases (e.g., for FnCpf1), the PAM sequence can be upstream of the 5′ end and have the sequence 5′-TTN-3′.
(164) Examples of gRNA recognition sequences include a DNA sequence complementary to the DNA-targeting segment of a gRNA, or such a DNA sequence in addition to a PAM sequence. For example, the target motif can be a 20-nucleotide DNA sequence immediately preceding an NGG motif recognized by a Cas9 protein, such as GN.sub.19NGG (SEQ ID NO: 25) or N.sub.20NGG (SEQ ID NO: 26). See, e.g., WO 2014/165825, herein incorporated by reference in its entirety for all purposes. The guanine at the 5′ end can facilitate transcription by RNA polymerase in cells. Other examples of guide RNA recognition sequences can include two guanine nucleotides at the 5′ end (e.g., GGN.sub.20NGG; SEQ ID NO: 27) to facilitate efficient transcription by T7 polymerase in vitro. See, e.g., WO 2014/065596, herein incorporated by reference in its entirety for all purposes. Other guide RNA recognition sequences can have between 4-22 nucleotides in length of SEQ ID NOs: 28-30, including the 5′ G or GG and the 3′ GG or NGG. Yet other guide RNA recognition sequences can have between 14 and 20 nucleotides in length of SEQ ID NOS: 31-58.
(165) The gRNA recognition sequence can be any nucleic acid sequence endogenous or exogenous to a cell. The gRNA recognition sequence can be a sequence coding a gene product (e.g., a protein) or a non-coding sequence (e.g., a regulatory sequence) or can include both.
(166) E. Repair Template
(167) In one aspect, the retroviral particle comprises a sequence corresponding to a repair template.
(168) As used herein, the terms "repair template", "RT", "recombination template", "donor nucleic

acid molecule" or "donor polynucleotide", which can be used interchangeably, refer to a segment of DNA that one desires to integrate at the target locus. In certain embodiments, the repair template comprises one or more polynucleotides of interest. In other embodiments, the repair template can comprise one or more expression cassettes. A given expression cassette can comprise a polynucleotide of interest, a polynucleotide encoding a selection marker and/or a reporter gene along with the various regulatory components that influence expression.

(169) In certain embodiments, the repair template can comprise a segment of genomic DNA, a cDNA, a regulatory region, or any portion or combination thereof. In certain embodiments, the repair template can comprise a nucleic acid from a eukaryote, a mammal, a human, a non-human mammal, a rodent, a rat, a non-rat rodent, a mouse, a hamster, a rabbit, a pig, a bovine, a deer, a sheep, a goat, a chicken, a cat, a dog, a ferret, a primate (e.g., marmoset, rhesus monkey), a domesticated mammal, or an agricultural mammal or any other organism of interest.

(170) In certain embodiments, the repair template comprises a knock-in allele of at least one exon of an endogenous gene. In certain embodiments, the repair template comprises a knock-in allele of the entire endogenous gene (i.e., "gene-swap knock-in").

(171) In further embodiments, the repair template comprises a conditional allele. In certain embodiments, the conditional allele is a multifunctional allele, as described in US 2011/0104799, which is incorporated by reference in its entirety. In certain embodiments, the conditional allele comprises: (a) an actuating sequence in sense orientation with respect to transcription of a target gene, and a drug selection cassette in sense or antisense orientation; (b) in antisense orientation a nucleotide sequence of interest (NSI) and a conditional by inversion module (COIN, which utilizes an exon-splitting intron and an invertible genetrap-like module; see, for example, US 2011/0104799, which is incorporated by reference in its entirety); and (c) recombinable units that recombine upon exposure to a first recombinase to form a conditional allele that (i) lacks the actuating sequence and the DSC, and (ii) contains the NSI in sense orientation and the COIN in antisense orientation.

(172) In certain embodiments, the repair template is under 10 kb in size.

(173) In certain embodiments, the repair template comprises a deletion of, for example, a eukaryotic cell, a mammalian cell, a human cell, or a non-human mammalian cell genomic DNA sequence.

(174) In certain embodiments, the repair template comprises an insertion or a replacement of a eukaryotic, a mammalian, a human, or a non-human mammalian nucleic acid sequence with a homologous or orthologous human nucleic acid sequence. In certain embodiments, the repair template comprises an insertion or replacement of a DNA sequence with a homologous or orthologous human nucleic acid sequence at an endogenous locus that comprises the corresponding DNA sequence.

(175) In certain embodiments, the genetic modification is an addition of a nucleic acid sequence.

(176) In certain embodiments, repair template comprises a genetic modification in a coding sequence. In certain embodiments, the genetic modification comprises a deletion mutation of a coding sequence. In certain embodiments, the genetic modification comprises a fusion of two endogenous coding sequences.

(177) In certain embodiments, the repair template comprises an insertion or a replacement of a eukaryotic, a non-rat eukaryotic, a mammalian, a human, or a non-human mammalian, nucleic acid sequence with a homologous or orthologous human nucleic acid sequence. In certain embodiments, the repair template comprises an insertion or replacement of a rat DNA sequence with a homologous or orthologous human nucleic acid sequence at an endogenous rat locus that comprises the corresponding rat DNA sequence.

(178) In certain embodiments, the genetic modification comprises a deletion of a non-protein-coding sequence, but does not comprise a deletion of a protein-coding sequence. In certain embodiments, the deletion of the non-protein-coding sequence comprises a deletion of a regulatory

element. In certain embodiments, the genetic modification comprises a deletion of a promoter. In certain embodiments, the genetic modification comprises an addition of a promoter or a regulatory element. In certain embodiments, the genetic modification comprises a replacement of a promoter or a regulatory element.
(179) In certain embodiments, the nucleic acid sequence of the repair template can comprise a polynucleotide that when integrated into the genome will produce a genetic modification of a region of the mammalian, human, or a non-human mammalian target locus (e.g., ApoE, IL-2, Rag1, or Rag2), wherein the genetic modification at the target locus results in a decrease in activity, increase in activity, or a modulation of activity of the target gene. In certain embodiments, a knockout ("null allele") is generated.
(180) In further embodiments, the repair template results in the replacement of a portion of the mammalian, human cell, or non-human mammalian target locus (e.g., ApoE locus, the interleukin-2 receptor gamma locus and/or Rag2 locus, and/or Rag1 locus and/or Rag2/Rag1 locus with the corresponding homologous or orthologous portion of an ApoE locus, an interleukin-2 receptor gamma locus, a Rag2 locus, a Rag1 locus and/or a Rag2/Rag1 locus from another organism).
(181) Still in other embodiments, the repair template comprises a polynucleotide sharing across its full length at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% to a portion of the locus it is replacing (e.g., an ApoE locus, an IL-2 receptor gamma locus, a Rag2 locus, a Rag1 locus and/or a Rag2/Rag).
(182) The given repair template and the corresponding region of the mammalian, human cell, or non-human mammalian locus being replaced can be a coding region, an intron, an exon, an untranslated region, a regulatory region, a promoter, or an enhancer or any combination thereof Moreover, the given repair template and/or the region of the mammalian, human cell, or non-human mammalian locus being deleted can be of any desired length, including for example, between 10-100 nucleotides in length, 100-500 nucleotides in length, 500-1 kb nucleotides in length, 1 Kb to 1.5 kb nucleotides in length, 1.5 kb to 2 kb nucleotides in length, 2 kb to 2.5 kb nucleotides in length, 2.5 kb to 3 kb nucleotides in length, 3 kb to 5 kb nucleotides in length, 5 kb to 8 kb nucleotides in length, 8 kb to 10 kb nucleotides in length or more. In other instances, the size of the insertion or replacement is from about 5 kb to about 10 kb, from about 10 kb to about 20 kb, from about 20 kb to about 40 kb, from about 40 kb to about 60 kb, from about 60 kb to about 80 kb, from about 80 kb to about 100 kb, from about 100 kb to about 150 kb, from about 150 kb to about 200 kb, from about 200 kb to about 250 kb, from about 250 kb to about 300 kb, from about 300 kb to about 350 kb, from about 350 kb to about 400 kb, from about 400 kb to about 800 kb, from about 800 kb to 1 Mb, from about 1 Mb to about 1.5 Mb, from about 1.5 Mb to about 2 Mb, from about 2 Mb, to about 2.5 Mb, from about 2.5 Mb to about 2.8 Mb, from about 2.8 Mb to about 3 Mb. In other embodiments, the given repair template and/or the region of the mammalian, human cell, or non-human mammalian locus being deleted is at least 100, 200, 300, 400, 500, 600, 700, 800, or 900 nucleotides or at least 1 kb, 2 kb, 3 kb, 4 kb, 5 kb, 6 kb, 7 kb, 8 kb, 9 kb, 10 kb, 11 kb, 12 kb, 13 kb, 14 kb, 15 kb, 16 kb or greater.
(183) DNA of the repair template can be stably integrated into the genome of the cell
(184) In certain embodiments, the promoter is a tissue-specific promoter. In certain embodiments, the promoter is a hepatocyte-specific promoter. In certain embodiments, the promoter is a neuron-specific promoter. In certain embodiments, the promoter is a glia-specific promoter. In certain embodiments, the promoter is a muscle cell-specific promoter. In certain embodiments, the promoter is a heart cell-specific promoter. In certain embodiments, the promoter is a kidney cell-specific promoter. In certain embodiments, the promoter is a bone cell-specific promoter. In certain embodiments, the promoter is an endothelial cell-specific promoter. In certain embodiments, the promoter is an immune cell-specific promoter. In certain embodiments, the immune cell promoter is a B cell promoter. In certain embodiments, the immune cell promoter is a T cell promoter.
(185) In certain embodiments, the promoter is a developmentally-regulated promoter. In certain

embodiments, the developmentally-regulated promoter is active only during an embryonic stage of development. In certain embodiments, the developmentally-regulated promoter is active only in an adult cell.
(186) In specific embodiments, the promoter may be selected based on the cell type. Thus the various promoters find use in a eukaryotic cell, a non-rat eukaryotic cell, a mammalian cell, a non-human mammalian cell, a pluripotent cell, a non-pluripotent cell, a non-human pluripotent cell, a human pluripotent cell, a human ES cell, a human adult stem cell, a developmentally-restricted human progenitor cell, a human iPS cell, a human cell, a rodent cell, a non-rat rodent cell, a rat cell, a mouse cell, a hamster cell, a fibroblast or a CHO cell.
(187) In some embodiments, the repair template comprises a nucleic acid flanked with site-specific recombination target sequences. It is recognized the while the entire nucleic acid can be flanked by such site-specific recombination target sequences, any region or individual polynucleotide of interest within the insert nucleic acid can also be flanked by such sites. The site-specific recombinase can be introduced into the cell by any means, including by introducing the recombinase polypeptide into the cell or by introducing a polynucleotide encoding the site-specific recombinase into the target cell. The polynucleotide encoding the site-specific recombinase can be located within the repair template or within a separate polynucleotide. The site-specific recombinase can be operably linked to a promoter active in the cell including, for example, an inducible promoter, a promoter that is endogenous to the cell, a promoter that is heterologous to the cell, a cell-specific promoter, a tissue-specific promoter, or a developmental stage-specific promoter. Site-specific recombination target sequences, which can flank the nucleic acid or any polynucleotide of interest in the nucleic acid can include, but are not limited to, loxP, lox511, lox2272, lox66, lox71, loxM2, lox5171, FRT, FRT11, FRT71, attp, att, FRT, rox, and a combination thereof.
(188) In certain embodiments, the site-specific recombination sites flank a polynucleotide encoding a selection marker and/or a reporter gene contained within the repair template. In such instances following integration of the repair template the targeted locus the sequences between the site-specific recombination sites can be removed.
(189) In certain embodiments, the repair template comprises a polynucleotide encoding a selection marker. The selection marker can be contained in a selection cassette. Such selection markers include, but are not limited, to neomycin phosphotransferase (neor), hygromycin B phosphotransferase (hygr), puromycin-N-acetyltransferase (puror), blasticidin S deaminase (bsrr), xanthine/guanine phosphoribosyl transferase (gpt), or herpes simplex virus thymidine kinase (HSV-k), or a combination thereof. In certain embodiments, the polynucleotide encoding the selection marker is operably linked to a promoter active in the cell, rat cell, pluripotent rat cell, the ES rat cell, a eukaryotic cell, a non-rat eukaryotic cell, a pluripotent cell, a non-pluripotent cell, a non-human pluripotent cell, a human pluripotent cell, a human ES cell, a human adult stem cell, a developmentally-restricted human progenitor cell, a human iPS cell, a mammalian cell, a non-human mammalian cell, a human cell, a rodent cell, a non-rat rodent cell, a mouse cell, a hamster cell, a fibroblast, or a CHO cell. When serially tiling polynucleotides of interest into a targeted locus, the selection marker can comprise a recognition site for a gene-editing molecule, as outlined above. In certain embodiments, the polynucleotide encoding the selection marker is flanked with a site-specific recombination target sequences.
(190) The repair template can further comprise a reporter gene operably linked to a promoter, wherein the reporter gene encodes a reporter protein selected from the group consisting of or comprising LacZ, mPlum, mCherry, tdTomato, mStrawberry, J-Red, DsRed, mOrange, mKO, mCitrine, Venus, YPet, enhanced yellow fluorescent protein (eYFP), Emerald, enhanced green fluorescent protein (EGFP), CyPet, cyan fluorescent protein (CFP), Cerulean, T-Sapphire, luciferase, alkaline phosphatase, and/or a combination thereof. Such reporter genes can be operably linked to a promoter active in the cell. Such promoters can be an inducible promoter, a promoter

that is endogenous to the reporter gene or the cell, a promoter that is heterologous to the reporter gene or to the cell, a cell-specific promoter, a tissue-specific promoter, or a developmental stage-specific promoter.
(191) In certain embodiments, the repair template can comprise a mammalian nucleic acid comprising a genomic locus that encodes a protein expressed in the nervous system, the skeletal system, the digestive system, the circulatory system, the muscular system, the respiratory system, the cardiovascular system, the lymphatic system, the endocrine system, the urinary system, the reproductive system, or a combination thereof. In certain embodiments, the mammalian nucleic acid comprises a genomic locus that encodes a protein expressed in a bone marrow or a bone marrow-derived cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a spleen cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a hepatocyte.
(192) In certain embodiments, the mammalian nucleic acid comprises a genomic locus that encodes a protein expressed in the immune system, the lymphatic system, the endocrine system, the nervous system, the skeletal system, the digestive system, the circulatory system, the muscular system, the respiratory system, the cardiovascular system, the urinary system, the reproductive system, or a combination thereof. In certain embodiments, the mammalian nucleic acid comprises a genomic locus that encodes a protein expressed in a bone marrow or a bone marrow-derived cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a spleen cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a hepatocyte.
(193) In certain embodiments, the genomic locus comprises a mouse genomic DNA sequence, a rat genomic DNA sequence, eukaryotic genomic DNA sequence, a non-rat eukaryotic genomic DNA sequence, a mammalian genomic DNA sequence, a human genomic DNA sequence, or non-human DNA sequence mammalian, or a combination thereof. In certain embodiments, the genomic locus comprises, in any order, rat and human genomic DNA sequences. In certain embodiments, the genomic locus comprises, in any order, mouse and human genomic DNA sequences. In certain embodiments, the genomic locus comprises, in any order, mouse and rat genomic DNA sequences. In certain embodiments, the genomic locus comprises, in any order, rat, mouse, and human genomic DNA sequences.
(194) In certain embodiments, the genetic modification comprises at least one human disease allele of a human gene. In certain embodiments, the human disease is a neurological disease. In certain embodiments, the human disease is a cardiovascular disease. In certain embodiments, the human disease is a kidney disease. In certain embodiments, the human disease is a muscle disease. In certain embodiments, the human disease is a blood disease. In certain embodiments, the human disease is a bleeding disorder. In certain embodiments, the human disease is a cancer. In certain embodiments, the human disease is an immune system disease.
(195) In certain embodiments, the human disease allele is a dominant allele. In certain embodiments, the human disease allele is a recessive allele. In certain embodiments, the human disease allele comprises a single nucleotide polymorphism (SNP) allele.
(196) In certain embodiments, the genetic modification produces a mutant form of a protein with an altered binding characteristic, altered localization, altered expression, and/or altered expression pattern.
(197) In certain embodiments, the repair template comprises a selection cassette. In certain embodiments, the selection cassette comprises a nucleic acid sequence encoding a selective marker, wherein the nucleic acid sequence is operably linked to a promoter active in rat ES cells. In certain embodiments, the selective marker is selected from or comprises a hygromycin resistance gene or a neomycin resistance gene.
(198) In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a B cell. In certain embodiments, the nucleic acid comprises a genomic locus that

encodes a protein expressed in an immature B cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a mature B cell.
(199) In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a T cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in an immature T cell. In certain embodiments, the nucleic acid comprises a genomic locus that encodes a protein expressed in a mature T cell.
(200) In certain embodiments, the repair template comprises a regulatory element. In certain embodiments, the regulatory element is a promoter. In certain embodiments, the regulatory element is an enhancer. In certain embodiments, the regulatory element is a transcriptional repressor-binding element.
(201) In certain embodiments, the genetic modification comprises a deletion of a non-protein-coding sequence, but does not comprise a deletion of a protein-coding sequence. In certain embodiments, the deletion of the non-protein-coding sequence comprises a deletion of a regulatory element. In certain embodiments, the genetic modification comprises a deletion of a regulatory element. In certain embodiments, the genetic modification comprises an addition of a promoter or a regulatory element. In certain embodiments, the genetic modification comprises a replacement of a promoter or a regulatory element.
(202) F. Retroviral Particles
(203) The viral particles described herein are derived from viruses of the family Retroviridae. The viral particles described herein can be derived from retroviruses such as, but not limited to, rous sarcoma virus, human and bovine T-cell leukaemia virus (HTLV and BLV), lentiviruses (e.g., human and simian immunodeficiency viruses (HIV and SIV), Mason-Pfizer monkey virus), foamy viruses (e.g., Human Foamy Virus (HFV)), and herpes viruses (herpes simplex virus (HSV), varicella-zoster virus, VZVEBV, HCMV, HHV). Compared to other gene transfer systems, retroviral vectors offer a wide range of advantages, including their ability to transduce a variety of cell types, to stably integrate transferred genetic material into the genome of the target cell, and to express the transduced gene at significant levels. Vectors derived from the gamma-retroviruses, for example, the murine leukemia virus (MLV), have been used in clinical gene therapy trials (Ross et al., Hum. Gen Ther. 7:1781-1790, 1996).
(204) In one specific embodiment, the retroviral particles described herein are lentiviral particles. In one specific embodiment, the retroviral particle does not contain gp120 surface envelope protein and/or gp41 transmembrane envelope protein. In another specific embodiment, the retroviral particle contains a mutant gp120 surface envelope protein and/or a mutant gp41 transmembrane envelope protein and is not capable of binding to a cell in the absence of a targeting moiety.
(205) In some embodiments, the retroviral particle described herein comprises components from a virus selected from the group consisting of Human Immunodeficiency Virus (e.g., HIV-1 or HIV-2), Bovine Immunodeficiency Virus (BIV), Feline Immunodeficiency Virus (FIV), Simian Immunodeficiency Virus (SIV), Equine Infectious Anemia Virus (EIAV), Murine Stem Cell Virus (MSCV), Murine Leukemia Virus (MLV), Avian leukosis virus (ALV), Feline leukemia virus (FLV), Bovine leukemia virus (BLV), Human T-lymphotropic virus (HTLV), feline sarcoma virus, avian reticuloendotheliosis virus, caprine arthritis encephalitis virus (CAEV), and Visna-Maedi virus (VMV). Retroviral vectors encoding MLV are widely available to those skilled in the art, such as PINCO (Grignani et al., 1998) or the pBabe vector series (Morgenstern and Land, 1990).
(206) In some embodiments, the retroviral particles described herein are replication deficient and only contain an incomplete genome of the virus from which they are derived. For example, in some embodiments, the retroviral particles do not comprise the genetic information of the gag, env, and/or pol genes (which may be involved in the assembly of the retroviral particle), which is a known minimal requirement for successful replication of a retrovirus. In these cases, the minimal set of retroviral proteins needed to assemble the vector particle are provided in trans by means of a packaging cell line. In one specific embodiment, for lentiviral particles derived from HIV-1, env,

tat, vif, vpu and nef genes are lacking and are not provided in trans.
(207) G. Target Cells
(208) In certain embodiments, the invention provides a target cell comprising the system described herein. In certain embodiments, the invention provides a target cell transduced with the retroviral particles comprising a gene-editing molecule fusion RNA, and optionally a gene-editing molecule fusion protein and/or a gRNA, and/or a repair template (RT) as described herein or a composition comprising the aforementioned retroviral particles. In certain embodiments, target cells are those that have well characterized expression systems, have reasonably fast growth rates, and can be infected, transformed, transfected, or transduced easily and efficiently with a retroviral vector. The target cell can be any suitable eukaryotic cell known in the art including, for example, yeast cells, insect cells, and mammalian cells. In certain embodiments, the target cell is a mammalian cell.
(209) Target cells can be “autologous” or “allogeneic”. In certain embodiments, autologous target cells are removed from a subject, stored (and optionally modified), and returned back to the same subject. In certain embodiments, allogeneic target cells are removed from a donor, stored (and optionally modified), and transplanted into a genetically similar, but not identical, subject (i.e., recipient). Preferably, the target cells are autologous to the subject.
(210) In certain embodiments, the target cell is a somatic cell. The term “somatic cell” refers to any cell of a living organism other than the reproductive cells (e.g., other than a gamete, genu line cell, gametocyte, or undifferentiated stem cell). Somatic cells can be harvested from the subject or donor and used as a target cell in the context of the invention. Examples of somatic cells include, but are not limited to, keratinizing epithelial cells (e.g., keratinized epidermal cells), mucosal epithelial cells (e.g., epithelial cells of the superficial layer of tongue), exocrine gland epithelial cells (e.g., mammary gland cells), hormone-secreting cells (e.g., adrenomedullary cells), cells for metabolism or storage (e.g., liver cells), intimal epithelial cells constituting interfaces (e.g., type I alveolar cells), intimal epithelial cells of the obturator canal (e.g., vascular endothelial cells), cells having cilia with transporting capability (e.g., airway epithelial cells), cells for extracellular matrix secretion (e.g., fibroblasts), constrictive cells (e.g., smooth muscle cells), cells of the blood and the immune system (e.g., T lymphocytes), sense-related cells (e.g., bacillary cells), autonomic nervous system neurons (e.g., cholinergic neurons), sustentacular cells of sensory organs and peripheral neurons (e.g., satellite cells), nerve cells and glia cells of the central nervous system (e.g., astroglia cells), pigment cells (e.g., retinal pigment epithelial cells), and progenitor cells thereof (tissue progenitor cells). In certain embodiments, undifferentiated progenitor cells (including somatic stem cells) and differentiated mature cells can be used as sources of somatic cells. Examples of undifferentiated progenitor cells include, but are not limited to, tissue stem cells (somatic stem cells) such as nerve stem cells, hematopoietic stem cells (discussed above), mesenchymal stem cells, and dental pulp stem cells.
III. Methods of Producing Retroviral Particles
(211) In a related aspect, described herein is a method of producing a single retroviral particle that is capable of delivering a gene-editing fusion molecule and optionally a guide RNA (gRNA) and/or a repair template (RT). In certain embodiments, the gene-editing fusion molecule can be a Cas fusion molecule.
(212) A. Methods of Generating and Purifying Retroviral Particles
(213) Disclosed herein is a method of producing a single retroviral particle that is capable of delivering a gene-editing fusion molecule comprising culturing a packaging cell (described below) in conditions sufficient for the production of a plurality of retroviral particles, wherein the packaging cell comprises one or more plasmids comprising (i) one or more retroviral elements involved in the assembly of the retroviral particle and (ii) a nucleic acid sequence encoding a gene-editing fusion molecule. In certain embodiments, the packaging cell further comprises a plasmid encoding one or more gRNA and/or comprising a sequence corresponding to one or more repair templates (RT).

(214) In certain embodiments, the method comprises culturing a packaging cell in conditions sufficient for the production of a plurality of retroviral particles, wherein the packaging cell comprises one or more plasmids comprising (i) one or more retroviral elements involved in the assembly of the retroviral particle and (ii) a nucleic acid sequence encoding a Cas fusion RNA and/or a Cas fusion protein. In certain embodiments, the packaging cell further comprises a plasmid encoding one or more gRNA and/or comprising a sequence corresponding to one or more repair templates (RT).
(215) In some embodiments of any of the above methods, the method further comprises collecting the retroviral particles. In one specific embodiment, the collecting step comprises one or more of the following steps: clearing cell debris, treating the supernatant containing the retroviral particles with DNase I and MgCl.sub.2, concentrating the retroviral particles, and purifying the retroviral particles.
(216) In the methods described herein, plasmids/vectors used for retroviral particle production can be introduced into the packaging cells using methods well known in the art such as, e.g., electroporation (using for example Multiporator (Eppendorf), Genepulser (BioRad), MaxCyte Transfection Systems (Maxcyte)), PEI (Polysciences Inc. Warrington, Eppelheim), Ca2+-mediated transfection or via liposomes (for example: "Lipofectamine" (Invitrogen)), non-liposomal compounds (for example: "Fugene" (Roche) or nucleofection (Lonza)) into cells.
(217) In certain embodiments, the packaging cells are present in an in vitro culture and can be cultured in a manner that allows for mass production of the retroviral particles so as to provide suitable titers useful for production of the retroviral particle preparations for various applications (e.g., for clinical application in gene therapy). In certain embodiments, where large-scale production of viral particles is desirable, the packaging cell is preferably easy to culture, stable in long term culture (e.g., healthy cells can be maintained at relatively high cell densities for several days to several weeks or months; the cells do not undergo any significant genetic changes during culturing), and allows easy isolation of the viral particles from the cell culture (e.g., by collection and concentration of cell culture supernatant to provide a crude retroviral particle preparation of an acceptable viral titer).
(218) In certain embodiments, retroviral particles can be generated by trans-complementation in packaging cells that are co-transfected with a plasmid containing the retroviral genome and the packaging constructs that encode only the proteins essential for lentiviral assembly and function. A self-inactivating (SIN) lentiviral vector can be generated by eliminating the intrinsic promoter/enhancer activity of the HIV-1 LTR, which reduces the likelihood of aberrant expression of cellular coding sequences located adjacent to the vector integration site. See e.g., Naldini et al, (1996) *Science* 272:263-267; and Matrai et al., (2010) *Molecular Therapy* 18:477-490. In certain embodiments, the method of producing a lentiviral particle comprising co-transfecting packaging cells (e.g., 293T human embryonic kidney cells) with a lentiviral vector plasmid and three packaging constructs encoding the viral Gag-Pol, Rev-Tat, and envelope (Env) proteins. In certain embodiments, the lentiviral vector can integrate its genome into a target cell genome. In certain embodiments, the lentiviral vector is not integrated its genome into a target cell genome. In such cases, the lentiviral vector particle does not integrate its genome into a target cell genome (also referred to as a "non-integrating" vector). Non-integrating lentiviral vectors typically are generated by mutating the lentiviral integrase gene or by modifying the attachment sequences of the LTRs. See e.g., Sarkis et al., (2008) *Curr. Gene. Ther.* 6:430-437. In certain embodiments, lentiviral vectors can be produced by co-transfecting 293T human embryonic kidney cells with several different plasmid constructs, which separately contain the lentiviral cis-acting sequences and trans-acting factors that are required for retroviral particle production, infection, and integration. Lentiviral production protocols are further described in, for example, Tiscornia et al., (2006) *Nature Protocols* 1:241-245; Stevenson, M., (2002) *Curr. Top Microbiol. Immunol.* 261:1-30; Cronin et al., (2005) *Curr. Gene Ther.* 5:387-398; Sandrin et al., (2003) *Curr. Top. Microbiol. Immunol.* 281:137-

178; Zufferey, R., (2002) *Curr. Top. Microbiol. Immunol.* 261:107-121; Sinn et al, (2005) *Gene Ther.* 12:1089-1098; and Saenz, D. T. and Poeschla, E. M., (2004) *J. Gene Med.* 6:S95-S104. Other methods for producing lentiviral vectors are known in the art and described in, for example, U.S. Patent Application Publications 2008/0254008 and 2010/0003746; and Yang et al, (2012) *Hum Gene Ther. Methods* 23:73-83.

(219) For additional packaging techniques see, for example, Polo, et al, Proc Natl Acad Sci USA, (1999) 96:4598-4603. Methods of packaging include using packaging cells that permanently express the retroviral components, or by transiently transfecting cells with plasmids.

(220) B. Packaging Cells and Vectors

(221) Also disclosed herein is a packaging cell for producing the retroviral particles described herein comprising one or more plasmids comprising (i) one or more retroviral elements involved in assembly of the retroviral particles and (ii) a nucleic acid sequence encoding a gene-editing fusion molecule. In certain embodiments, the packaging cell further comprises one or more plasmids encoding one or more gRNA and/or comprising a sequence corresponding to one or more repair templates (RT).

(222) In certain embodiments, the invention provides a packaging cell for producing the retroviral particles described herein comprising one or more plasmids comprising (i) one or more retroviral elements involved in assembly of the retroviral particles and (ii) a nucleic acid sequence encoding a Cas fusion RNA and/or a Cas fusion protein. In certain embodiments, the packaging cell further comprises one or more plasmids encoding one or more gRNA and/or comprising one or more sequences corresponding to one or more repair templates (RT).

(223) Packaging cells useful for production of the retroviral particles described herein include, e.g., animal cells permissive for the virus, or cells modified so as to be permissive for the virus; or the packaging cell construct, for example, with the use of a transformation agent such as calcium phosphate. Non-limiting examples of packaging cell lines useful for producing retroviral particles described herein include, e.g., human embryonic kidney 293 (HEK-293) cells (e.g., American Type Culture Collection [ATCC] No. CRL-1573), HEK-293 cells that contain the SV40 Large T-antigen (HEK-293T or 293T), HEK293T/17 cells, human sarcoma cell line HT-1080 (CCL-121), lymphoblast-like cell line Raj i (CCL-86), glioblastoma-astrocytoma epithelial-like cell line U87-MG (HTB-14), T-lymphoma cell line HuT78 (TIB-161), NIH/3T3 cells, Chinese Hamster Ovary cells (CHO) (e.g., ATCC Nos. CRL9618, CCL61, CRL9096), HeLa cells (e.g., ATCC No. CCL-2), Vero cells, NIH 3T3 cells (e.g., ATCC No. CRL-1658), Huh-7 cells, BHK cells (e.g., ATCC No. CCL10), PC12 cells (ATCC No. CRL1721), COS cells, COS-7 cells (ATCC No. CRL1651), RATI cells, mouse L cells (ATCC No. CCLI.3), HLHepG2 cells, CAP cells, CAP-T cells, and the like.

(224) L929 cells, the FLY viral packaging cell system outlined in Cosset et al (1995) J Virol 69,7430-7436, NS0 (murine myeloma) cells, human amniocytic cells (e.g., CAP, CAP-T), yeast cells (including, but not limited to, *S. cerevisiae*, *Pichia pastoris*), plant cells (including, but not limited to, Tobacco NT1, BY-2), insect cells (including but not limited to SF9, S2, SF21, Tni (e.g. High 5)) or bacterial cells (including, but not limited to, *E. coli*).

(225) When generating retroviral (e.g., lentiviral) particles of the invention, 2-4 basic components, usually (but not necessarily) provided on separate plasmids, are used: (i) sequences encoding molecules involved in assembly of the lentiviral particle (e.g., a psi-negative gag/pol gene) provided on a packaging plasmid, (ii) sequences encoding a retroviral env protein or a binding molecule to replace the env protein (or as a fusion molecule with a retroviral env protein) provided on an envelope expression plasmid, optionally (iii) sequence(s) encoding one or more gRNA and/or sequence(s) corresponding to one or more repair templates (RT), e.g., provided on a transfer vector together, optionally with one or more retroviral elements needed for facilitating transfer of the transfer vector (e.g., psi packaging signal and LTR), and, optionally, (iv) a sequence encoding a fusogen provided on a fusogen encoding plasmid. The term “fusogen” or “fusogenic molecule” is used herein to refer to any molecule that can trigger membrane fusion when present on the surface

of a virus particle. A fusogen can be, for example, a protein (e.g., a viral glycoprotein) or a fragment, mutant or derivative thereof.

(226) Nucleic acids encoding a gene-editing molecule fusion protein (e.g., a Cas fusion RNA) can be transiently expressed in a target cell. A nucleic acid encoding a gene-editing molecule fusion protein can be operably linked to a promoter in an expression construct. Promoters that can be used in an expression construct include promoters active, for example, in one or more of a eukaryotic cell, a human cell, a non-human cell, a mammalian cell, a non-human mammalian cell, a rodent cell, a mouse cell, a rat cell, a hamster cell, a rabbit cell, a pluripotent cell, an embryonic stem (ES) cell, an adult stem cell, a developmentally restricted progenitor cell, an induced pluripotent stem (iPS) cell, or a one-cell stage embryo. Such promoters can be, for example, conditional promoters, inducible promoters, constitutive promoters, or tissue-specific promoters.

(227) In certain embodiments of any of the described methods, the packaging cell comprises an expression plasmid for expressing the gene-editing fusion molecule. In some embodiments of any of the described methods, the packaging cell comprises an expression plasmid for expressing the Cas fusion RNA and/or Cas fusion protein. Suitable expression plasmids are known to those of skill in the art. In certain embodiments, the expression plasmid is pRG984.

(228) In some embodiments of any of the described methods, the packaging cell further comprises one or more transfer vectors or an RNA molecule(s) encoded by the transfer vector, wherein the transfer vector or RNA molecule comprises at least one retroviral element and a sequence encoding one or more gRNA(s) and/or one or more sequences corresponding to one or more repair templates (RT) and, wherein the retroviral particles comprise the transfer vector(s) or RNA molecule(s). In another specific embodiment, the at least one retroviral element is a lentiviral element. In one specific embodiment, the at least one retroviral element is a Psi (ψ) packaging signal. In one specific embodiment, in addition to a Psi (ψ) packaging signal, the retroviral element further comprises a 5′ Long Terminal Repeat (LTR) and/or a 3′ LTR, or a derivative or mutant thereof. In one specific embodiment, the at least one retroviral element is selected from the group consisting of a 5′ Long Terminal Repeat (LTR), a Psi (ψ) packaging signal, a Rev Response Element (RRE), a promoter that drives expression of a gRNA (e.g., H1 or U6), a Central Polypurine Tract (cPPT), a Woodchuck hepatitis virus post-transcriptional regulatory element (WPRE), a Unique 3′ (U3), a Repeat (R) region, a Unique 5′ (U5), a 3′ LTR, a 3′ LTR with the U3 element deleted (e.g., to make the lentivirus non-replicative), a Trans-activating response element (TAR), and any combination thereof.

(229) In some embodiments the transfer vector preferably comprises at least one RNA Polymerase II or III promoter. The RNA Polymerase II or III promoter is operably linked to the nucleic acid sequence of interest and can also be linked to a termination sequence. RNA polymerase II and III promoters are well known to one of skill in the art. A suitable range of RNA polymerase III promoters can be found, for example, in Paule and White. Nucleic Acids Research., Vol 28, pp 1283-1298 (2000); Ohkawa and Taira Human Gene Therapy, Vol. 11, pp 577-585 (2000); Meissner et al. Nucleic Acids Research, Vol. 29, pp 1672-1682 (2001). Non-limiting examples of useful promoters include, e.g., cytomegalovirus (CMV)-promoter, the spleen focus forming virus (SFFV)-promoter, the elongation factor 1 alpha (EF1a)-promoter (the 1.2 kb EF1a-promoter or the 0.2 kb EF1a-promoter), the chimeric EF 1 a/IF4-promoter, and the phospho-glycerate kinase (PGK)-promoter. An internal enhancer may also be present in the retroviral construct to increase expression of the gene of interest. For example, the CMV enhancer (Karasuyama et al. 1989. J. Exp. Med. 169:13) may be used. In some embodiments, the CMV enhancer can be used in combination with the chicken β-actin promoter. One of skill in the art will be able to select the appropriate enhancer based on the desired expression pattern. In addition, transfer vector may contain one or more genetic elements designed to enhance expression of the gene of interest. For example, a woodchuck hepatitis virus responsive element (WRE) may be placed into the construct (Zufferey et al. 1999. J. Virol. 74:3668-3681; Deglon et al. 2000. Hum. Gene Ther. 11:179-190).

(230) In some embodiments of any of the described methods, the packaging cell further comprises one or more packaging vectors. In certain embodiments, a packaging plasmid comprises (a) GAG, (b) POL, (c) TAT and/or REV retroviral (e.g., lentiviral) elements, each of which may be considered involved with the assembly of the retroviral particle. In certain embodiments, the packaging plasmid is psPAX2.
(231) In some embodiments of any of the above methods, the one or more plasmids comprise an envelope plasmid. In certain embodiments, an envelope plasmid comprises (a) but not limit to VSV-G, Ebola virus envelope, MLV envelope, LCMV envelope, Rabies virus envelope and/or (b) PolyA. In certain embodiments, the envelope plasmid can be pMD2.G.
(232) In certain embodiments, the envelope of the retroviral particle can be pseudotyped. Pseudotyping is to alter the tropism of the retroviral particle or for generating an increased or decreased stability of a retrovirus particle. As such, foreign viral envelope proteins (heterologous envelope proteins) are introduced into retroviral particle and are typically glycoproteins derived from portions of the membrane of the virus infected host cells or glycoproteins encoded by the virus genome. The structural envelope proteins (e.g., Env, VP1, VP2, or VP3) can determine the range of target cells that can ultimately be infected and transformed by recombinant retroviruses. In the case of lentiviruses, such as HIV-1, HIV-2, SIV, FIV and EIV, the Env proteins include gp41 and gp120. When producing recombinant retroviruses (e.g., recombinant lentiviruses), a wild type retroviral (e.g., lentiviral)) env, vp1, vp2, or vp3 gene can be used, or can be substituted with any other viral env, vp1, vp2, or vp3 gene from another lentivirus or AAV or other virus (such as vesicular stomatitis virus GP (VSV-G)). Methods of pseudotyping recombinant viruses with envelope proteins from other viruses in this manner are well known in the art (see, e.g., WO 99/61639, WO 98/05759, Mebatsion et al., Cell 90:841-847 (1997); Cronin et al., Curr. Gene Ther. 5:387-398 (2005)).
(233) In some embodiments of any of the above methods, the plasmids present in the packaging cell do not comprise a retroviral ENV gene or comprise a mutant non-functional ENV gene. In another embodiment, the one or more plasmids comprise a mutant lentiviral ENV gene, which does not produce gp120 surface envelope protein or gp41 transmembrane envelope or which encodes a mutant gp120 surface envelope protein and/or a mutant gp41 transmembrane envelope protein and wherein the resulting retroviral particle is not capable of binding to a target cell in the absence of an exogenous protein that specifically binds to the target cell.
(234) In some embodiments of any of the above methods, retroviral particle further comprises a fusogen. Many different protein and non-protein fusogens can be used. In some embodiments, the fusogen is a protein. In one specific embodiment, the fusogen is a viral protein. Non-limiting examples of useful viral fusogens include, e.g., vesiculovirus fusogens (e.g., vesicular stomatitis virus G glycoprotein (VSVG)), alphavirus fusogens (e.g., a Sindbis virus glycoprotein), orthomyxovirus fusogens (e.g., influenza HA protein), paramyxovirus fusogens (e.g., a Nipah virus F protein or a measles virus F protein) as well as fusogens from Dengue virus (DV), Lassa fever virus, tick-borne encephalitis virus, Dengue virus, Hepatitis B virus, Rabies virus, Semliki Forest virus, Ross River virus, Aura virus, Borna disease virus, Hantaan virus, SARS-CoV virus, and various fragments, mutants and derivatives thereof. Other exemplary fusogenic molecules and related methods are described, for example, in U.S. Pat. Appl. Pub. 2005/0238626 and 2007/0020238.
(235) In one specific embodiment, the fusogen is heterologous to the virus from which the particle is derived. In one specific embodiment, the fusogen is a mutated protein which does not bind its natural ligand.
(236) There are two recognized classes of viral fusogens and both can be used as targeting molecules (D. S. Dimitrov, Nature Rev. Microbio. 2, 109 (2004)). The class I fusogens trigger membrane fusion using helical coiled-coil structures, whereas the class II fusogens trigger fusion with 13 barrels. In some embodiments, class I fusogens are used. In other embodiments, class II

fusogens are used. In still other embodiments, both class I and class II fusogens are used. See, e.g., Skehel and Wiley, Annu. Rev. Biochem. 69, 531-569 (2000); Smit, J. et al. J. Virol. 73, 8476-8484 (1999), Morizono et al. J. Virol. 75, 8016-8020 (2005), Mukhopadhyay et al. (2005) Rev. Microbiol. 3, 13-22.

(237) In some specific embodiments, a form of hemagglutinin (HA) from influenza A/fowl plague virus/Rostock/34 (FPV), a class I fusogen, is used (Hatziioannou et al., J. Virol. 72, 5313 (1998)). In some specific embodiments, a form of FPV HA is used (Lin et al., Hum. Gene. Ther. 12, 323 (2001)). HA-mediated fusion is generally considered to be independent of receptor binding (Lavillette et al., Cosset, Curr. Opin. Biotech. 12, 461 (2001)).

(238) In other embodiments, the Sindbis virus glycoprotein (a class II fusogen) from the alphavirus family is used (Wang et al., J. Virol. 66, 4992 (1992); Mukhopadhyay et al., Nature Rev. Microbio. 3, 13 (2005), Morizono et al., Nature Med. 11, 346 (2005)).

(239) In some embodiments, mutant fusogens are used which maintain their fusogenic ability but have a decreased or eliminated binding ability or specificity. Functional properties of mutant fusogens can be tested, e.g., in cell culture or by determining their ability to stimulate an immune response without causing undesired side effects in vivo.

(240) To select most effective and non-toxic combinations of pseudotyping protein and fusogens (either wild type or mutant), retroviral particles bearing these molecules can be tested for their selectivity and/or their ability to facilitate penetration of the target cell membrane. Retroviral particles that display wild type fusogens can be used as controls for examining titer effects in mutants. For example, cells can be transduced by the retroviral particles using a standard infection assay. After a specified time, for example 48 hours post-transduction, cells can be collected, and the percentage of transduced cells can be determined by, for example, monitoring reporter gene expression (e.g., using FACS analysis). The selectivity can be scored by calculating the percentage of cells infected by the retroviral particles. Similarly, the effect of mutations on viral titer can be quantified by dividing the percentage of cells infected by retroviral particles comprising a mutant targeting molecule by the percentage of cells infected by retrovirus comprising the corresponding wild type targeting molecule. The titers of retroviral particles can be determined, e.g., by limited dilution of the stock solution and transduction of cells expressing the proteins of interest. A preferred mutant will give the best combination of selectivity and infectious titer.

(241) To investigate whether fusogen-mediated cell penetration is dependent upon pH, and select fusogens with the desired pH dependence, NH.sub.4Cl or other compound that alters pH can be added at the infection step (NH.sub.4Cl will neutralize the acidic compartments of endosomes). In the case of NH.sub.4Cl, the disappearance of cells expressing the reporter will indicate that penetration of viruses is low pH-dependent. In addition, to confirm that the activity is pH-dependent, lysosomotropic agents, such as ammonium chloride, chloroquine, concanamycin, bafilomycin Al, monensin, nigericin, etc., may be added into the incubation buffer. These agents can elevate the pH within the endosomal compartments (Drose and Altendorf, J. Exp. Biol. 200, 1-8, 1997). The inhibitory effect of these agents will reveal the role of pH for viral fusion and entry. The different entry kinetics between retroviruses displaying different fusogenic molecules may be compared and the most suitable selected for a particular application.

(242) PCR-based retroviral particle entry assays may be utilized to measure kinetics of viral DNA synthesis as an indication of the kinetics of retroviral particle entry. For example, retroviral particles comprising a particular pseudotyping molecule and fusogen can be incubated with target cells, unbound retroviruses can be then removed, and aliquots of the cells can be analyzed by extracting DNA and performing semi-quantitative PCR (e.g., using LTR-specific primers for retroviral particles). The appearance of LTR-specific DNA products will indicate the success of retroviral particle entry and uncoating.

(243) In some embodiments, the pseudotyping molecule and a fusogen are two separate molecules. In other embodiments, they form a single fusion protein.

(244) In preferred embodiments, the fusogen is a viral glycoprotein that mediates fusion or otherwise facilitates delivery of the nucleic acid of interest to the target cell. The fusogen preferably exhibits fast enough kinetics that the retroviral particle contents can empty into the cytosol before the degradation of the retroviral particle. In addition, the fusogen can be modified to reduce or eliminate any binding activity and thus reduce or eliminate any non-specific binding. That is, by reducing the binding ability of the fusogen, binding of the retroviral particles to the target cell is determined predominantly or entirely by the pseudotyping molecule, allowing for high target specificity and reducing undesired side-effects.
(245) The measles virus (MeV), a prototype morbillivirus of the genus Paramyxovi{acute over (η)}dae, utilizes two envelope glycoproteins (the fusion protein (F) and the hemagglutinin protein (H)) to gain entry into the target cell. Protein F is a type I transmembrane protein, while protein H is a type II transmembrane domain, i.e., its amino-terminus is exposed directly to the cytoplasmic region. Both proteins thus comprise a transmembrane and a cytoplasmic region. One known function of the F protein is mediating the fusion of viral membranes with the cellular membranes of the target cell. Functions attributed to the H protein include recognizing the receptor on the target membrane and supporting F protein in its membrane fusion function. The direct and highly efficient membrane fusion at the cellular surface membrane is a particular property of measles virus and the morbilliviruses, thus distinguishing themselves from many other enveloped viruses that become endocytosed and will only fuse upon pH drop upon endocytosis. Both proteins are organized on the viral surface in a regular array of tightly packed spikes, H tetramers, and F trimers (Russell et al., Virology 199:160-168, 1994).
(246) In certain embodiments, the fusogenic molecule is a Sindbis virus envelope protein (SIN). The SINdbis virus transfers its RNA into the cell by low pH mediated membrane fusion. SIN contains five structural proteins, E1, E2, E3, 6K and capsid. E2 contains the receptor binding sequence that allows the wild type SIN to bind, while E1 is known to contain the properties necessary for membrane fusion (Konoochik et al., Virology Journal 2011, 8:304). E1, E2, and E3 are encoded by a polyprotein, the amino acid sequence of which is provided, e.g., by Accession No. VHWVB, VHWVB2, and P03316: the nucleic acid sequence is provided, e.g., by Accession No. SVU90536 and V01403 (see also Rice & Strauss, Proc. Nat'l Acad. Sci USA 78:2062-2066 (1981); and Strauss et al., Virology 133:92-110 (1984)).
(247) In certain embodiments, the Sindbis virus envelope protein is mutated (SINmu). In certain embodiments, the mutation reduces the natural tropism of the Sindbis virus. In certain embodiments, a SINmu comprising SIN proteins E1, E2, and E3, wherein at least one of E1, E2, or E3 is mutated as compared to a wild type sequence. For example, one or more of the E1, E2, or E3 proteins can be mutated at one or more amino acid positions. In addition, combinations of mutations in E1, E2, and E3 are encompassed by fusogen as described herein, e.g., mutations in E1 and E2, or in E2 and E3, or E3 and E1, or E1, E2, and E3. In certain embodiments, at least E2 is mutated.
(248) In certain embodiments, the SINmu comprises the following envelope protein mutations in comparison to wild type Sindbis virus envelope proteins: (i) deletion of E3 amino acids 61-64; (ii) E2 KE159-160AA; and (iii) E2 SLKQ68-71AAAA (SEQ ID NOS 144 and 145). In a further embodiment, the SINmu additionally comprises the envelope protein mutation E1 AK226-227SG. Examples of SINmu may be found in, for example, in U.S. Pat. No. 9,163,248; WO2011011584; Cronin et. al., Curr Gene Ther. 2005 August; 5(4): 387-398.
(249) Other Togaviridae family envelopes, e.g., from the Alphavirus genus, e.g., Semliki Forest Virus, Ross River Virus, and equine encephalitis virus, can also be used to pseudotype the vectors described herein. The envelope protein sequences for such Alphaviruses are known in the art.
(250) In certain embodiments, the fusogen is a vesicular stomatitis virus (VSV) envelope protein. In certain embodiments, the fusogen is the G protein of VSV (VSV-G; Burns et al., Proc. Natl. Acad. Sci. U.S.A. 1993, vol. 90, no. 17, p. 1833-′7) or a fragment, mutant, derivative or homolog

thereof. VSV-G interacts with a phospholipid component of the cell (e.g., T cell) membrane to mediate viral entry by membrane fusion (Mastromarino et al., J Gen Virol. 1998, vol. 68, no. 9, p. 2359-69; Marsh et al., Adv Virus Res. 1989, vol. 107, no. 36, p. 107-51. Examples of VSV-G may be found in, for example, WO2008058752.
(251) The constructs described herein may also contain additional genetic elements. The types of elements that may be included in the constructs are not limited in any way and will be chosen by the skilled practitioner to achieve a particular result. For example, a signal that facilitates nuclear entry of the RNA corresponding to the transfer vector in the target cell may be included. An example of such a signal is the HIV-1 flap signal.
(252) For additional packaging cells and systems and vectors for packaging the nucleic acid genome into the retroviral particle (including pseudotyped retroviral particles) see, for example, Polo, et al, Proc Natl Acad Sci USA, (1999) 96:4598-4603.
IV. Methods of Treatment and Use
(253) The retroviral particles described herein can be used for various therapeutic applications (in vivo and ex vivo) and as research tools. In one aspect, described herein is a method for treating a disease in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of the retroviral particle of the invention or the pharmaceutical composition thereof.
(254) In certain embodiments, target cells may be transduced with the lentiviral particle in vivo, ex vivo or in vitro.
(255) In certain embodiments, the method is for targeted editing of a gene in a target cell comprising introducing into the target cell the retroviral particle of the invention by infection. In certain embodiments, the method is for targeted deletion or addition of a gene in a target cell comprising introducing into the target cell the retroviral particle of the invention. In certain embodiments, the method is for modulating an activity of a gene in a target cell comprising introducing into the target cell the retroviral particle of the invention. In certain embodiments, the method inhibits, suppresses, down regulates, knocks down, knocks out, or silences the expression of a gene product. In certain embodiments, the gene product is a protein. In certain embodiments, the gene product is an RNA. In certain embodiments, the retroviral particle is administered to a subject, wherein the target cell is in the subject. In certain embodiments, the target cell is harvested from the subject prior to introducing the retroviral particle into the target cell, wherein the retroviral particle is introduced to the target cell ex vivo.
(256) In certain embodiments, the method is for treating a disease in a subject in need thereof, the method comprising administering to the subject a therapeutically effective amount of the retroviral particle of the invention or the pharmaceutical composition of the invention. In certain embodiments, the retroviral particle or the pharmaceutical composition is administered, by way of example and not limitation, intravenously, subcutaneously, intramuscularly, transdermally, intranasally, orally, or mucosally.
(257) In certain embodiments, the method is for treating a disease in a subject in need thereof, the method comprising: a) harvesting a target cell from the subject; b) introducing into the target cell from step a) a therapeutically effective amount of the retroviral particle of the invention or the pharmaceutical composition of the invention; and c) returning the target cell from step b) to the subject.
(258) In certain embodiments, the target cell is a HSC. In certain embodiments, the HSCs are transduced in vitro with the retroviral particle or composition comprising the retroviral particle followed by infusion of the transduced stem cells into the subject. For example, the subject's (e.g., human's) stem cell can be removed from the subject using methods well known in the art and transduced as described above. The transduced HSCs are then reintroduced into the same (autologous) or different subject (allogeneic).
(259) Once harvested and transduced with the retroviral particle or composition in vitro, HSCs can

be cultured under suitable conditions known in the art (e.g., Csaszar et al., *Cell Stem Cell*, 10(2): 218-29 (2012); Madlambayan et al., *Biol Blood Marrow Transplant.*, 72(10): 1020-1030 (2006); Woods et al., *Stem Cells*, 29(1): 1 158-1 164 (2011); U.S. Patent Application Publications 2002/0061293 and 2012/0071397; and International Patent Application Publication WO 2014/043131; or using commercially available systems (e.g., Life Technologies Corp. and Stem Cell Technologies, Inc.).

(260) In certain embodiments, the target cell can be a somatic cell. In certain embodiments, the somatic cell can be transduced in vitro with the retroviral particle or composition comprising the retroviral particle and cultured under conditions to generate induced pluripotent stem cells (also known as iPS cells or iPSCs). An iPS cell is a type of pluripotent stem cell that can be generated directly from somatic (e.g., adult) cells by introducing appropriate reprogramming factors into somatic cells. A “reprogramming factor” refers to any substance(s) capable of inducing an iPS cell from a somatic cell, and can be a proteinaceous molecule, a nucleic acid sequence encoding same, or a low-molecular compound. Reprogramming factors typically used to generate iPS cells include, but are not limited to, the four genes Oct3/4, Sox2, Klf4, and c-Myc. See e.g., U.S. Pat. No. 8,951,801; International Patent Application Publication WO 2007/069666; and Takahashi, K. and Yamanaka, S., Cell, 126: 663-676 (2006)).

(261) In certain embodiments, the target cell can be a mammalian induced pluripotent stem (iPS) cell that can be derived from various types of somatic cells. The iPS cell preferably is transduced in vitro with the inventive system or composition comprising the inventive system and can be differentiated into hematopoietic stem cells, red blood cells, or other suitable cell type.

(262) In certain embodiments, the somatic cell is a hepatocyte. For example, the present invention also relates to the delivery to the liver, for gene therapy of liver conditions or the creation of liver models. Liver or liver tissue includes parenchymal cells commonly referred to as hepatocytes. Liver or Liver tissue can also be liver cells that are non-parenchymal cells (e.g., sinusoidal hepatic endothelial cells, upffer cells and hepatic stellate cells). Cells of the liver express one or more liver gene product(s). In certain embodiments, the invention is directed to the liver, whether that is the organ per se or a tissue within it or simply one or more liver cells, e.g., hepatocytes. Primary hepatocytes are preferred.

(263) Hepatic targets include, but are not limited to amyloid neuropathy (TTR, PALB); Amyloidosis (APOA1, APP, AAA, CVAP, AD1, GSN, FGA, LYZ, TTR, PALB); Cirrhosis (RT18, T8, CIRH1A, NAIC, TEX292, KJAA1988); cystic fibrosis (CFTR, ABCC7, CF, MRP7); glycogen storage diseases (SLC2A2, GLUT2, G6PC, G6PT, G6PT1, GAA, LAMP2, LAMPB, AGL, GDE, GBE1, GYS2, PYGL, PFKM); hepatic adenoma, 142330 (TCF1, HNF1A, MODY3), hepatic failure, early onset, and neurologic disorder (SCOD1, SCO1), Hepatic lipase deficiency (LIPC), hepatoblastoma, cancer and carcinomas (CTNNB1, PDGFRL, PDGRL, PRLTS, AXIN1, AXFN, CTNNB1, TP53, P53, LFS1, IGF2R, MPRI, MET, CASP8, MCH5; Medullary cystic kidney disease (UMOD, HNFJ, FJHN, MCKD2, ADMCKD2); phenylketonuria (PAH, PKU1, QDPR, DHPR, PTS); Polycystic kidney and hepatic disease (FCYT, PKHD1, ARPKD, PKD1, PKD2, PKD4, PKDTS, PRKCSH, G19P1, PCLD, SEC63); blood clotting factors (factor V, factor VII, factor VIII, factor IX, factor X, factor XI, factor XII, factor XIII, prothrombin, fibrinogen, von Willebrand factor or recombinant soluble tissue factor (rsTF) or activated forms of any of the preceding); PCSK9; Hmgcr; SERPINA1; ApoB; and/or LDL.

(264) In certain embodiments, the somatic cell is a lymphocyte. In certain embodiments, the lymphocyte is a T cell. In certain embodiments, the lymphocyte is a B cell. Non-limiting examples of lymphocyte targets include TCRs, BCRs, regulatory elements, artificial TCR like molecules, CARs, etc.

(265) Somatic cells harvested from a subject such as human or mouse can be pre-cultured using any suitable medium known in the art, depending on the cell type. Examples of such media include, but are not limited to, a minimal essential medium (MEM) comprising about 5 to 20% fetal calf

serum, Dulbecco's modified Eagle medium (DMEM), RPMI1640 medium, 199 medium, F12 medium, and the like. Methods for culturing somatic cells to produce iPS cells are described in, for example, U.S. Pat. No. 8,951,801, International Patent Application Publication WO 2007/069666; and Takahashi, K. and Yamanaka, S., supra.

(266) In certain aspects, the invention provides a method of altering a DNA sequence in a target cell, the method comprising contacting a target cell comprising a DNA sequence comprising a target locus with the retroviral particle described herein. In certain embodiments, at least one gRNA sequence binds to the DNA sequence in the target cell genome, and the Cas9 protein, or functional fragment or derivative thereof, induces a double strand break in the DNA sequence, thereby altering a DNA sequence in a target cell. In certain embodiments, at least one ZFns and/or TALENs, or functional fragments or derivatives thereof, induces a double strand break in the DNA sequence, thereby altering a DNA sequence in a target cell. In certain embodiments, the method further comprises a repair template. Descriptions of the retroviral particle, the guide RNA sequence, repair template, the target cell, target locus, and gene-editing molecules, or functional fragment or derivative thereof, (e.g., Cas protein, or functional fragment or derivative thereof) are set forth above.

(267) In certain embodiments, the gene-editing molecule cleaves a target locus, to produce double strand DNA breaks. The double strand breaks can be repaired by the target cell by either non-homologous end joining (NHEJ) or homologous recombination. For example, in NHEJ, the double-strand breaks can be repaired by direct ligation of the broken ends to one another. As such, no new nucleic acid material is inserted into the target locus—although, some nucleic acid material may be lost, resulting in a deletion. Homologous recombination entails a repair in which a repair template comprising a second DNA sequence with homology to the cleaved target locus sequence is used as a template for repair of the cleaved target locus sequence, resulting in the transfer of genetic information from the repair template to the target locus. As a result, new nucleic acid material is inserted/copied into the DNA break site. These methods lead to, for example but not limited to, gene correction, gene replacement, gene tagging, transgene insertion, nucleotide deletion, gene disruption, gene mutation, and/or gene knockdown.

(268) In certain embodiments, the invention further comprises a repair template comprising a second DNA sequence, which can be different from the first DNA sequence of the target locus. For example, the first DNA sequence of the target cell can be replaced with the second DNA sequence by homologous recombination following gene-editing-induced cleavage of the first DNA sequence of the target locus. When the retroviral particles are used to correct one or more mutations or defects in a gene, the target locus contains a first DNA sequence that encodes a defective protein and the repair template contains a second DNA sequence that encodes a wild type or corrective version of the defective protein. For example, the first DNA sequence can be a gene associated with a disease, which refers to any gene or polynucleotide whose gene products are expressed at an abnormal level or in an abnormal form in the cells obtained from a subject affected by the disease. In certain embodiments, the disease-associated gene may be expressed at an abnormally high level, where the altered expression correlates with the occurrence and/or progression of the disease. In certain embodiments, the disease-associated gene may be expressed at an abnormally low level, where the altered expression correlates with the occurrence and/or progression of the disease. In certain embodiments, a disease-associated gene can be a mutation or genetic variation of the gene, which is directly or indirectly responsible for the etiology of a disease.

(269) In certain embodiments, the method can be used to delete nucleic acids from a target locus in a target cell by cleaving the target sequence and allowing the target cell to repair the cleaved sequence in the absence of a repair template. Deletion of a nucleic acid sequence in this manner can be used to, as an example but not a limitation, create gene knock-outs or knock-downs, knock-ins and generate mutations for disease models in research.

(270) In certain embodiments, the method can be used to knock-in a nucleic acid that encodes by

way of example but not limitation, a protein, an siRNA, an miRNA, etc or a tag (e.g., FLAG, HA, His, GPF), a regulatory sequence to a gene (e.g., a promoter, polyadenylation signal, internal ribosome entry sequence (IRES), 2A peptide, start codon, stop codon, splice signal, localization signal, etc.), or to modify a nucleic acid sequence (e.g., introduce a mutation).

V. Pharmaceutical Compositions, Dosage Forms and Administration

(271) Also disclosed herein are pharmaceutical compositions comprising the retroviral particles described herein and a pharmaceutically acceptable carrier and/or excipient. In addition, disclosed herein are pharmaceutical dosage forms comprising the retroviral particle described herein.

(272) Pharmaceutical compositions based on the vector particles disclosed herein can be formulated in any conventional manner using one or more physiologically acceptable carriers and/or excipients. The vector particles may be formulated for administration by, for example, injection, inhalation or insulation (either through the mouth or the nose) or by oral, buccal, parenteral or rectal administration, or by administration directly to a tumor.

(273) The pharmaceutical compositions can be formulated for a variety of modes of administration, including systemic, topical or localized administration. Techniques and formulations can be found in, for example, Remmington's Pharmaceutical Sciences, Meade Publishing Co., Easton, Pa. For systemic administration, injection is preferred, including intramuscular, intravenous, intraperitoneal, and subcutaneous. For the purposes of injection, the pharmaceutical compositions can be formulated in liquid solutions, preferably in physiologically compatible buffers, such as Hank's solution or Ringer's solution. In addition, the pharmaceutical compositions may be formulated in solid form and redissolved or suspended immediately prior to use. Lyophilized forms of the pharmaceutical composition are also suitable.

(274) For oral administration, the pharmaceutical compositions may take the form of, for example, tablets or capsules prepared by conventional means with pharmaceutically acceptable excipients such as binding agents (e.g. pregelatinized maize starch, polyvinylpyrrolidone or hydroxypropyl methylcellulose); fillers (e.g. lactose, microcrystalline cellulose or calcium hydrogen phosphate); lubricants (e.g. magnesium stearate, talc or silica); disintegrants (e.g. potato starch or sodium starch glycolate); or wetting agents (e.g. sodium lauryl sulfate). The tablets can also be coated by methods well known in the art. Liquid preparations for oral administration may take the form of, for example, solutions, syrups or suspensions, or they may be presented as a dry product for constitution with water or other suitable vehicle before use. Such liquid preparations may be prepared by conventional means with pharmaceutically acceptable additives such as suspending agents (e.g. sorbitol syrup, cellulose derivatives or hydrogenated edible fats); emulsifying agents (e.g. lecithin or acacia); non-aqueous vehicles (e.g. ationd oil, oily esters, ethyl alcohol or fractionated vegetable oils); and preservatives (e.g. methyl or propyl-p-hydroxybenzoates or sorbic acid). The preparations can also contain buffer salts, flavoring, coloring and sweetening agents as appropriate.

(275) The pharmaceutical compositions can be formulated for parenteral administration by injection, e.g. by bolus injection or continuous infusion. Formulations for injection can be presented in a unit dosage form, e.g. In ampoules or in multi-dose containers, with an optionally added preservative. The pharmaceutical compositions can further be formulated as suspensions, solutions or emulsions in oily or aqueous vehicles, and may contain other agents including suspending, stabilizing and/or dispersing agents.

(276) Additionally, the pharmaceutical compositions can also be formulated as a depot preparation. These long acting formulations can be administered by implantation (e.g. subcutaneously or intramuscularly) or by intramuscular injection. Thus, for example, the compounds may be formulated with suitable polymeric or hydrophobic materials (e.g. as an emulsion in an acceptable oil) or ion exchange resins, or as sparingly soluble derivatives, for example, as a sparingly soluble salt. Other suitable delivery systems include microspheres, which offer the possibility of local noninvasive delivery of drugs over an extended period of time. This technology can include

microspheres having a precapillary size, which can be injected via a coronary catheter into any selected part of an organ without causing inflammation or ischemia. The administered therapeutic is men slowly released from the microspheres and absorbed by the surrounding cells present in the selected tissue.
(277) Systemic administration can also be by transmucosal or transdermal means. For transmucosal or transdermal administration, penetrants appropriate to the barrier to be permeated are used in the formulation. Such penetrants are generally known in the art, and include, for example, for transmucosal administration, bile salts, and fusidic acid derivatives. In addition, detergents may be used to facilitate permeation. Transmucosal administration can occur using nasal sprays or suppositories. For topical administration, the vector particles described herein can be formulated into ointments, salves, gels, or creams as generally known in the art. A wash solution can also be used locally to treat an injury or inflammation in order to accelerate healing.
(278) Pharmaceutical forms suitable for injectable use can include sterile aqueous solutions or dispersions; formulations including sesame oil, peanut oil or aqueous propylene glycol; and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. In all cases, the form must be sterile and must be fluid. It must be stable under the conditions of manufacture and certain storage parameters (e.g. refrigeration and freezing) and must be preserved against the contaminating action of microorganisms, such as bacteria and fungi.
(279) If formulations disclosed herein are used as a therapeutic to boost an immune response in a subject, a therapeutic agent can be formulated into a composition in a neutral or salt form. Pharmaceutically acceptable salts, include the acid addition salts (formed with the free amino groups of the protein) and which are formed with inorganic acids such as, for example, hydrochloric or phosphoric acids, or such organic acids as acetic, oxalic, tartaric, mandelic, and the like. Salts formed with the free carboxyl groups can also be derived from inorganic bases such as, for example, sodium, potassium, ammonium, calcium, or ferric hydroxides, and such organic bases as isopropylamine, trimethylamine, histidine, procaine and the like.
(280) A carrier can also be a solvent or dispersion medium containing, for example, water, ethanol, polyol (for example, glycerol, propylene glycol, and liquid polyethylene glycol, and the like), suitable mixtures thereof, and vegetable oils. The proper fluidity can be maintained, for example, by the use of a coating, such as lecithin, by the maintenance of the required particle size in the case of dispersion and by the use of surfactants. The prevention of the action of microorganisms can be brought about by various antibacterial and antifungal agents known in the art. In many cases, it will be preferable to include isotonic agents, for example, sugars or sodium chloride. Prolonged absorption of the injectable compositions can be brought about by the use in the compositions of agents delaying absorption, for example, aluminum monostearate and gelatin.
(281) Sterile injectable solutions can be prepared by incorporating the active compounds or constructs in the required amount in the appropriate solvent with various of the other ingredients enumerated above, as required, followed by filtered sterilization.
(282) Upon formulation, solutions can be administered in a manner compatible with the dosage formulation and in such amount as is therapeutically effective. The formulations are easily administered in a variety of dosage forms, such as the type of injectable solutions described above, but slow release capsules or microparticles and microspheres and the like can also be employed.
(283) For parenteral administration in an aqueous solution, for example, the solution should be suitably buffered if necessary and the liquid diluent first rendered isotonic with sufficient saline or glucose. These particular aqueous solutions are especially suitable for intravenous, intratumorally, intramuscular, subcutaneous and intraperitoneal administration. In this context, sterile aqueous media that can be employed will be known to those of skill in the art in light of the present disclosure. For example, one dosage could be dissolved in 1 ml of isotonic NaCl solution and either added to 1000 ml of hypodermoclysis fluid or injected at the proposed site of infusion.
(284) The person responsible for administration will, in any event, determine the appropriate dose

for the individual subject. For example, a subject may be administered retroviral particles described herein on a daily or weekly basis for a time period or on a monthly, bi-yearly or yearly basis depending on need or exposure to a pathogenic organism or to a condition in the subject (e.g. cancer).

(285) In addition to the compounds formulated for parenteral administration, such as intravenous, intratumorally, intradermal or intramuscular injection, other pharmaceutically acceptable forms include, e.g., tablets or other solids for oral administration; liposomal formulations; time release capsules; biodegradable and any other form currently used.

(286) One may also use intranasal or inhalable solutions or sprays, aerosols or inhalants. Nasal solutions can be aqueous solutions designed to be administered to the nasal passages in drops or sprays. Nasal solutions can be prepared so that they are similar in many respects to nasal secretions. Thus, the aqueous nasal solutions usually are isotonic and slightly buffered to maintain a pH of 5.5 to 7.5. In addition, antimicrobial preservatives, similar to those used in ophthalmic preparations, and appropriate drug stabilizers, if required, may be included in the formulation. Various commercial nasal preparations are known and can include, for example, antibiotics and antihistamines and are used for asthma prophylaxis.

(287) Oral formulations can include excipients as, for example, pharmaceutical grades of mannitol, lactose, starch, magnesium stearate, sodium saccharine, cellulose, magnesium carbonate and the like. These compositions take the form of solutions, suspensions, tablets, pills, capsules, sustained release formulations or powders. In certain defined embodiments, oral pharmaceutical compositions will include an inert diluent or assimilable edible carrier, or they may be enclosed in hard or soft shell gelatin capsule, or they may be compressed into tablets, or they may be incorporated directly with the food of the diet. For oral therapeutic administration, the active compounds may be incorporated with excipients and used in the form of ingestible tablets, buccal tablets, troches, capsules, elixirs, suspensions, syrups, wafers, and the like.

(288) The tablets, troches, pills, capsules and the like may also contain the following: a binder, as gum tragacanth, acacia, cornstarch, or gelatin; excipients, such as dicalcium phosphate; a disintegrating agent, such as corn starch, potato starch, alginic acid and the like; a lubricant, such as magnesium stearate; and a sweetening agent, such as sucrose, lactose or saccharin may be added or a flavoring agent, such as peppermint, oil of wintergreen, or cherry flavoring. When the dosage unit form is a capsule, it may contain, in addition to materials of the above type, a liquid carrier. Various other materials may be present as coatings or to otherwise modify the physical form of the dosage unit. For instance, tablets, pills, or capsules may be coated with shellac, sugar or both. A syrup of elixir may contain the active compounds sucrose as a sweetening agent methyl and propylparabens as preservatives, a dye and flavoring, such as cherry or orange flavor.

(289) Further embodiments disclosed herein can concern kits for use with methods and compositions. Kits can also include a suitable container, for example, vials, tubes, mini- or microfuge tubes, test tube, flask, bottle, syringe or other container. Where an additional component or agent is provided, the kit can contain one or more additional containers into which this agent or component may be placed. Kits herein will also typically include a means for containing the retroviral particles and any other reagent containers in close confinement for commercial sale. Such containers may include injection or blow-molded plastic containers into which the desired vials are retained. Optionally, one or more additional active agents such as, e.g., anti-inflammatory agents, anti-viral agents, anti-fungal or anti-bacterial agents or anti-tumor agents may be needed for compositions described.

(290) Dose ranges and frequency of administration can vary depending on the nature of the retroviral particles and the medical condition as well as parameters of a specific patient and the route of administration used. In some embodiments, retroviral particle compositions can be administered to a subject at a dose ranging from about 1×10.sup.5 plaque forming units (pfu) to about 1×10.sup.15 pfu, depending on mode of administration, the route of administration, the

nature of the disease and condition of the subject. In some cases, the retroviral particle compositions can be administered at a dose ranging from about 1×10.sup.6 pfu to about 1×10.sup.8 pfu, or from about 1×10.sup.8 pfu to about 1×10.sup.12 pfu. A more accurate dose can also depend on the subject in which it is being administered. For example, a lower dose may be required if the subject is juvenile, and a higher dose may be required if the subject is an adult human subject. In certain embodiments, a more accurate dose can depend on the weight of the subject.
(291) Compositions disclosed herein may be administered by any means known in the art.
(292) For example, compositions may include administration to a subject intravenously, intratumorally, intradermally, intraarterially, intraperitoneally, intralesionally, intracranially, intraarticularly, intraprostaticaly, intrapleurally, intratracheally, intranasally, intravitreally, intravaginally, intrarectally, topically, intratumorally, intramuscularly, intrathecally, subcutaneously, subconjunctival, intravesicularlly, mucosally, intrapericardially, intraumbilically, intraocularly, orally, locally, by inhalation, by injection, by infusion, by continuous infusion, by localized perfusion, via a catheter, via a lavage, in a cream, or in a lipid composition.
(293) Any method known to one skilled in the art maybe used for large scale production of retroviral particles, packaging cells and vector constructs described herein. For example, master and working seed stocks may be prepared under GMP conditions in qualified primary CEFs or by other methods. Packaging cells may be plated on large surface area flasks, grown to near confluence and retroviral particles purified. Cells may be harvested, and retroviral particles released into the culture media isolated and purified, or intracellular retroviral particles released by mechanical disruption (cell debris can be removed by large-pore depth filtration and packaging cell DNA digested with endonuclease). retrovirus particles may be subsequently purified and concentrated by tangential-flow filtration, followed by diafiltration. The resulting concentrated bulk maybe formulated by dilution with a buffer containing stabilizers, filled into vials, and lyophilized. Compositions and formulations may be stored for later use. For use, lyophilized retroviral particles may be reconstituted by addition of diluent.
(294) Certain additional agents used in the combination therapies can be formulated and administered by any means known in the art.
(295) Compositions as disclosed herein can also include adjuvants such as aluminum salts and other mineral adjuvants, tensoactive agents, bacterial derivatives, vehicles and cytokines. Adjuvants can also have antagonizing immunomodulating properties. For example, adjuvants can stimulate Th1 or Th2 immunity. Compositions and methods as disclosed herein can also include adjuvant therapy.

EXAMPLES

(296) The present invention is also described and demonstrated by way of the following examples. However, the use of these and other examples anywhere in the specification is illustrative only and in no way limits the scope and meaning of the invention or of any exemplified term. Likewise, the invention is not limited to any particular preferred embodiments described here. Indeed, many modifications and variations of the invention may be apparent to those skilled in the art upon reading this specification, and such variations can be made without departing from the invention in spirit or in scope. The invention is therefore to be limited only by the terms of the appended claims along with the full scope of equivalents to which those claims are entitled.

Example 1: Analysis of the Ability of CypA and Vpr Fusions to Effectively Enrich Cas9 Protein into Lentiviral Particles

(297) This example investigated the ability of CyclophilinA (CypA) and Vpr fusions to effectively enrich Cas9 protein into lentiviral particles. FIG. **1**A depicts retroviral particle structure with Cas9 protein fusions and lentiviral genome carrying a gRNA that targets the GFP nucleotide sequence (SEQ ID NO: 59) driven by hU6 promoter (pLVX hU6 GFP gRNA IRES puro, SEQ ID NO: 60). CypA is a host protein interacting with lentivirus Gag proteins and highly enrich in lentiviral particles (Human immunodeficiency virus type 1 Gag protein binds to cyclophilins A and B Cell.

1993 Jun. 18; 73(6):1067-78). Vpr is a lentiviral accessory protein. Two Cas9 constructs, pRG984 CypA-Cas9 (SEQ ID NO: 61) and pRG984 Cas9-Vpr (SEQ ID NO: 62) were made to express CypA-Cas9 and Cas9-Vpr fusion proteins. The ability of CypA and Vpr to enrich functional Cas9 proteins into lentiviral particles was assayed by measuring the level of eGFP knockout in a cell, wherein eGFP was introduced by lentivirus and named 293 T pLVX EF1a eGFP IRES puro c12 cell (SEQ ID NO: 63). Once the functional Cas9/GFP gRNA targets the GFP nucleotide sequence, the cells modified to express GFP will not fluoresce when detected by FACS. Constructs

(298) pLVX EF1a eGFP IRES puro (SEQ ID NO: 63). The eGFP DNA was inserted into pLVX EF1a IRES puro (Clontech) vector using Spe1 and Not 1 restriction sites.

(299) pLVX hU6 GFP gRNA EF1a IRES puro (SEQ ID NO: 60). Gblock containing the hU6 promoter and GFP guide RNA (gRNA) sequence (GGGCGAGGAGCTGTTCACCG; SEQ ID NO: 59) was ordered and cloned into pLVX EF1a IRES puro plasmid at the Cla1 restriction site using sequence and ligation independent cloning.

(300) pRG984 Cas9 (SEQ ID NO: 64). 5 g blocks (from Integrated DNA technologies) of Cas9 gene with overhangs were ordered from IDT and cloned into pRG984 plasmid using Gibson assembly methods.

(301) pRG984 CypA-Cas9 (SEQ ID NO: 61). Gblock containing CypA nucleotide sequences was ordered from IDT and inserted into pRG984 Cas9 plasmid using Xho1 restriction site and sequence and ligation-independent cloning (Methods Mol Biol., 2012; 852:51-9).

(302) pRG984 Cas9-Vpr (SEQ ID NO: 62). Gblock containing Vpr nucleotide sequences were ordered from IDT and inserted into pRG984 Cas9 plasmid using Not1 restriction site and sequence and ligation-independent cloning.

(303) TABLE-US-00004 Plasmids used in making lentiviral particles in FIG. 2 Panel Envelope Helper Genome Cas9 B pMD2.G psPAX2 pLVX EF1a IRES pRG984 Cas9 puro SEQ ID NO: 68 SEQ ID NO: 64 (Clontech Cat# 631988) C pMD2.G psPAX2 pLVX hU6 gGFP pRG984 Cas9 EF1a IRES puro SEQ ID NO: 64 SEQ ID NO: 60 D pMD2.G psPAX2 pLVX hU6 gGFP pRG984 CypA-Cas9 EF1a IRES puro SEQ ID NO: 61 SEQ ID NO: 60 E pMD2.G psPAX2 pLVX hU6 gGFP pRG984 Cas9-Vpr EF1a IRES puro SEQ ID NO: 62 SEQ ID NO: 60

Lentiviral Particle Production

(304) Lentiviral production was performed using three- or four-plasmid transfection method. Cells are plated one day prior to PEFpro (Polyplus transfection, New York, NY)-mediated transfection with appropriate vectors, one envelope plasmid, pMD2.G, one plasmid containing lentiviral gag/pol psPAX2 and one genome plasmid. When preparing a lentivirus carrying Cas9 mRNA, one more plasmid expressing Cas9 was added. For each 10 cm plate, 4 μg of envelope and gag/pol plasmids, 8 μg of genome plasmid and 8 μg of Cas9 plasmid (if required) were mixed with PEIpro at 1:1 ratio (1 μg DNA: 1 μPEIpro). 48 hours after transfection, the culture medium was collected, filtered through 0.45 μm filter, and treated with DNase I at 37° C. for 1 hour. Then the medium was ultra-centrifuged at 25000 rpm for 90 min. Pellet was suspended with PBS buffer. Virol titer was measured by quantitative PCR (qPCR) using Lenti-X™ qRT-PCR Titration Kit (Takara) according to the manufacturer's protocol.

(305) Construction of 293T pLVX EF1a eGFP IRES Puro c12 Cell Line

(306) Lentivirus with LVX EF1a eGFP IRES (SEQ ID NO: 60) puro genome was produced and titrated. 1E7 vg of virus was added to 1E7 cells in 15 cm cells. 24 hours after infection, cells were put in 1 μg/ml selection for 7 days. Single colonies were picked and analyzed by FACS. Clone 12 with low GFP intensity was used in the following experiment.

(307) 1E10 vg of lentiviral particles were added to 50,000 293T pLVX EF1a eGFP IRES puro c12 cells. Six days after infection, the GFP signal was analyzed by FACS. Without GFP gRNA, Cas9/non-gRNA lentiviral particles did not knock out GFP (FIG. **2**B). In cells infected with Cas9/GFP gRNA lentiviral particles, about 16.4% of cells were GFP negative which indicates that Cas9 protein could be randomly packaged into lentiviral particles. Cells infected with CypA-

Cas9/GFP gRNA lentiviral particles were 82.2% GFP negative (FIG. **2**D), which indicates that fusion to CypA sequence enriches Cas9 protein in lentiviral particles. Cells infected with Cas9-Vpr/GFP gRNA lentiviral particles were 94.9% GFP negative (FIG. **2**E), which indicates that fusion to Vpr sequence enriches Cas9 protein in lentiviral particles.

Example 2: Recruitment of Cas9 Proteins and mRNAs in Lentiviral Particles

(308) This example investigated the use of 7SL RNA-Cas9 mRNA fusion RNA molecules (FIGS. **1**A and B) to enrich Cas9 mRNA in lentiviral particles. This study took advantage of the fact that 7SL RNA, highly enriched in retroviral particles via interacting with Gag protein (J Virol. 2010 September; 84(18): 9070-9077), is the most abundant non-viral RNA found in lentiviruses (see, e.g., Eckwahl et al., mBio, 2016, 7(1):e02025-15). In particular, lentiviral particles were produced in a cell culture in the presence of the Cas9 mRNA (SEQ ID NO: 2), 7SL RNA-Cas9 mRNA fusion RNA (SEQ ID NO: 3) or Cas9 mRNA-7SL RNA fusion RNA (SEQ ID NO: 4). The ability of 7SL RNA to enrich Cas9 mRNA into lentiviral particles was assayed by measuring the level of eGFP knockout in a cell, wherein eGFP was introduced by lentivirus and named 293 T pLVX EF1a GFP IRES puro c12 cell (SEQ ID NO: 63). Once the functional Cas9/GFP gRNA targets the GFP nucleotide sequence, the cells modified to express GFP will not fluoresce when detected by FACS.

(309) Constructs

(310) pRG984 7SL RNA-Cas9 (SEQ ID NO: 66). Gblocks containing 7SL RNA sequence were ordered and inserted into pRG984 Cas9 plasmid using Spe1 restriction site and sequence and ligation-independent cloning (SLIC) to make pRG984 7SL RNA-Cas9, in which the 7SL sequence is located before the start codon of Cas9.

(311) pRG984 Cas9-7SL RNA (SEQ ID NO: 67). Gblocks containing 7SL RNA sequence were ordered and inserted into pRG984 Cas9 plasmid using Not1 restriction site and sequence and ligation-independent cloning (SLIC) to make pRG984 Cas9-7SL RNA, in which the 7SL sequence is located at after the stop codon of Cas9.

(312) TABLE-US-00005 TABLE 3 Plasmids used in making lentiviral particles in FIG. 2 Panels Envelope Helper Genome Cas9 B pMD2.G psPAX2 pLVX EF1a IRES puro pRG984 Cas9 SEQ ID NO: 68 SEQ ID NO: 64 C pMD2.G psPAX2 pLVX hU6 gGFPEF1a pRG984 Cas9 IRES puro SEQ ID NO: 64 SEQ ID NO: 60 F pMD2.G psPAX2 pLVX hU6 gGFPEF1a pRG984 IRES puro 7SL-Cas9 SEQ ID NO: 60 SEQ ID NO: 66 G pMD2.G psPAX2 pLVX hU6 gGFPEF1a pRG984 IRES puro Cas9-7SL SEQ ID NO: 60 SEQ ID NO: 67

(313) The amount of Cas9 mRNA recruited in the lentiviral particles was assayed using qPCR and primers specific for Cas9 [Cas9F GACAGGCACAGCATCAAGAA (SEQ ID NO: 69) Cas9R TTCTGGCGGTTCTCTTCAGT (SEQ ID NO: 70)]. Assembly of lentiviral particles in the presence of the Cas9 mRNA-7SL RNA fusion RNA surprisingly resulted in a 4-fold increase in the amount of Cas9 mRNA in lentiviral particles as compared to the amount of Cas9 mRNA in lentiviral particles assembled in the presence of Cas9 mRNA without 7SL RNA fusion (FIG. **3**).

(314) 1E10 vg of lentiviral particles were added to 50,000 293T pLVX EF1a eGFP IRES puro c12 cells. Six days after infection, the GFP signal was analyzed by FACS. Without GFP gRNA, Cas9/non-gRNA lentiviral particles did not knock out GFP (FIG. **2**B). In cells infected with Cas9/GFP gRNA lentiviral particles, about 17% of cells were GFP negative, which indicates that Cas9 proteins or mRNA could be randomly packaged into lentiviral particles. Cells infected with 7SL Cas9/GFP gRNA or Cas9 7SL/GFP gRNA lentiviral particles were 93% or 97% GFP negative (FIGS. **2**F and G), which indicates that fusion to 7SL RNA sequence enriches Cas9 mRNA in lentiviral particles.

Example 3: Testing of Functionality of the Cas9 mRNA 7SL RNA Fusions Incorporated in Lentiviral Particles by Assaying Rescue of Zombie LacZ

(315) Zombie LacZ is a defective lacZ which contains a mutation of Q530E (FIG. **4**). 293T cells were transfected with pShuttle CMV zombie lacZ (SEQ ID NO: 71). 24 hours after transfection, lentiviral particles which contain zombie lacZ gRNA (SEQ ID NO: 72 and repair template (SEQ ID

NO: 73) in their genome and Cas9 mRNA-7SL RNA fusion RNA (SEQ ID NO: 4) were added to the cells (FIG. 5A). 96 hours after infection, cells were lysed and lacZ activity was analyzed by the Beta-Glo® Assay System (Promega). Infection with lentiviral particles carrying lacZ gRNA and repair template resulted in slightly higher lacZ activity than the background which indicates background recombination between repair template and zombie lacZ. Infection with lentiviral particles carrying zombie lacZ gRNA and repair template (RT) in their genome and Cas9 mRNA-7SL RNA fusion RNA significantly increased the lacZ activity.
(316) Constructs
(317) pShuttle CMV Zombie lacZ (SEQ ID NO: 71). LacZ gene was cloned into pShuttle CMV vector using XhoI and HindIII restriction site to make pShuttle CMV lacZ. Mutation Q530 (GAA) was introduced into pShuttle CMV lacZ to make pShuttle CMV zombie lacZ by SLIC methods. Two fragments with overhangs were amplified by PCR using primers listed below and put back into pShuttle CMV lacZ vector digested by XhoI and HindIII site.
(318) TABLE-US-00006 lacZF (SEQ ID NO: 74) CGACGCGGCCGCTCGAG (XhoI) QtoER (SEQ ID NO: 75) TGGGCGTATTGGCAAAGGAT QtoEF (SEQ ID NO: 76) ATCCTTTGCCAATACGCCCA lacZR (SEQ ID NO: 77) CGGATATCTTATCTAGAAGCTT
(319) pLVX hU6 gZlacZ repair template (RT) (SEQ ID NO: 78). Gblock containing hU6 promoter and zombie lacZ gRNA was ordered form IDT and cloned into pLVX EF1a IRES Puro vector at ClaI site. 2.8 kb of lacZ repair template with PAM site mutated was synthesized and replaced the EF1a promoter in pLVX EF1a IRES puro.
(320) TABLE-US-00007 TABLE 4 Plasmids used in making lentiviral particles Construct Envelope Helper Genome Cas9 (ii) pMD2.G psPAX2 pLVX hU6 zombie N/A lacZ gRNA repair template SEQ ID NO: 78 (iii) pMD2.G psPAX2 pLVX hU6 zombie pRG984 lacZ gRNA Cas9-7SL repair template SEQ ID SEQ ID NO: 78 NO: 67 (iv) - pMD2.G psPAX2 pLVX hU6 zombie pRG984 gRNA/RT/Cas9 lacZ gRNA Cas9-7SL 7S particle repair template SEQ ID SEQ ID NO: 78 NO: 67 (iv) - Cas9 pMD2.G psPAX2 pLVX CAGG particle Cas9 SEQ ID NO: 79
Example 4: Dual Enrichment Methods to Enrich Cas9 Proteins and mRNA
(321) As described in example 3, providing Cas9 in trans could increase homologous recombination activity. To introduce more Cas9 into a target cell, dual enrichment methods could be used. CypA-Cas9-7SL RNA fusion or Cas9-Vpr-7SL RNA fusion is made to enrich both Cas9 proteins and mRNA in lentiviral particles to introduce more Cas9 into a target cell. Such a method can provide extended Cas9 activity in the host cell by providing two phases of Cas9 protein—direct from the Cas9 fusion protein and a second phase of Cas9 protein via translation from the Cas9 fusion mRNA.
Example 5: Engineering T Cells for Immunotherapy of Cancer
(322) This example outlines the use of the lentiviral particles of the invention for cancer immunotherapy by engineering T cells to express chimeric antigen receptors (CARs) while eliminating the T cells' native T cell receptor (TCR) and/or programmed cell death-1 (PD1) receptor. To promote the specific targeting of cancer cells, T cells can be transduced with CARs which redirect the T cells against antigens expressed at the surface of target cancer cells. The native TCR can be eliminated from the engineered T cells to prevent recondition of host tissue as foreign by the TCR and to avoid graft versus host disease. The PD1 expressed in the transduced T cell can be eliminated to prevent suppression of the T cell's anti-tumor activity. T cells are harvested from a blood sample from an individual patient or from a blood bank and activated using anti-CD3/CD28 activator beads. The cells are then transduced with lentiviral particles containing a multi-chain CAR derived from FcεRI directed towards cancer antigen CD20 driven by EF1α promoter in their genome and CypA-Cas9 fusion protein and CypA-Cas9-7SL RNA fusion RNA and a gRNA targeting TCRα, a gRNA targeting PD1, or both. 96 hours after infection, the TCR inactivated and CAR positive T cells are sorted using FACS and expanded in vitro prior to administration to the patient (or in vivo following administration to the patient) through stimulation of CD3 complex and

are administered to the patient for the treatment of cancer.

(323) T cells could also be transduced with lentiviral particles containing gRNA targeting TCRα and/or PD1 and a homologous recombination template with a multi-chain CAR derived from FcεRI directed towards cancer antigen CD20 flanked by homologous arms targeting TCRα in their genome and CypA-Cas9 fusion protein and CypA-Cas9-7SL RNA fusion RNA. CAR is integrated into TCRα locus and expressed by endogenous TCR promoter. The expression of TCRα is disrupted simultaneously. 96 hours after infection, the TCR inactivated and CAR positive T cells are sorted using FACS and expanded in vitro prior to administration to the patient (or in vivo following administration to the patient) through stimulation of CD3 complex and are administered to the patient for the treatment of cancer.

Example 6: Treatment of Sickle Cell Disease

(324) This example outlines the use of the lentiviral particles of the invention for repairing sickle cell anemia-causing mutations within the β-globin (HBB) gene. The presence of atypical HBB gene cluster haplotypes within the red blood cells results in sickle cell anemia. Subject derived haematopoietic stem cells are harvested and incubated with lentiviral particles which contain HBB gRNA and repair template in their genome and CypA-Cas9 fusion protein and CypA-Cas9-7SL RNA fusion RNA. 96 hours after infection, cells are tested for the presence of anemia-causing mutations (or their reparation) by sequencing. Upon achieving 90% or more mutation reparation, the cells are transplanted back into the subject. Subject's bone marrow is tested for the presence of the repaired cells after 16 weeks.

(325) The claimed subject matter is not to be limited in scope by the specific embodiments described herein. Indeed, various modifications of the claimed subject matter in addition to those described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are intended to fall within the scope of the appended claims.

(326) All patents, applications, publications, test methods, literature, and other materials cited herein are hereby incorporated by reference in their entirety as if physically present in this specification.

Claims

1. A recombinant RNA molecule comprising (i) a sequence of a gene-editing molecule mRNA, or a sequence of a functional fragment thereof, and (ii) a sequence of a non-coding enrichment RNA, wherein said non-coding enrichment RNA comprises 7SL RNA, or a functional fragment thereof, and is capable of enhancing inclusion of said gene-editing molecule mRNA, or functional fragment thereof, into a retroviral particle.
2. The RNA molecule of claim 1, wherein the gene-editing molecule is a Cas protein.
3. The RNA molecule of claim 2, wherein the Cas protein is a Cas9 protein.
4. The RNA molecule of claim 1, wherein the 7SL RNA fragment comprises the Alu domain, the S domain, the 5c helix, or a combination thereof.
5. The RNA molecule of claim 1, wherein the retroviral particle is a lentiviral particle.
6. An isolated nucleic acid molecule encoding the RNA molecule of claim 1.
7. A vector comprising the nucleic acid molecule of claim 6, wherein said nucleic acid molecule is operably linked to a promoter.
8. An isolated host cell comprising the RNA molecule of claim 1.
9. An isolated host cell comprising the nucleic acid molecule of claim 6.
10. An isolated host cell comprising the vector of claim 7.
11. A recombinant retroviral particle comprising the RNA molecule of claim 1.
12. The retroviral particle of claim 11, wherein the retroviral particle is a lentiviral particle.
13. The retroviral particle of claim 11, further comprising a nucleic acid molecule encoding one or more guide RNAs (gRNA) and/or a nucleic acid molecule comprising one or more nucleic acid

sequences corresponding to one or more repair templates (RT).
14. The retroviral particle of claim 11, further comprising a gene-editing molecule fusion protein comprising (i) a sequence of a gene-editing protein, or a sequence of a functional fragment thereof, and (ii) a sequence of an enrichment protein, or a sequence of a functional fragment thereof, wherein said enrichment protein, or functional fragment thereof, is capable of enhancing inclusion of said gene-editing protein, or functional fragment thereof, into the retroviral particle.
15. The retroviral particle of claim 14, wherein the gene-editing molecule is a Cas protein.
16. The retroviral particle of claim 15, wherein the Cas protein is a Cas9 protein.
17. The retroviral particle of claim 14, wherein the enrichment protein is cyclophilin A (CypA) protein and/or a viral protein R (Vpr).
18. A method of producing a recombinant retroviral particle comprising the RNA molecule of claim 1, said method comprising culturing a packaging cell in conditions sufficient for the production of a plurality of retroviral particles, wherein the packaging cell comprises one or more plasmids comprising (i) one or more retroviral elements involved in the assembly of the retroviral particle, and (ii) a nucleic acid sequence encoding the RNA molecule of claim 1.
19. The method of claim 18, wherein the packaging cell further comprises a plasmid encoding one or more guide RNAs (gRNA) and/or comprising one or more sequences corresponding to one or more repair templates (RT).
20. The method of claim 18, wherein the packaging cell comprises (a) GAG, (b) POL, and (c) TAT and/or REV retroviral elements.
21. The method of claim 18, comprising one or more of the following steps: a) clearing cell debris, b) treating a supernatant containing the retroviral particles with DNase I and MgCl.sub.2, c) concentrating the retroviral particles, and d) purifying the retroviral particles.
22. A pharmaceutical composition comprising the retroviral particle of claim 11 and a pharmaceutically acceptable carrier or excipient.
23. A pharmaceutical dosage form comprising the retroviral particle of claim 11.
24. A method for modifying a genome of a target cell or modulating an activity of a gene in a target cell comprising introducing into said cell the retroviral particle of claim 11.
25. The method of claim 24, wherein the target cell is in a subject and the retroviral particle is administered to the subject.
26. The method of claim 24, further comprising harvesting the target cell from a subject prior to introducing the retroviral particle into the target cell, introducing the retroviral particle into the target cell ex vivo and returning the target cell to the subject.
27. A method for treating a disease in a subject in need thereof, said method comprising administering to the subject a therapeutically effective amount of the retroviral particle of claim 11, wherein the retroviral particle targets a cell in the subject.
28. A method for treating a disease in a subject in need thereof, said method comprising: a) harvesting a target cell from the subject; b) introducing into the target cell from step (a) ex vivo a therapeutically effective amount of the retroviral particle of claim 11; and c) returning the target cell from step (b) to the subject.